



SMART MANAGER ERP

MASTER SYSTEM BOOK
ENGLISH AND TANZANIAN SWAHILI EDITION

Your Business. Connected. Controlled. Intelligent.

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The Vision Behind SMART MANAGER



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SMART MANAGER is documented here as a repository-audited product whose source, database, authentication, security, subscription, and operational boundaries are explicitly described.

Dar es Salaam, Tanzania

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1 Editorial and Evidence Note

This is a repository-audited master book. English is the primary technical reference; the complete Tanzanian Swahili edition follows the English reference. Status labels are evidence boundaries, not marketing claims. The live Supabase audit was read-only and its snapshots are retained in the evidence directory.

Metric	Value	Interpretation	Evidence
Project version	1.0.0	Package version	package.json
Live tables	542	519 public; 23 auth	Read-only Supabase inventory
RLS	535	7 not enabled in returned inventory	Read-only Supabase inventory
Migrations	133	Records in live ledger	Read-only Supabase migration list
Security lints	119	WARN/INFO advisor records	Supabase security advisor
Performance lints	851	WARN/INFO advisor records	Supabase performance advisor

2 Part I — Product Introduction

2.1 I.1 Executive Purpose

The book is a repository-audited reference, not a generic ERP brochure. It distinguishes implemented source and database evidence from configuration requirements, partial contracts, external services, and planned work.

2.1.1 Purpose, audience, evidence boundary, and how to use this book.

Kitabu hiki kinaeleza bidhaa, uendeshaji, usanifu, usalama na mipaka ya ushahidi kwa wamiliki, watumiaji, wasimamizi, developers, support teams na auditors.

2.2 I.2 Product Philosophy

The design intent is to make business work measurable, responsibilities visible, and decisions easier to explain. The evidence boundary is part of the product quality model.

2.2.1 The platform connects records, people, workflows, control surfaces, and decision views without claiming that every problem is automated.

Falsafa ya bidhaa ni kuunganisha rekodi, watu, mitiririko, udhibiti na maoni ya uongozi bila kudai kwamba kila tatizo litatatuliwa kiotomatiki.

2.3 I.3 Business Value

Value is created when a real operational fact is captured once, validated at the server boundary, enforced in the database, and surfaced through the right role-aware view.

2.3.1 Smart Manager helps organizations reduce fragmented records, improve visibility, protect access, and create repeatable operational paths.

Smart Manager husaidia kupunguza rekodi zilizotawanyika, kuongeza mwonekano, kulinda ufikiaji na kujenga mitiririko inayoweza kurudiwa.

2.4 I.4 Audience and Adoption

Adoption should start with a bounded workflow, a real company context, verified data, and a measured review rhythm rather than a promise to activate every module at once.

2.4.1 Different roles use different evidence: owners review control, operators execute workflows, accountants reconcile, administrators govern, and developers extend.

Kila jukumu hutumia ushahidi tofauti: wamiliki huangalia udhibiti, operators hutekeleza workflows, accountants hufanya reconciliation, admins hutawala na developers huendeleza.

2.5 I.5 Evidence and Status Language

A status never means that a feature is universally complete for every deployment. It describes the strongest evidence found during this audit window.

2.5.1 Status words are delivery controls: IMPLEMENTED, TESTING, PARTIALLY IMPLEMENTED, CONFIGURATION REQUIRED, EXTERNAL SERVICE REQUIRED, and BLOCKED.

Maneno ya hali ni udhibiti wa delivery: IMEJENGWA, INAJARIBIWA, IMEJENGWA KWA SEHEMU, INAHITAJI USANIDI, INAHITAJI HUDUMA YA NJE na IMEZUIWA.

3 Part II — System Overview

3.1 II.1 What is SMART MANAGER?

SMART MANAGER is an authenticated business operating platform whose verified repository surfaces connect commercial, operational, financial, people, sector, intelligence, and control workflows.

3.1.1 Kiswahili

SMART MANAGER ni platform ya uendeshaji wa biashara yenye uthibitishaji inayounganisha workflows za biashara, operations, fedha, watu, sekta, intelligence na udhibiti.

3.2 II.2 ERP concept

ERP is treated as a connected record and control model: a sale may relate to stock, finance, customer history, receipt, and report when relevant contracts are enabled.

3.2.1 Kiswahili

ERP inaeleweka kama mfumo wa rekodi na udhibiti uliunganishwa: mauzo yanaweza kuhusishwa na stock, fedha, historia ya mteja, receipt na ripoti pale contracts zimewezeshwa.

3.3 II.3 System ecosystem

The ecosystem includes public entry, secure authentication, company/workspace context, role-aware modules, Supabase persistence, server APIs, scheduled handlers, storage, external providers, and audit evidence.

3.3.1 Kiswahili

Ecosystem ina entry ya umma, authentication, company/workspace, modules za role, Supabase, APIs za server, scheduled handlers, storage, providers wa nje na audit.

3.4 II.4 Supported business types

The verified module registry spans universal SME operations and specialist contexts such as property, healthcare, school, pharmacy, hospitality, restaurant, fleet, microfinance, banking, VICOBA, and money-agent operations.

3.4.1 Kiswahili

Rejista ya modules inahusisha SME operations na sekta kama property, afya, shule, famasi, hoteli, mgahawa, fleet, microfinance, benki, VICOBA na wakala wa fedha.

3.5 II.5 Lifecycle model

A safe lifecycle is discover → authenticate → establish company scope → record → validate → approve/reconcile → report → audit → improve.

3.5.1 Kiswahili

Lifecycle salama ni gundua → authenticate → weka company scope → rekodi → thibitisha → approve/reconcile → ripoti → audit → boresha.

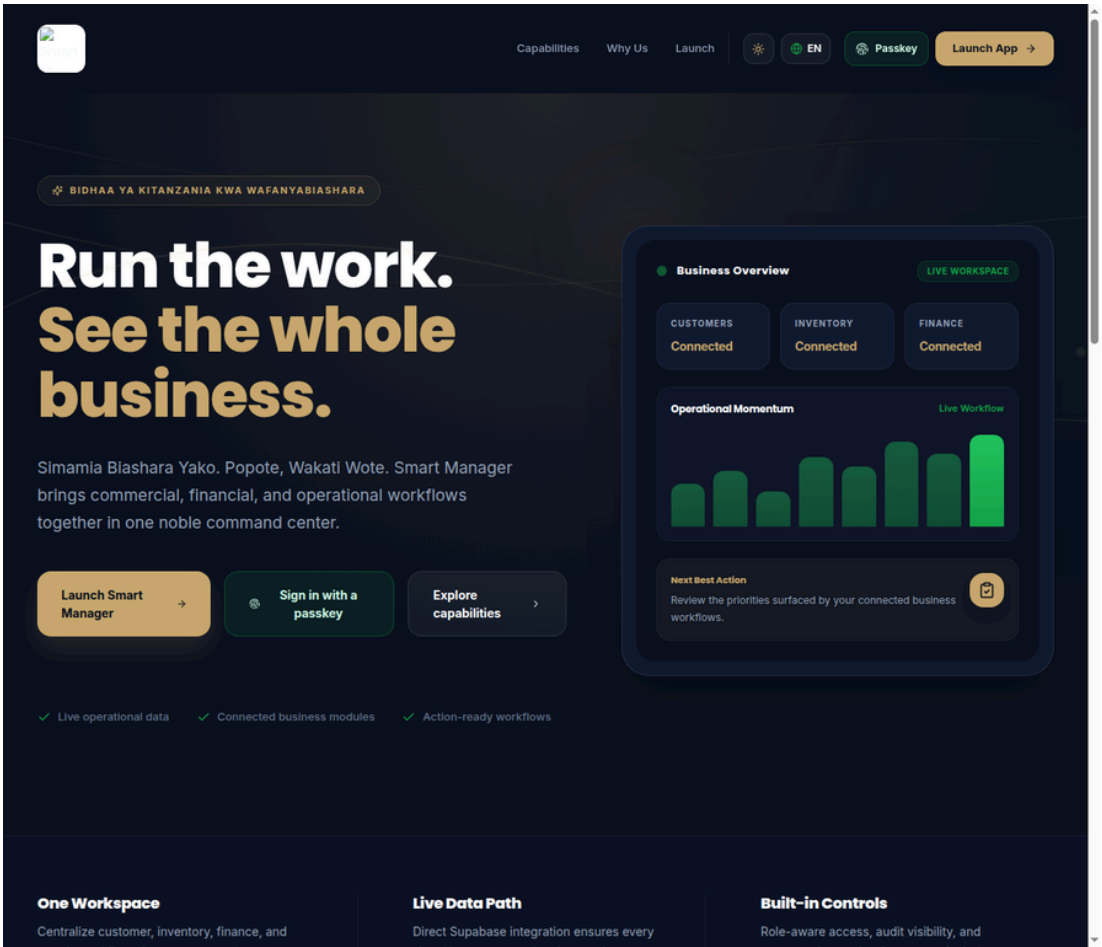


Figure 1: Verified captured public landing page; used as product-introduction evidence.

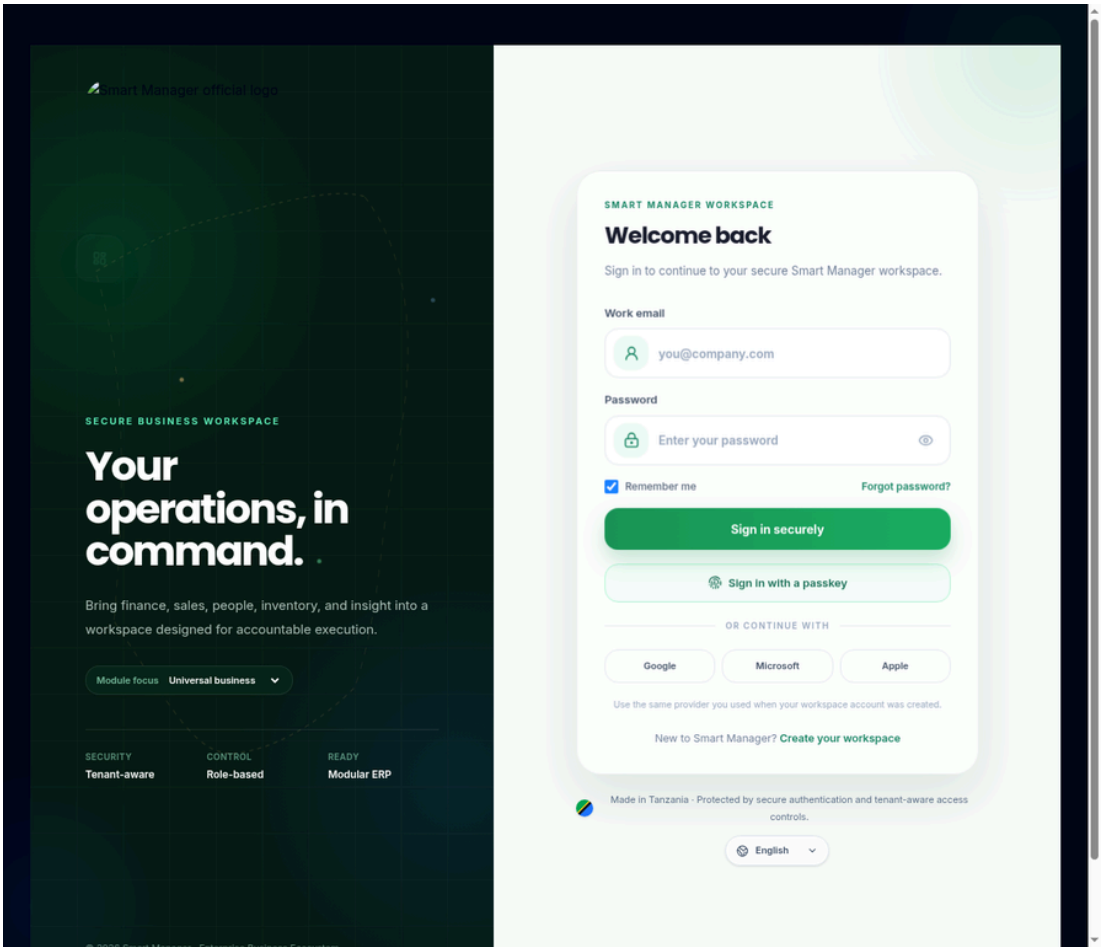


Figure 2: Verified captured workspace-entry and authentication experience.

4 Part III — Technical Architecture

4.1 III.1 Technology Stack

Vite, React 19, TypeScript/JavaScript, Tailwind, Express 5, tRPC 11, Supabase Auth/Postgres/REST/RLS, S3-compatible storage, and Vercel-compatible serverless runtime are present in the project. The package manifest is the source for exact dependency versions.

4.1.1 Kiswahili

Stack ya teknolojia ina Vite, React 19, TypeScript/JavaScript, Tailwind, Express 5, tRPC 11, Supabase Auth/Postgres/REST/RLS, storage inayolingana na S3 na runtime ya Vercel.

Layer	Implementation	Evidence	Boundary
Frontend	Vite + React 19 + Tailwind + wouter	package.json; App.tsx	Browser is not database authority
Backend	Express 5 + tRPC 11 + handlers	apiApp.ts; routers.ts	Server validates and scopes
Data	Supabase Auth/Postgres/REST/RLS	migrations and live audit	RLS and constraints
Runtime	Vercel-compatible import plus local server	index.ts; build scripts	Environment configuration

4.2 III.2 Frontend Architecture

App.tsx owns the top-level route switch and providers. BusinessSphereDashboard.jsx is the authenticated module composition boundary with lazy workspaces, a direct Supabase REST client path, shared persistence guards, subscription access adapter, and role-aware navigation.

4.2.1 Kiswahili

App.tsx inamiliki routes na providers. BusinessSphereDashboard.jsx ndiyo mpaka mkubwa wa workspace yenye lazy modules, Supabase REST, persistence guards, subscription access na urambazaji wa role.

4.3 III.3 Backend Architecture

Express API bootstrap separates API-only behavior from SPA hosting. HTTP routes cover public configuration, billing, HarakaPay, scheduled handlers, webhooks, storage proxy, OAuth, and the tRPC middleware. Feature routers and operations live in server files.

4.3.1 Kiswahili

Express API hutenganisha API na SPA. Routes za HTTP zinahusu public config, billing, HarakaPay, scheduled handlers, webhooks, storage proxy, OAuth na tRPC. Operations za moduli ziko server-side.

4.4 III.4 Database Architecture

The repository uses a source-versioned Supabase migration history. The live audit on 24 August 2026 returned 542 tables: 519 public and 23 auth; 535 tables reported RLS enabled and 7 did not. Table definitions, indexes, constraints, functions, grants, and policies are migration-owned.

4.4.1 Kiswahili

Repository ina migration history ya Supabase yenye version. Audit ya 24 Agosti 2026 ilirudisha tables 542: 519 public na 23 auth; tables 535 zilionesha RLS na 7 hazikuonesha.

Metric	Value	Meaning	Evidence
Total tables	542	Live schema meta-data	Read-only list_tables
Public	519	Public schema tables	Read-only list_tables
Auth	23	Auth schema tables	Read-only list_tables
RLS enabled	535	Reported enabled	Read-only list_tables
RLS not enabled	7	Specific review required	Read-only list_tables
Migrations	133	Ledger records	Read-only list_migrations

4.5 III.5 Supabase Authority

Supabase is authoritative for authenticated profiles, company scope, business records, subscription plans, payment state, invoices, entitlements, and access snapshots. Browser storage can hold session tokens under the supported auth flow, but it cannot grant subscription access or represent a durable business write.

4.5.1 Kiswahili

Supabase ndiyo mamlaka ya profiles, company scope, rekodi za biashara, packages, payment state, invoices, entitlements na access snapshots. Browser haiwezi kutoa ruhusa ya subscription au kudai business write imehifadhiwa.

4.6 III.6 Authentication Architecture

The public gateway obtains public Supabase configuration from `/api/config/public` when needed, calls Supabase Auth endpoints for password and recovery operations, supports Google/Azure/Apple provider selection, handles OAuth hash callbacks, and persists access/refresh tokens through the supported session adapter.

4.6.1 Kiswahili

Gateway hupata public config kupitia `/api/config/public`, hutumia Supabase Auth kwa password/recovery, ina Google/Azure/Apple na hushughulikia OAuth callbacks pamoja na token persistence kupitia adapter.

4.7 III.7 Authorization and RLS

Authorization is layered: session extraction, verified profile resolution, role checks, company scope, protected server procedures, database functions, grants, constraints, and RLS policies. A visible button is not proof of permission. Global administration has a separate policy boundary.

4.7.1 Kiswahili

Uidhinishaji una tabaka: kikao, verified profile, role, company scope, procedures za server, functions, grants, constraints na RLS. Button inayoonekana si ushahidi wa permission. Global admin ina mpaka tofauti.

4.8 III.8 Multi-Tenant Architecture

The effective model is platform → organization/company → workspace context → authenticated profile/membership → role/module entitlement → row-scoped data. The client does not select an arbitrary company ID as authority; Supabase profile and RLS scope control what can be read or changed.

4.8.1 Kiswahili

Mfumo ni platform → organization/company → workspace → profile/membership → role/entitlement → data ya row. Client haipaswi kuchagua company ID kiholela; profile na RLS ndiyo hudhibiti data.

4.9 III.9 API Architecture

Critical flows use server HTTP handlers or protected tRPC procedures. Subscription routes call user-scoped or service-scoped Supabase RPCs. Specialist operations validate Zod inputs and forward authenticated tokens. Service credentials remain server-side.

4.9.1 Kiswahili

Flows muhimu hutumia HTTP handlers za server au protected tRPC. Billing hutumia user/service RPCs za Supabase. Operations maalumu huthibitisha Zod input na kutuma token iliyothibitishwa. Siri hubaki server-side.

4.10 III.10 Storage Architecture

File metadata and references are separated from file bytes. Avatar and property-document paths are scoped by company/entity; storage proxy and download boundaries enforce controlled access. Missing storage configuration must be shown as unavailable rather than successful.

4.10.1 Kiswahili

Metadata na references zimetenganishwa na bytes za file. Avatar na property documents zina keys zenye company/entity scope; storage proxy na download boundaries zinalinda access.

4.11 III.11 AI Architecture

The assistant uses a built-in model boundary with a default gpt-5-mini model, bounded history/context, JSON Schema response format, safe module targets, a limited operation proposal list, and an explicit requirement for independent approval before mutation.

4.11.1 Kiswahili

Assistant hutumia built-in model, history/context yenye mipaka, JSON Schema, safe targets za modules na proposal cache. Mutation inahitaji approval huru ya role husika.

4.12 III.12 Deployment Architecture

Local development mounts Vite and the API; production bundles client and API separately. When VERCEL=1 the runtime exposes the API app for serverless import rather than opening a local listener. Builds run Supabase schema verification when server credentials are available and skip it explicitly for credentialless Vercel builds.

4.12.1 Kiswahili

Development hutumia Vite na API; production hubundle client na API. VERCEL=1 huwezesha import ya API ya serverless badala ya listener wa local. Build huendesha schema verification pale credentials za server zipo.

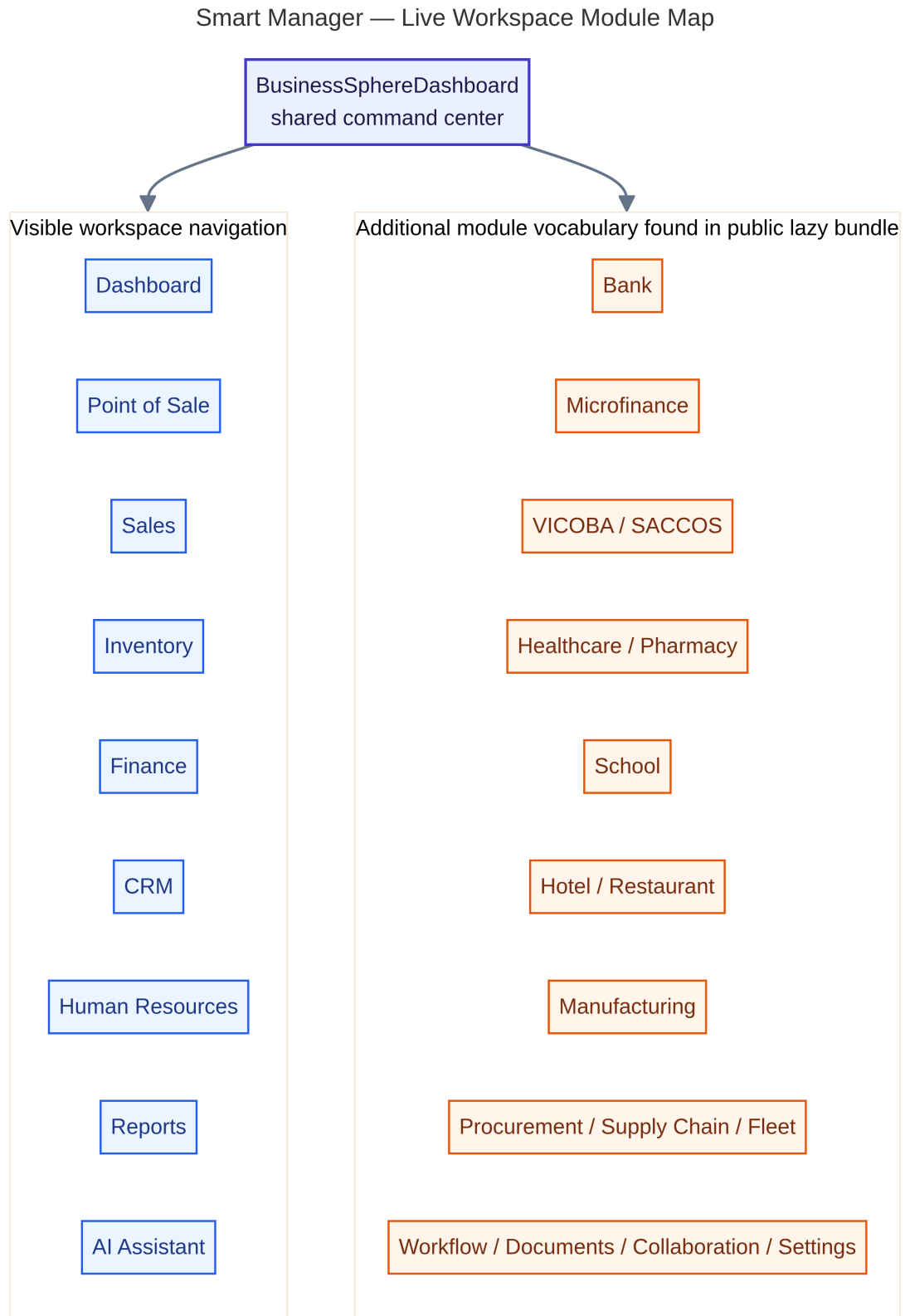


Figure 3: Live module-map evidence from the repository architecture review.

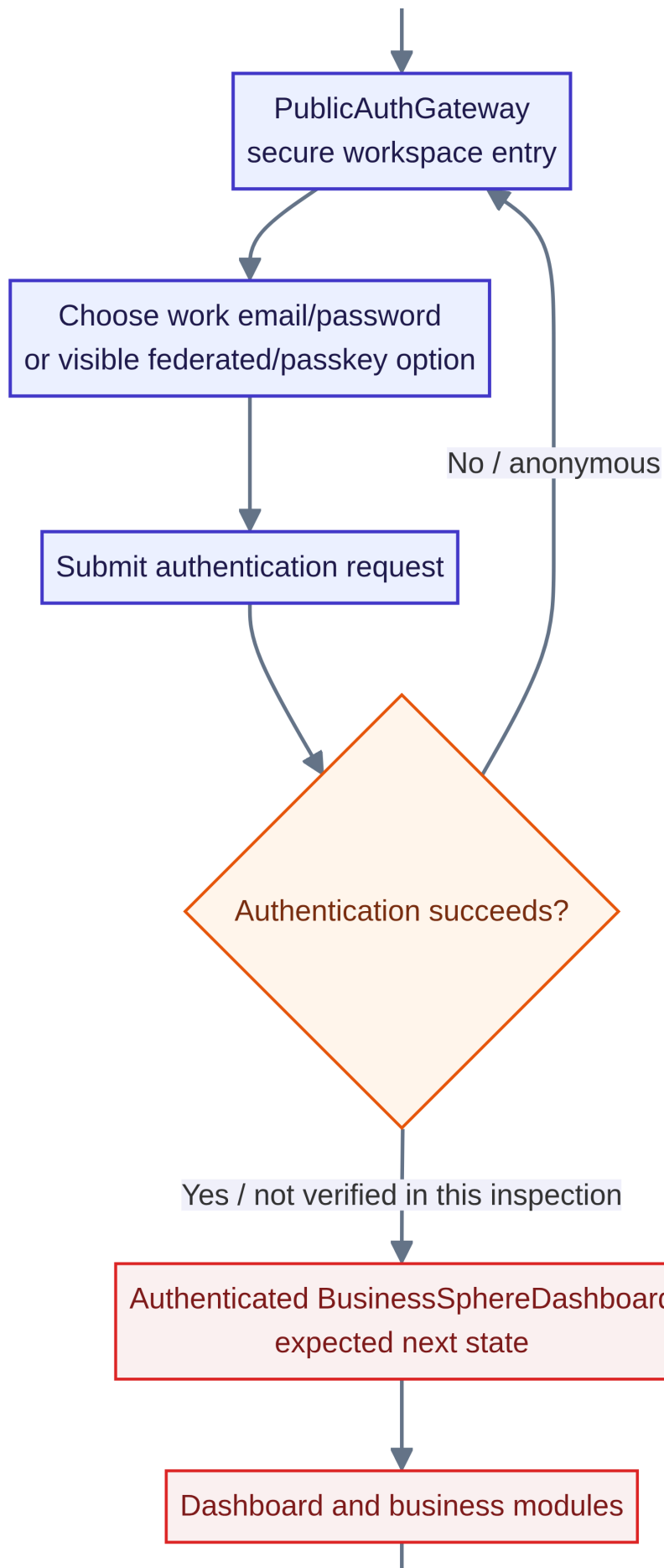


Figure 4: Authentication and tenant-aware workflow evidence.

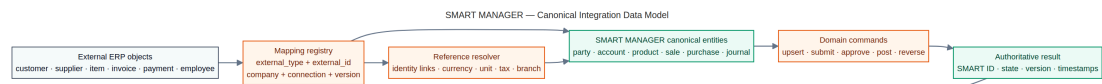


Figure 5: Canonical data-model relationship evidence.

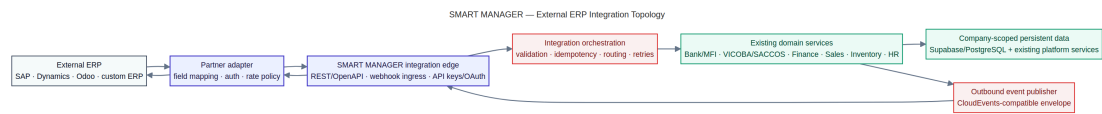


Figure 6: Integration topology evidence.

5 Part IV — Complete Module Book

Each module chapter follows the required documentation structure and is based on real UI, server, migration, and live inventory evidence. A module status is not a guarantee that every deployment has every provider or data prerequisite configured.

5.1 IV.1 Public Brand and Marketing Entry

IMPLEMENTED

5.1.1 Module overview

The public landing page communicates the Smart Manager proposition, links to the application, exposes language and theme controls, and provides a passkey entry point without treating the marketing preview as operational data. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Public Brand and Marketing Entry is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes companies. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Discover product → review capabilities → choose launch or authentication → enter the secure workspace. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Marketing delivery and conversion analytics are not the same as a signed-in company workflow; external campaign channels require configuration.

5.1.2 Purpose, users, navigation, and features

Primary roles: Public visitor; prospective customer. UI evidence: client/src/pages/Home.tsx. The public landing page communicates the Smart Manager proposition, links to the application, exposes language and theme controls, and provides a passkey entry point without treating the marketing preview as operational data.

5.1.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-terpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: client/src/pages/Home.tsx; server: server/_core/apiApp.ts
Purpose	Public Brand and Marketing Entry	Primary users	Public visitor; prospective customer
Navigation	Authenticated module registry or dedicated workspace	Workflow	Discover product → review capabilities → choose launch or authentication → enter the secure workspace.
Database	companies	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in-interpretation	Limitation
Known limitation	Marketing delivery and conversion analytics are not the same as a signed-in company workflow; external campaign channels require configuration.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.1.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.1.5 Known limitations and future improvements

Marketing delivery and conversion analytics are not the same as a signed-in company workflow; external campaign channels require configuration. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.



Figure 7: Public Brand and Marketing Entry evidence image from repository or captured application assets.

5.2 IV.2 Authentication and Secure Onboarding

IMPLEMENTED

5.2.1 Module overview

The public gateway supports password sign-in, password recovery, email confirmation, approved OAuth providers, passkeys, session persistence, and the transition into a company-aware workspace. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Authentication and Secure Onboarding is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `companies`, `company\u{005F}memberships`, `profiles`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Load public Supabase configuration → authenticate or recover → persist token/session → resolve verified profile → enter /app. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Actual OAuth delivery, email delivery, and passkey availability remain dependent on Supabase/provider configuration and browser capability.

5.2.2 Purpose, users, navigation, and features

Primary roles: Unauthenticated visitor; verified user; company owner. UI evidence: `client/src/components/PublicAuthGateway.jsx`; `client/src/App.tsx`. The public gateway supports password sign-in, password recovery, email confirmation, approved OAuth providers, passkeys, session persistence, and the transition into a company-aware workspace.

5.2.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in- interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/PublicAuthGateway.jsx</code> ; <code>client/src/App.tsx</code> ; server: <code>server/_core/apiApp.ts</code> ; <code>server/_core/oauth.ts</code> ; <code>server/authHeaders.ts</code>
Purpose	Authentication and Secure Onboarding	Primary users	Unauthenticated visitor; verified user; company owner
Navigation	Authenticated module registry or dedicated workspace	Workflow	Load public Supabase configuration → authenticate or recover → persist token/session → resolve verified profile → enter /app.
Database	<code>companies</code> , <code>company_memberships</code> , <code>profiles</code>	Subunits	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
Known limitation	Actual OAuth delivery, email delivery, and passkey availability remain dependent on Supabase/provider configuration and browser capability.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.2.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.2.5 Known limitations and future improvements

Actual OAuth delivery, email delivery, and passkey availability remain dependent on Supabase/provider configuration and browser capability. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

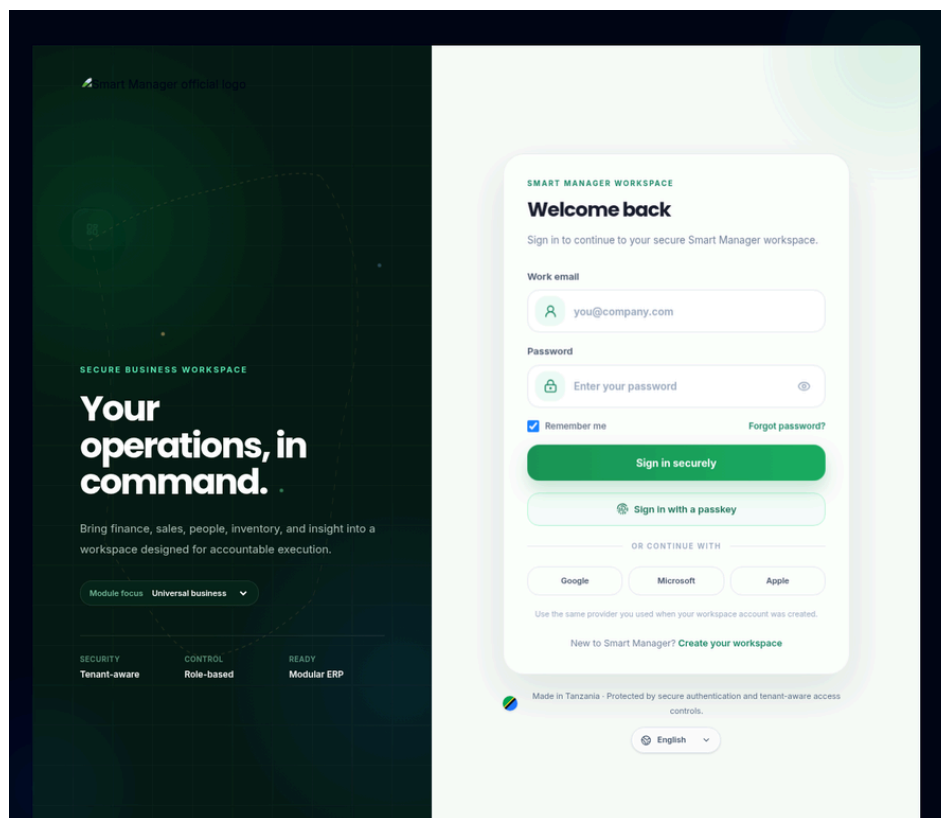


Figure 8: Authentication and Secure Onboarding evidence image from repository or captured application assets.

5.3 IV.3 Master Application Shell and Navigation

IMPLEMENTED

5.3.1 Module overview

The application shell provides route handling, error boundaries, theme and language providers, dashboard preferences, lazy workspace loading, command navigation, and fail-closed subscription access checks. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Master Application Shell and Navigation is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `company\{005F}modules`, `workspaces`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Route match → auth decision → lazy workspace load → role/entitlement-aware navigation → module action. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. The dashboard remains a large legacy single-file boundary and produces a non-fatal bundle-size warning; safe decomposition is a future engineering workstream.

5.3.2 Purpose, users, navigation, and features

Primary roles: All authenticated users, filtered by role and entitlement. UI evidence: `client/src/App.tsx`; `client/src/BusinessSphereDashboard.jsx`. The application shell provides route handling, error boundaries, theme and language providers, dashboard preferences, lazy workspace loading, command navigation, and fail-closed subscription access checks.

5.3.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-terpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/App.tsx</code> ; <code>client/src/BusinessSphereDashboard.jsx</code> ; server: <code>server/routers.ts</code> ; <code>server/_core/apiApp.ts</code>
Purpose	Master Application Shell and Navigation	Primary users	All authenticated users, filtered by role and entitlement
Navigation	Authenticated module registry or dedicated workspace	Workflow	Route match → auth decision → lazy workspace load → role/entitlement-aware navigation → module action.
Database	<code>company_modules</code> , <code>workspaces</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
Known limitation	The dashboard remains a large legacy single-file boundary and produces a non-fatal bundle-size warning; safe decomposition is a future engineering workstream.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.3.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.3.5 Known limitations and future improvements

The dashboard remains a large legacy single-file boundary and produces a non-fatal bundle-size warning; safe decomposition is a future engineering workstream. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

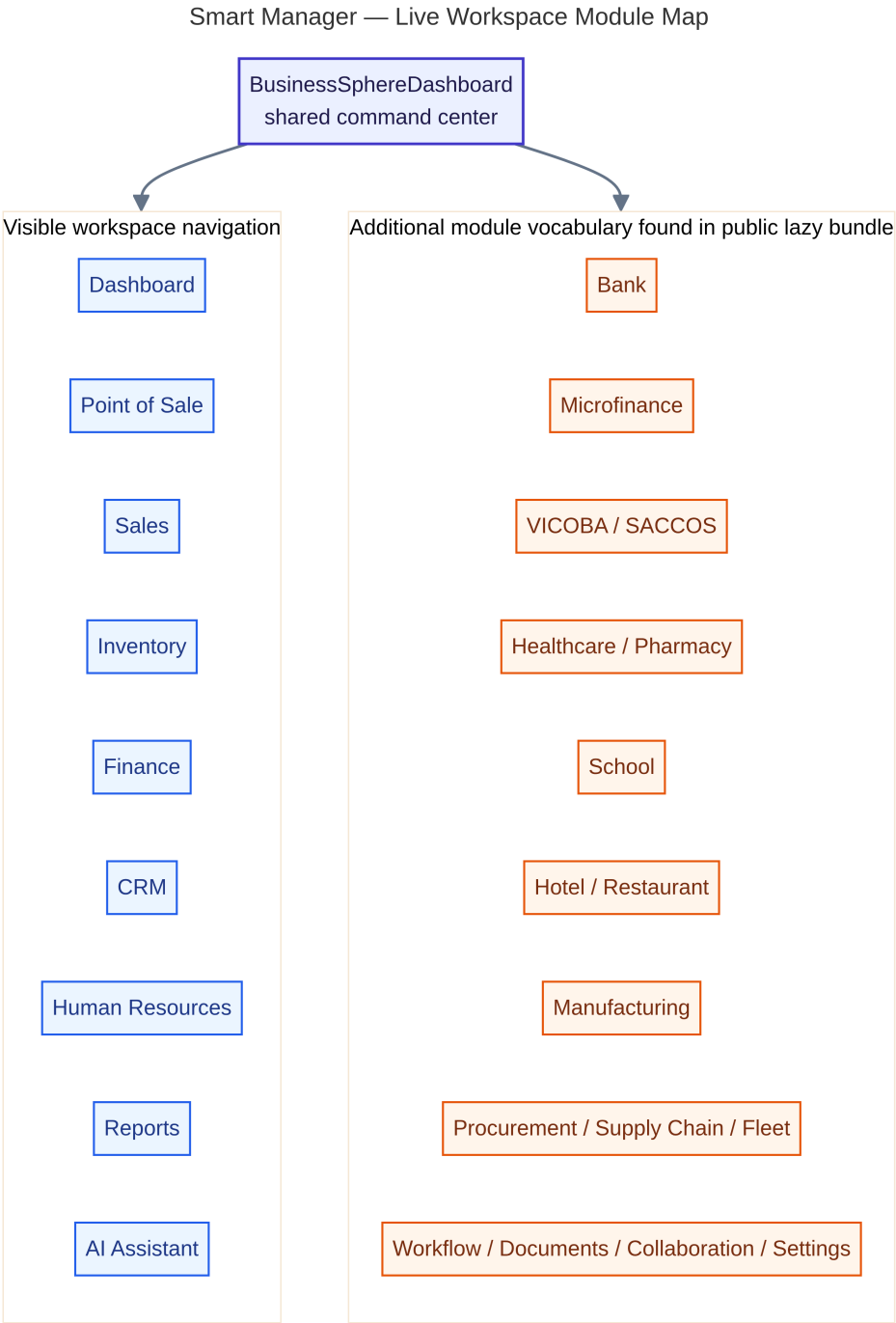


Figure 9: Master Application Shell and Navigation evidence image from repository or captured application assets.

5.4 IV.4 Profile Identity Center

IMPLEMENTED

5.4.1 Module overview

The profile center separates personal identity, work identity, security, preferences, notifications, and activity while keeping protected identity fields server-authoritative. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Profile Identity Center is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `profiles`, `user\{005F}table\{005F}preferences`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Open profile → read verified profile → update permitted self-service fields → upload or remove avatar through controlled storage → refresh confirmed state. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Device/session listing and workspace switching are explicitly unavailable unless supported by a verified backend contract.

5.4.2 Purpose, users, navigation, and features

Primary roles: Authenticated user; workspace administrator for linked context. UI evidence: `client/src/components/ProfileIdentityCenter.jsx`. The profile center separates personal identity, work identity, security, preferences, notifications, and activity while keeping protected identity fields server-authoritative.

5.4.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/ProfileIdentityCenter.jsx</code> ; server: <code>server/profileIdentity.ts</code> ; <code>server/routers.ts</code>
Purpose	Profile Identity Center	Primary users	Authenticated user; workspace administrator for linked context
Navigation	Authenticated module registry or dedicated workspace	Workflow	Open profile → read verified profile → update permitted self-service fields → upload or remove avatar through controlled storage → refresh confirmed state.
Database	<code>profiles</code> , <code>user_table_preferences</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Device/session listing and workspace	Future improvement	Complete controlled authenticated CRUD and provider accep-

Area	Verified reference	Operational in- terpretation	Limitation
	switching are explicitly unavailable unless supported by a verified backend contract.		tance where the status is not PASSED.

5.4.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.4.5 Known limitations and future improvements

Device/session listing and workspace switching are explicitly unavailable unless supported by a verified backend contract. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

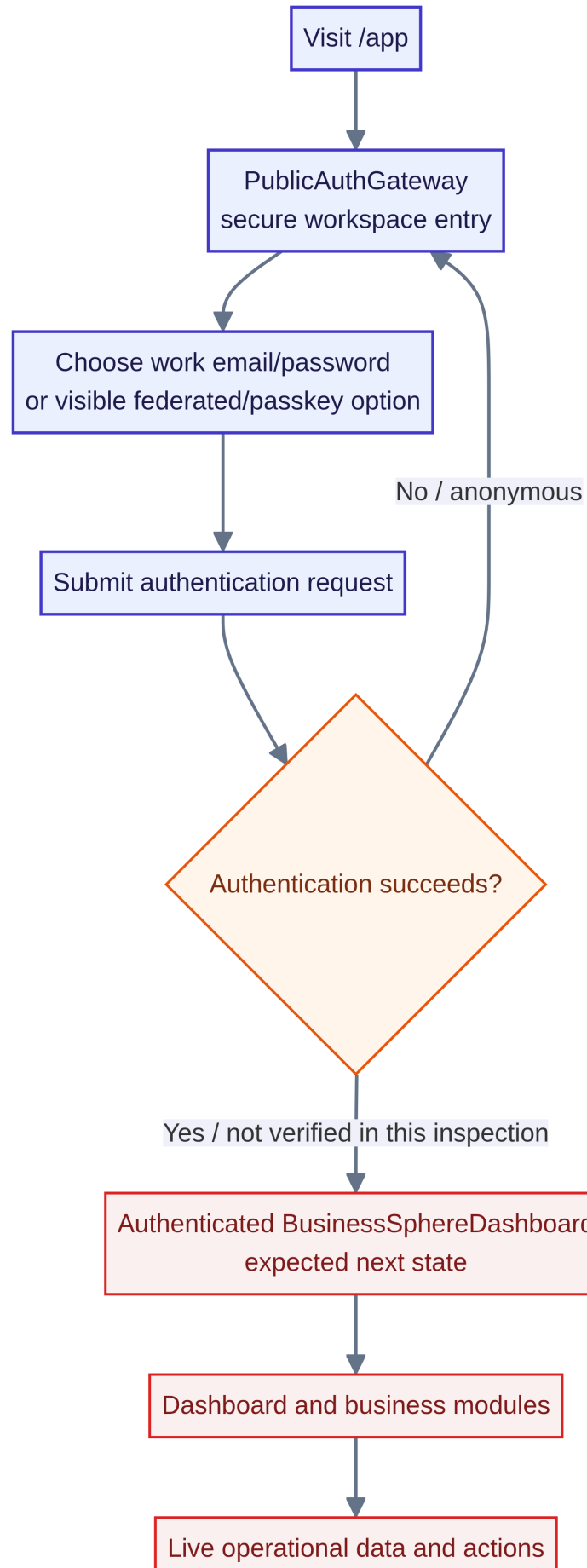


Figure 10: Profile Identity Center evidence image from repository or captured application assets.

5.5 IV.5 Executive Dashboard IMPLEMENTED

5.5.1 Module overview

The executive surface consolidates operational signals from connected modules. It is intended for review and prioritization, not as an independent source of financial truth. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Executive Dashboard is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `custom\kpi`s, `financial` benchmarks, `scheduled` reports. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Select company context → load confirmed metrics → inspect trends/exceptions → navigate to source module → approve or act through the protected boundary. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. A dashboard is only as complete as the connected records, schedules, and integrations that feed it; empty or unavailable states must remain visible.

5.5.2 Purpose, users, navigation, and features

Primary roles: CEO; COO; CFO; organization owner; manager roles. UI evidence: `client/src/components/ExecutiveCommandCenter.jsx`; `client/src/BusinessSphereDashboard.jsx`. The executive surface consolidates operational signals from connected modules. It is intended for review and prioritization, not as an independent source of financial truth.

5.5.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/ExecutiveCommandCenter.jsx</code> ; <code>client/src/BusinessSphereDashboard.jsx</code> ; server: <code>server/dashboardReports.ts</code> ; <code>server/routers.ts</code>
Purpose	Executive Dashboard	Primary users	CEO; COO; CFO; organization owner; manager roles
Navigation	Authenticated module registry or dedicated workspace	Workflow	Select company context → load confirmed metrics → inspect trends/exceptions → navigate to source module → approve or act through the protected boundary.

Area	Verified reference	Operational interpretation	in-	Limitation
Database	custom_kpis, financial_scheduled_reports	Security,		Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	A dashboard is only as complete as the connected records, schedules, and integrations that feed it; empty or unavailable states must remain visible.	Future improvement		Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.5.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.5.5 Known limitations and future improvements

A dashboard is only as complete as the connected records, schedules, and integrations that feed it; empty or unavailable states must remain visible. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.



Figure 11: Executive Dashboard evidence image from repository or captured application assets.

5.6 IV.6 CRM and Customer Pipeline

TESTING

5.6.1 Module overview

CRM organizes contacts, leads, interactions, customer feedback, and pipeline signals so commercial activity can be related to sales and support records. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of CRM and Customer Pipeline is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `crm\contacts`, `crm\interactions`, `crm\leads`, `customer\feedback`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Create or import lead → qualify → record interaction → convert or follow up → connect to quotation/invoice/support. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. The repository contains strong persistence and truthfulness contracts, but broader authenticated CRUD walkthroughs remain required for every legacy screen.

5.6.2 Purpose, users, navigation, and features

Primary roles: Sales Manager; sales user; account owner; support user. UI evidence: `client/src/components/CommercialCommandCenters.jsx`; `client/src/BusinessSphereDashboard.jsx`. CRM organizes contacts, leads, interactions, customer feedback, and pipeline signals so commercial activity can be related to sales and support records.

5.6.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/CommercialCommandCenters.jsx</code> ; <code>client/src/BusinessSphereDashboard.jsx</code> ; server: <code>server/routers.ts</code> ; <code>server/salesInteractions.ts</code>
Purpose	CRM and Customer Pipeline	Primary users	Sales Manager; sales user; account owner; support user
Navigation	Authenticated module registry or dedicated workspace	Workflow	Create or import lead → qualify → record interaction → convert or follow up → connect to quotation/invoice/support.
Database	<code>crm_contacts</code> , <code>crm_interactions</code> ,	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
	crm_leads, customer_feedback		
Known limitation	The repository contains strong persistence and truthfulness contracts, but broader authenticated CRUD walkthroughs remain required for every legacy screen.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.6.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.6.5 Known limitations and future improvements

The repository contains strong persistence and truthfulness contracts, but broader authenticated CRUD walkthroughs remain required for every legacy screen. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

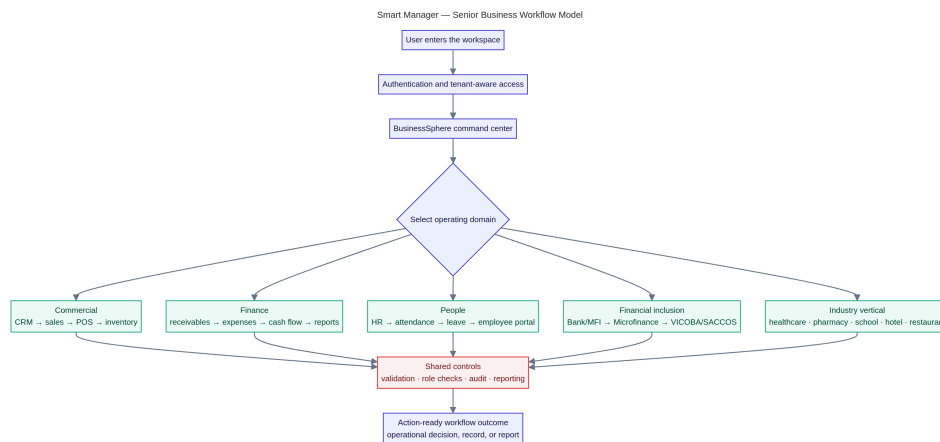


Figure 12: CRM and Customer Pipeline evidence image from repository or captured application assets.

5.7 IV.7 Sales and Billing

TESTING

5.7.1 Module overview

Sales surfaces cover quotations, invoices, subscriptions, payments, returns, and customer interactions. The customer-facing amount and status must come from confirmed server/database results. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Sales and Billing is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `ecommerce\orders`, `ecommerce\products`, `sales\invoice\items`, `sales\invoices`, `sales\order\items`, `sales\order\return\items`, `sales\order\returns`, `sales\orders`, `sales\payments`, `sales\quotation\items`, `sales\quotations`, `sales\subscriptions`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Customer → quotation/order → invoice → payment or collection → receipt/ledger/report → follow-up. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Legacy invoice payment posting still requires an atomic database RPC and durable idempotency constraint before concurrent-safe payment posting can be claimed.

5.7.2 Purpose, users, navigation, and features

Primary roles: Sales Manager; finance manager; sales user; billing manager. UI evidence: `client/src/components/SalesDetailWorkspace.jsx`; `client/src/components/CommercialCommandCenters.jsx`. Sales surfaces cover quotations, invoices, subscriptions, payments, returns, and customer interactions. The customer-facing amount and status must come from confirmed server/database results.

5.7.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/SalesDetailWorkspace.jsx</code> ; <code>client/src/components/CommercialCommandCenters.jsx</code> ; server: <code>server/routers.ts</code> ; <code>server/subscriptionBilling.ts</code>
Purpose	Sales and Billing	Primary users	Sales Manager; finance manager; sales user; billing manager
Navigation	Authenticated module registry or dedicated workspace	Workflow	Customer → quotation/order → invoice → payment or collection → receipt/ledger/report → follow-up.

Area	Verified reference	Operational in- interpretation	Limitation
Database	ecommerce_orders, ecommerce_products, sales_invoice_items, sales_invoices, sales_order_items, sales_order_return_items, sales_order_returns, sales_orders, sales_payments, sales_quotation_items, sales_quotations, sales_subscriptions	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Legacy invoice payment posting still requires an atomic database RPC and durable idempotency constraint before concurrent-safe payment posting can be claimed.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.7.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.7.5 Known limitations and future improvements

Legacy invoice payment posting still requires an atomic database RPC and durable idempotency constraint before concurrent-safe payment posting can be claimed. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

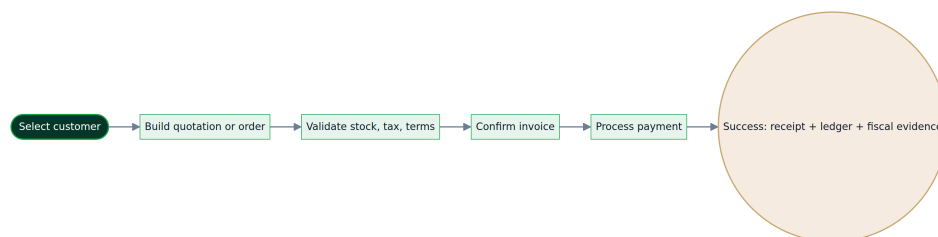


Figure 13: Sales and Billing evidence image from repository or captured application assets.

5.8 IV.8 Point of Sale

PASSED

5.8.1 Module overview

POS supports registers, terminals, shifts, sale headers and lines, tenders, returns, tax, promotions, loyalty, pending offline transport, reconciliation, and audit paths. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Point of Sale is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `pos\cash\movements`, `pos\discount\rules`, `pos\loyalty\ledger`, `pos\loyalty\members`, `pos\loyalty\programs`, `pos\loyalty\redemptions`, `pos\loyalty\rwards`, `pos\promotion\items`, `pos\promotions`, `pos\registers`, `pos\return\commits`, `pos\return\headers`, `pos\return\items`, `pos\return\lines`, `pos\returns`, `pos\sale\adjustments`, `pos\sale\headers`, `pos\sale\lines`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Open register/shift → scan or select item → validate price and stock → collect tender → confirm sale → receipt → reconcile or return. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Controlled staging credentials and real device/provider behavior remain operational acceptance requirements; client pending queues are not treated as durable server saves.

5.8.2 Purpose, users, navigation, and features

Primary roles: Cashier; finance manager; warehouse manager; administrator. UI evidence: `client/src/components/OperationsCommandCenters.jsx`; `client/src/BusinessSphereDashboard.jsx`. POS supports registers, terminals, shifts, sale headers and lines, tenders, returns, tax, promotions, loyalty, pending offline transport, reconciliation, and audit paths.

5.8.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	PASSED	Evidence	UI: <code>client/src/components/OperationsCommandCenters.jsx</code> ; <code>client/src/BusinessSphereDashboard.jsx</code> ; server: <code>server/posWorkforceRpcAdapters.ts</code> ; <code>server/posTransactionEngine.ts</code>

Area	Verified reference	Operational interpretation	in-	Limitation
Purpose	Point of Sale	Primary users		Cashier; finance manager; warehouse manager; administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow		Open register/shift → scan or select item → validate price and stock → collect tender → confirm sale → receipt → reconcile or return.
Database	pos_cash_movements, pos_discount_rules, pos_loyalty_ledger, pos_loyalty_members, pos_loyalty_programs, pos_loyalty_redemptions, pos_loyalty_rewards, pos_promotion_items, pos_promotions, pos_registers, pos_return_commits, pos_return_headers, pos_return_items, pos_return_lines, pos_returns, pos_sale_adjustments	Security		Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Controlled staging credentials and real device/provider behavior remain operational acceptance requirements; client pending queues are not treated as durable server saves.	Future improvement		Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.8.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.8.5 Known limitations and future improvements

Controlled staging credentials and real device/provider behavior remain operational acceptance requirements; client pending queues are not treated as durable server saves. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

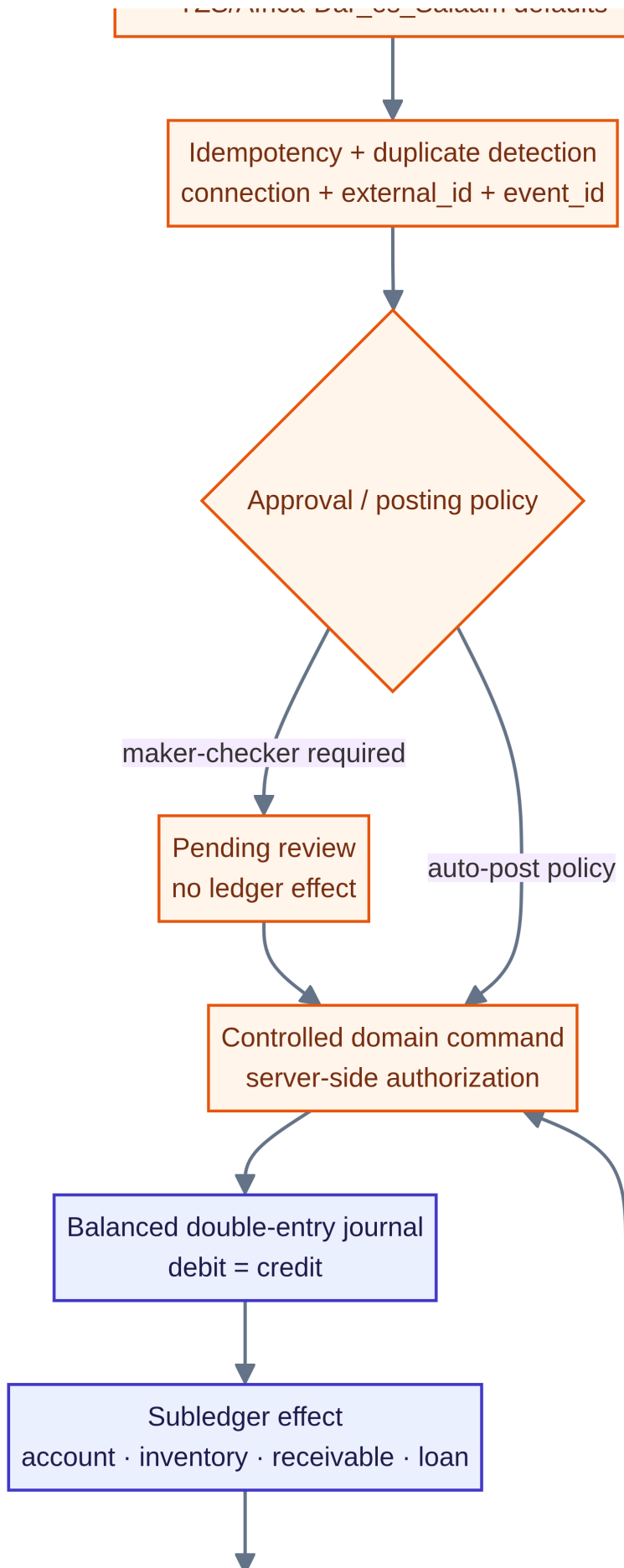


Figure 14: Point of Sale evidence image from repository or captured application assets.

5.9 IV.9 Inventory and Warehouse Management

TESTING

5.9.1 Module overview

Inventory records cover items, batches, stock movement, suppliers, warehouses, transfers, audits, and the operational connection from stock to sales and procurement. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Inventory and Warehouse Management is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `inventory\batches`, `inventory\items`, `inventory\stock\movements`, `inventory\suppliers`, `inventory\transfers`, `inventory\warehouses`, `stock\audit\items`, `stock\audits`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Create item → receive or adjust stock → transfer or sell → record movement → inspect balance/audit → replenish. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Full authenticated import/export and concurrency walkthroughs remain part of the broader acceptance backlog.

5.9.2 Purpose, users, navigation, and features

Primary roles: Warehouse Manager; procurement officer; sales user; administrator. UI evidence: `client/src/components/OperationsCommandCenters.jsx`. Inventory records cover items, batches, stock movement, suppliers, warehouses, transfers, audits, and the operational connection from stock to sales and procurement.

5.9.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/OperationsCommandCenters.jsx</code> ; server: <code>server/supabasePersistence.ts</code> ; <code>server/routers.ts</code>
Purpose	Inventory and Warehouse Management	Primary users	Warehouse Manager; procurement officer; sales user; administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Create item → receive or adjust stock → transfer or sell → record movement → inspect balance/audit → replenish.

Area	Verified reference	Operational in- interpretation	Limitation
Database	inventory_batches, inventory_items, inventory_stock_movements, inventory_suppliers, inventory_transfers, inventory_warehouses, stock_audit_items, stock_audits	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Full authenticated import/export and concurrency walk- throughs remain part of the broader accep- tance backlog.	Future improve- ment	Complete controlled authenti- cated CRUD and provider accep- tance where the status is not PASSED.

5.9.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.9.5 Known limitations and future improvements

Full authenticated import/export and concurrency walkthroughs remain part of the broader acceptance backlog. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

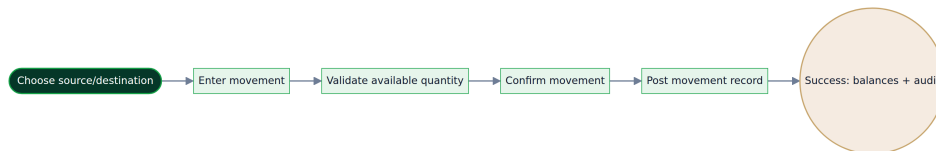


Figure 15: Inventory and Warehouse Management evidence image from repository or captured application assets.

5.10 IV.10 Procurement and Vendor Management

TESTING

5.10.1 Module overview

Procurement relates suppliers, purchase orders, contracts, receiving, stock, approvals, and payables evidence. The workflow must preserve what was requested, approved, received, and owed. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Procurement and Vendor Management is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `procurement\contracts`, `procurement\purchase\orders`, `purchase\order\items`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Supplier → request/order → approval → receipt → stock update → payable/finance evidence → report. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Downstream integrations and external supplier channels are configuration-dependent; recorded integration evidence must not be replaced with fabricated acknowledgements.

5.10.2 Purpose, users, navigation, and features

Primary roles: Procurement Officer; warehouse manager; finance manager; administrator. UI evidence: `client/src/components/OperationsCommandCenters.jsx`. Procurement relates suppliers, purchase orders, contracts, receiving, stock, approvals, and payables evidence. The workflow must preserve what was requested, approved, received, and owed.

5.10.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/OperationsCommandCenters.jsx</code> ; server: <code>server/routers.ts</code> ; <code>server/procurementPersistence.ts</code>
Purpose	Procurement and Vendor Management	Primary users	Procurement Officer; warehouse manager; finance manager; administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Supplier → request/order → approval → receipt → stock update → payable/finance evidence → report.
Database	<code>procurement_contracts</code> , <code>procurement_purchase_orders</code> , <code>purchase_order_items</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational interpretation	in- Limitation
Known limitation	Downstream integrations and external supplier channels are configuration-dependent; recorded integration evidence must not be replaced with fabricated acknowledgements.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.10.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.10.5 Known limitations and future improvements

Downstream integrations and external supplier channels are configuration-dependent; recorded integration evidence must not be replaced with fabricated acknowledgements. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

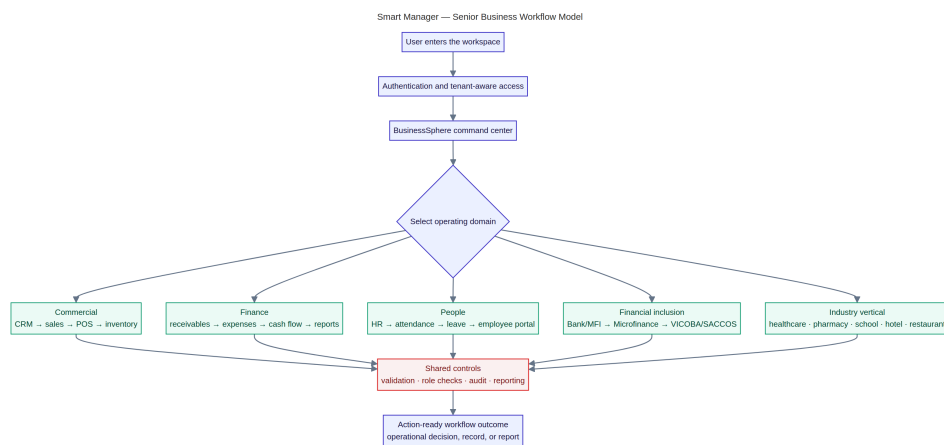


Figure 16: Procurement and Vendor Management evidence image from repository or captured application assets.

5.11 IV.11 Finance and Accounting

TESTING

5.11.1 Module overview

Finance includes accounts, journals, income and expenses, budgets, tax/VAT, period closes, bank reconciliation, assets, and management reports. It is the control plane for monetary evidence rather than a decorative dashboard. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Finance and Accounting is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `expense\budgets`, `fin\accounts`, `fin\approval\requests`, `fin\idempotency\keys`, `fin\journal\batches`, `fin\journal\lines`, `fin\periods`, `fin\posting\links`, `fin\reconciliation\batches`, `fin\reconciliation\items`, `finance\assets`, `finance\expenses`, `journal\entries`, `period\closes`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Capture transaction → validate company scope → post or approve → reconcile → close period → report. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Production reporting depends on intended schema alignment and real company data; tax filing itself is not implied by a local summary.

5.11.2 Purpose, users, navigation, and features

Primary roles: CFO; Finance Manager; accountant; organization owner. UI evidence: `client/src/components/FinanceCommandCenters.jsx`. Finance includes accounts, journals, income and expenses, budgets, tax/VAT, period closes, bank reconciliation, assets, and management reports. It is the control plane for monetary evidence rather than a decorative dashboard.

5.11.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/FinanceCommandCenters.jsx</code> ; server: <code>server/financeCommandCenters.ts</code> ; <code>server/financePersistence.ts</code>
Purpose	Finance and Accounting	Primary users	CFO; Finance Manager; accountant; organization owner
Navigation	Authenticated module registry or dedicated workspace	Workflow	Capture transaction → validate company scope → post or ap-

Area	Verified reference	Operational interpretation	in-	Limitation
				prove → reconcile → close period → report.
Database	expense_budgets, fin_accounts, fin_approval_requests, fin_idempotency_keys, fin_journal_batches, fin_journal_lines, fin_periods, fin_posting_links, fin_reconciliation_batches, fin_reconciliation_items, finance_assets, finance_expenses, journal_entries, period_closes	Security		Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Production reporting depends on intended schema alignment and real company data; tax filing itself is not implied by a local summary.	Future improvement		Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.11.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.11.5 Known limitations and future improvements

Production reporting depends on intended schema alignment and real company data; tax filing itself is not implied by a local summary. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

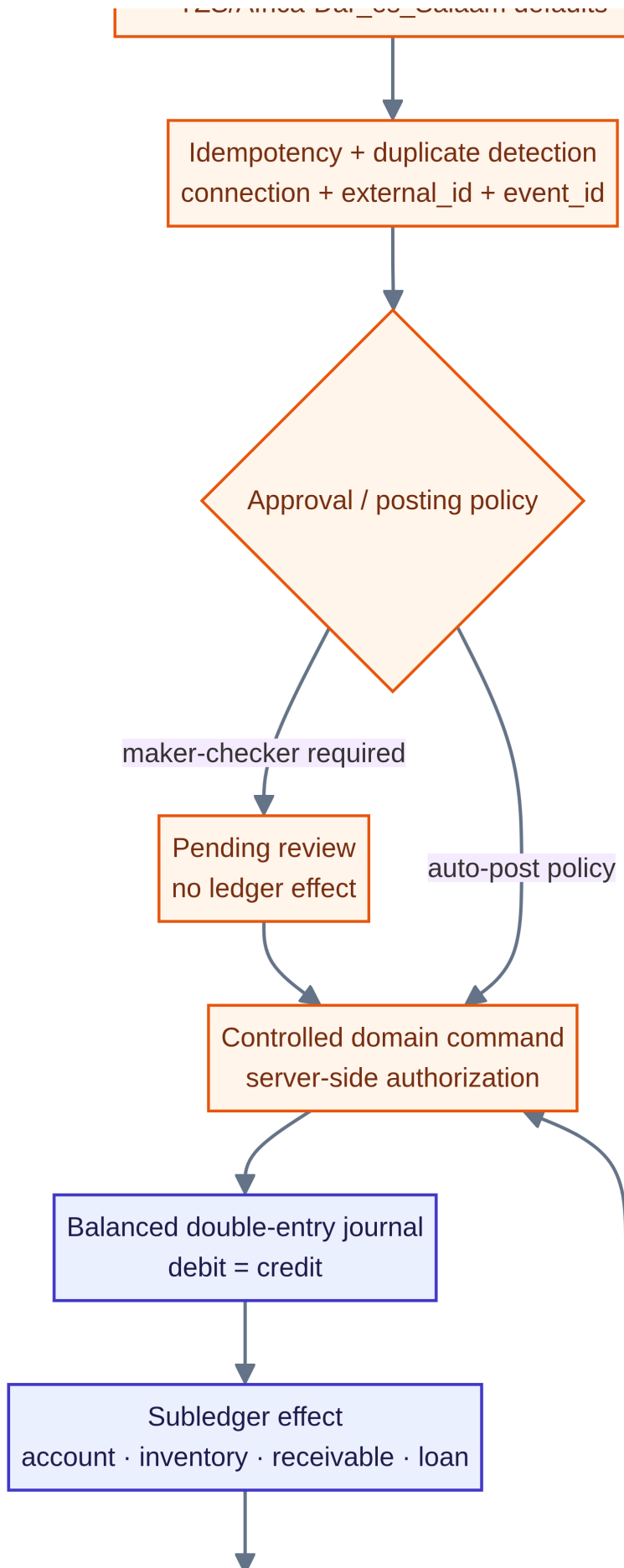


Figure 17: Finance and Accounting evidence image from repository or captured application assets.

5.12 IV.12 Reports and Scheduled Reporting

TESTING

5.12.1 Module overview

Reporting provides trend views, exports, schedules, delivery states, and compliance-oriented summaries. A generated report is evidence of recorded data and configured delivery, not a guarantee that an external message was delivered. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Reports and Scheduled Reporting is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `scheduled\reports`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Choose report → apply company/time scope → compute from source records → export or schedule → inspect delivery history. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Email delivery remains provider-configured; CSV is the dependency-safe export path when spreadsheet dependencies are intentionally absent.

5.12.2 Purpose, users, navigation, and features

Primary roles: Executive roles; manager; administrator. UI evidence: `client/src/components/FinanceCommandCenters.jsx`; `client/src/BusinessSphereDashboard.jsx`. Reporting provides trend views, exports, schedules, delivery states, and compliance-oriented summaries. A generated report is evidence of recorded data and configured delivery, not a guarantee that an external message was delivered.

5.12.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/FinanceCommandCenters.jsx</code> ; <code>client/src/BusinessSphereDashboard.jsx</code> ; server: <code>server/dashboardReports.ts</code> ; <code>server/reportSchedules.ts</code>
Purpose	Reports and Scheduled Reporting	Primary users	Executive roles; manager; administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Choose report → apply company/time scope → compute from source records → export or schedule → inspect delivery history.

Area	Verified reference	Operational in- interpretation	Limitation
Database	scheduled_reports	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Email delivery remains provider-configured; CSV is the dependency-safe export path when spreadsheet dependencies are intentionally absent.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.12.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.12.5 Known limitations and future improvements

Email delivery remains provider-configured; CSV is the dependency-safe export path when spreadsheet dependencies are intentionally absent. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

Smart Manager — Live Deployment Observation

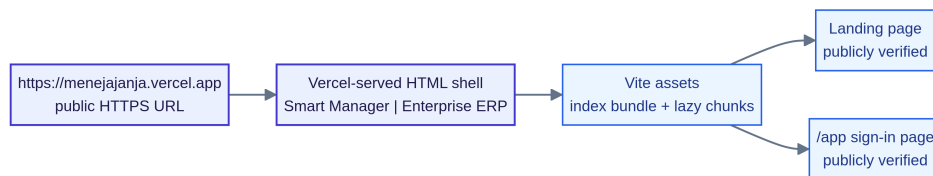


Figure 18: Reports and Scheduled Reporting evidence image from repository or captured application assets.

5.13 IV.13 Human Resources and Payroll

TESTING

5.13.1 Module overview

HR covers employees, attendance, benefits, leave, goals, approvals, payroll, announcements, and secure invitations. Tenant-scoped persistence replaces browser-only identity and invite paths. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Human Resources and Payroll is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `hr\announcement\reads`, `hr\announcements`, `hr\approval\requests`, `hr\approval\steps`, `hr\attendance`, `hr\benefit\enrollments`, `hr\benefit\plans`, `hr\benefits`, `hr\candidates`, `hr\duties`, `hr\employee\documents`, `hr\employees`, `hr\expense\claims`, `hr\goal\updates`, `hr\goals`, `hr\holidays`, `hr\invite\codes`, `hr\leave\balances`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Invite employee → accept securely → assign role → record attendance/leave → approve → calculate payroll → report. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Payroll and employee records need a full authenticated tenant-data walkthrough; external salary/payment rails are outside the repository's verified contract.

5.13.2 Purpose, users, navigation, and features

Primary roles: HR Manager; manager; employee; payroll approver; administrator. UI evidence: `client/src/components/PeopleCommandCenters.jsx`; `client/src/components/EmployeePortalWorkspace.jsx`. HR covers employees, attendance, benefits, leave, goals, approvals, payroll, announcements, and secure invitations. Tenant-scoped persistence replaces browser-only identity and invite paths.

5.13.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/PeopleCommandCenters.jsx</code> ; <code>client/src/components/EmployeePortalWorkspace.jsx</code> ; server: <code>server/teamWorkforce.ts</code> ; <code>server/tanzaniaPayroll.ts</code> ; <code>server/routers.ts</code>
Purpose	Human Resources and Payroll	Primary users	HR Manager; manager; employee; payroll approver; administrator

Area	Verified reference	Operational in- interpretation	Limitation
Navigation	Authenticated module registry or dedicated workspace	Workflow	Invite employee → accept securely → assign role → record attendance/leave → approve → calculate payroll → report.
Database	hr_announcement_read, hr_announcements, hr_approval_requests, hr_approval_steps, hr_attendance, hr_benefit_enrollments, hr_benefit_plans, hr_benefits, hr_candidates, hr_duties, hr_employee_documents, hr_employees, hr_expense_claims, hr_goal_updates, hr_goals, hr_holidays	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Payroll and employee records need a full authenticated tenant-data walkthrough; external salary/payment rails are outside the repository's verified contract.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.13.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.13.5 Known limitations and future improvements

Payroll and employee records need a full authenticated tenant-data walkthrough; external salary/payment rails are outside the repository's verified contract. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

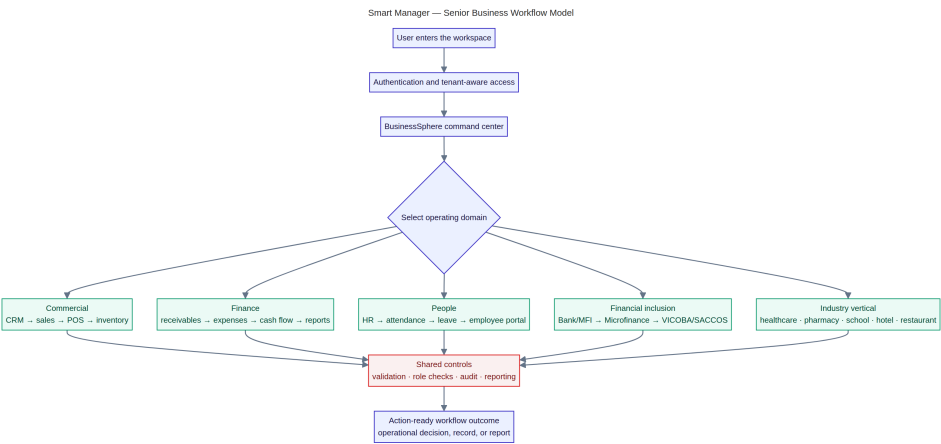


Figure 19: Human Resources and Payroll evidence image from repository or captured application assets.

5.14 IV.14 Manufacturing and Work Orders

PARTIALLY IMPLEMENTED

5.14.1 Module overview

The repository contains manufacturing and work-order persistence boundaries and navigation signals. The master book treats operational depth as partial unless a complete deployment-backed flow is demonstrated. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Manufacturing and Work Orders is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes manufacturing\bom\components, manufacturing\boms, manufacturing\machines, manufacturing\maintenance, manufacturing\qc\inspections, manufacturing\work\orders. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Define work order → allocate or consume material → update status → record output → review cost and exceptions. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. The visible surface is broader than the independently verified end-to-end contract for every manufacturing scenario.

5.14.2 Purpose, users, navigation, and features

Primary roles: Production manager; warehouse manager; project manager. UI evidence: client/src/BusinessSphereDashboard.jsx; client/src/components/OperationsCommandCenters.jsx. The repository contains manufacturing and work-order persistence boundaries and navigation signals. The master book treats operational depth as partial unless a complete deployment-backed flow is demonstrated.

5.14.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: client/src/BusinessSphereDashboard.jsx; client/src/components/OperationsCommandCenters.jsx; server: server/manufacturingPersistence.ts
Purpose	Manufacturing and Work Orders	Primary users	Production manager; warehouse manager; project manager
Navigation	Authenticated module registry or dedicated workspace	Workflow	Define work order → allocate or consume material → update status → record output → review cost and exceptions.

Area	Verified reference	Operational interpretation	in- Limitation
Database	manufacturing_bom_components, manufacturing_boms, manufacturing_machines, manufacturing_maintenance, manufacturing_qc_inspections, manufacturing_work_orders	Security,	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	The visible surface is broader than the independently verified end-to-end contract for every manufacturing scenario.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.14.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.14.5 Known limitations and future improvements

The visible surface is broader than the independently verified end-to-end contract for every manufacturing scenario. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

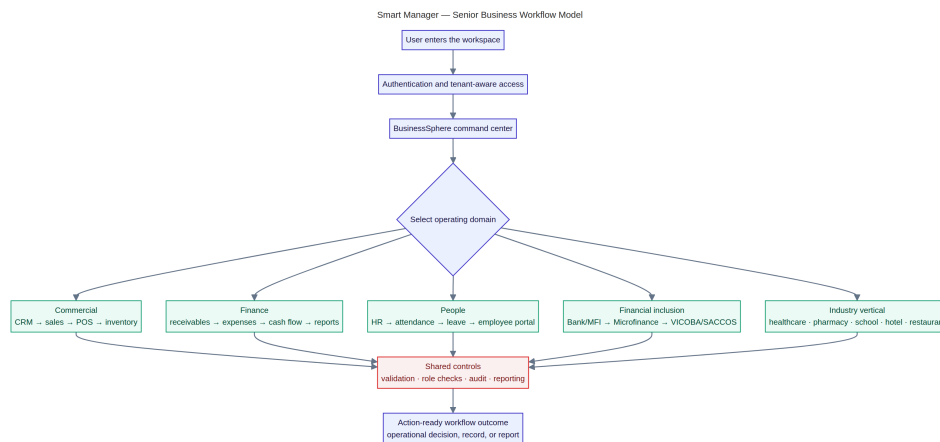


Figure 20: Manufacturing and Work Orders evidence image from repository or captured application assets.

5.15 IV.15 Supply Chain and Fleet

TESTING

5.15.1 Module overview

Supply-chain and fleet surfaces connect shipments, vehicles, trips, drivers, fuel, maintenance, incidents, routes, telematics events, and alerts. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Supply Chain and Fleet is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `fleet\alerts`, `fleet\audit\events`, `fleet\driver\assignments`, `fleet\drivers`, `fleet\fuel\cards`, `fleet\fuel\transactions`, `fleet\incidents`, `fleet\maintenance\jobs`, `fleet\maintenance\plans`, `fleet\routes`, `fleet\service\records`, `fleet\spare\parts`, `fleet\telematics\events`, `fleet\trips`, `fleet\tyres`, `fleet\vehicle\categories`, `fleet\vehicle\documents`, `fleet\vehicles`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Plan route/trip → assign driver/vehicle → record movement or fuel → ingest event → inspect alert → close maintenance or incident. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Telematics and external vehicle providers require approved webhook configuration and secrets; no external movement is claimed from a local screen alone.

5.15.2 Purpose, users, navigation, and features

Primary roles: Fleet manager; logistics user; administrator. UI evidence: `client/src/components/OperationsCommandCenters.jsx`; `client/src/components/FleetWorkspace.jsx`. Supply-chain and fleet surfaces connect shipments, vehicles, trips, drivers, fuel, maintenance, incidents, routes, telematics events, and alerts.

5.15.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/OperationsCommandCenters.jsx</code> ; <code>client/src/components/FleetWorkspace.jsx</code> ; server: <code>server/fleetManagement.ts</code>
Purpose	Supply Chain and Fleet	Primary users	Fleet manager; logistics user; administrator

Area	Verified reference	Operational interpretation	in- Limitation
Navigation	Authenticated module registry or dedicated workspace	Workflow	Plan route/trip → assign driver/vehicle → record movement or fuel → ingest event → inspect alert → close maintenance or incident.
Database	fleet_alerts, fleet_audit_events, fleet_driver_assignments, fleet_drivers, fleet_fuel_cards, fleet_fuel_transactions, fleet_incidents, fleet_maintenance_jobs, fleet_maintenance_plans, fleet_routes, fleet_service_records, fleet_spare_parts, fleet_telematics_events, fleet_trips, fleet_tyres, fleet_vehicle_categories	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Telematics and external vehicle providers require approved webhook configuration and secrets; no external movement is claimed from a local screen alone.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.15.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.15.5 Known limitations and future improvements

Telematics and external vehicle providers require approved webhook configuration and secrets; no external movement is claimed from a local screen alone. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

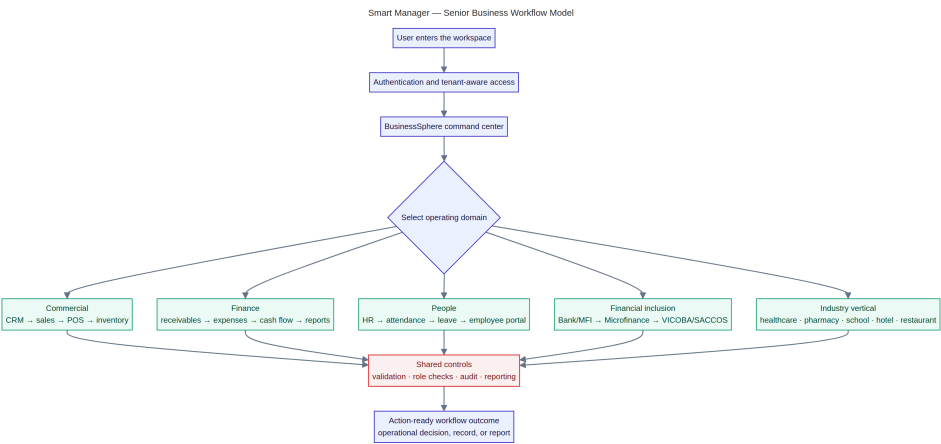


Figure 21: Supply Chain and Fleet evidence image from repository or captured application assets.

5.16 IV.16 Marketing Campaigns

TESTING

5.16.1 Module overview

Marketing provides campaign-oriented surfaces, customer segmentation signals, message templates, and analytics boundaries. Delivery is kept separate from the platform's verified persistence contract. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Marketing Campaigns is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `crm\u{005F}contacts`, `crm\u{005F}interactions`, `crm\u{005F}leads`, `emails`, `marketing\u{005F}campaigns`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Define audience → draft message → validate template/link → obtain approval → dispatch through configured channel → measure response. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. External email, SMS, WhatsApp, and campaign providers remain configuration-dependent; the UI must not claim dispatch when only a draft exists.

5.16.2 Purpose, users, navigation, and features

Primary roles: CMO; marketing manager; sales manager. UI evidence: `client/src/components/CommercialCommandCenters.jsx`. Marketing provides campaign-oriented surfaces, customer segmentation signals, message templates, and analytics boundaries. Delivery is kept separate from the platform's verified persistence contract.

5.16.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in- terpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/CommercialCommandCenters.jsx</code> server: <code>server/routers.ts</code> ; <code>server/emailTemplateWorkflow.ts</code>
Purpose	Marketing Campaigns	Primary users	CMO; marketing manager; sales manager
Navigation	Authenticated module registry or dedicated workspace	Workflow	Define audience → draft message → validate template/link → obtain approval → dispatch through configured channel → measure response.
Database	<code>crm_contacts</code> , <code>crm_interactions</code> , <code>crm_leads</code> , <code>emails</code> , <code>marketing_campaigns</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in-terpretation	Limitation
Known limitation	External email, SMS, WhatsApp, and campaign providers remain configuration-dependent; the UI must not claim dispatch when only a draft exists.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.16.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.16.5 Known limitations and future improvements

External email, SMS, WhatsApp, and campaign providers remain configuration-dependent; the UI must not claim dispatch when only a draft exists. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.



Figure 22: Marketing Campaigns evidence image from repository or captured application assets.

5.17 IV.17 E-Commerce Storefront

PARTIALLY IMPLEMENTED

5.17.1 Module overview

The repository names e-commerce products and orders and includes storefront-oriented surfaces. The evidence boundary does not invent a live Shopify or third-party commerce activation. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of E-Commerce Storefront is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `ecommerce\u{005F}orders`, `ecommerce\u{005F}products`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Publish catalog → receive order → validate customer/order → connect fulfilment and finance → report. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. External commerce provider credentials and production storefront activation are not verified by this book.

5.17.2 Purpose, users, navigation, and features

Primary roles: Store manager; sales manager; customer. UI evidence: `client/src/components/CommercialCommandCenters.jsx`. The repository names e-commerce products and orders and includes storefront-oriented surfaces. The evidence boundary does not invent a live Shopify or third-party commerce activation.

5.17.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: <code>client/src/components/CommercialCommandCenters.jsx</code> server: <code>server/routers.ts</code>
Purpose	E-Commerce Storefront	Primary users	Store manager; sales manager; customer
Navigation	Authenticated module registry or dedicated workspace	Workflow	Publish catalog → receive order → validate customer/order → connect fulfilment and finance → report.
Database	<code>ecommerce_orders</code> , <code>ecommerce_products</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	External commerce provider credentials and production storefront activation are	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

Area	Verified reference	Operational interpretation	in- Limitation
	not verified by this book.		

5.17.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.17.5 Known limitations and future improvements

External commerce provider credentials and production storefront activation are not verified by this book. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

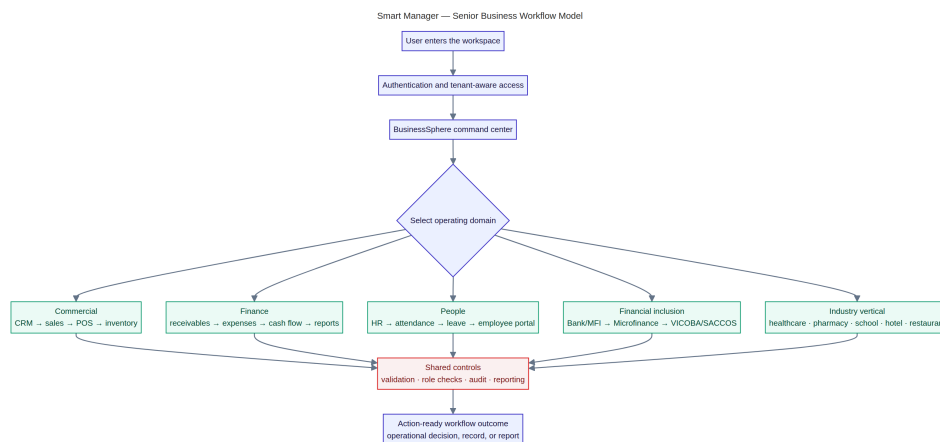


Figure 23: E-Commerce Storefront evidence image from repository or captured application assets.

5.18 IV.18 Documents and Secure Files

TESTING

5.18.1 Module overview

Document records, storage references, signatures, exports, and download boundaries keep file bytes separate from transactional metadata and preserve tenant scope. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Documents and Secure Files is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `approval\{005F}` signatures, `digital\{005F}` signatures, documents, signatures. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Create metadata → upload through storage boundary → verify reference → share or sign under role control → download/export evidence. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. S3/storage configuration is required for live file bytes; metadata persistence alone is not evidence that a file was uploaded successfully.

5.18.2 Purpose, users, navigation, and features

Primary roles: All authenticated users within permitted company scope; approver. UI evidence: `client/src/components/PeopleCommandCenters.jsx`; `client/src/components/ProfileIdentityCenter.jsx`. Document records, storage references, signatures, exports, and download boundaries keep file bytes separate from transactional metadata and preserve tenant scope.

5.18.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/PeopleCommandCenters.jsx</code> ; <code>client/src/components/ProfileIdentityCenter.jsx</code> ; server: <code>server/storage.ts</code> ; <code>server/documentDownloadBounda</code>
Purpose	Documents and Secure Files	Primary users	All authenticated users within permitted company scope; approver
Navigation	Authenticated module registry or dedicated workspace	Workflow	Create metadata → upload through storage boundary → verify reference → share or sign under role control → download/export evidence.

Area	Verified reference	Operational in- interpretation	Limitation
Database	approval_signatures, digital_signatures, documents, signatures	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	S3/storage configura- tion is required for live file bytes; meta- data persistence alone is not evidence that a file was uploaded suc- cessfully.	Future improve- ment	Complete controlled authenti- cated CRUD and provider accep- tance where the status is not PASSED.

5.18.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.18.5 Known limitations and future improvements

S3/storage configuration is required for live file bytes; metadata persistence alone is not evidence that a file was uploaded successfully. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

SMART MANAGER — Integration Authentication and Tenant Isolation

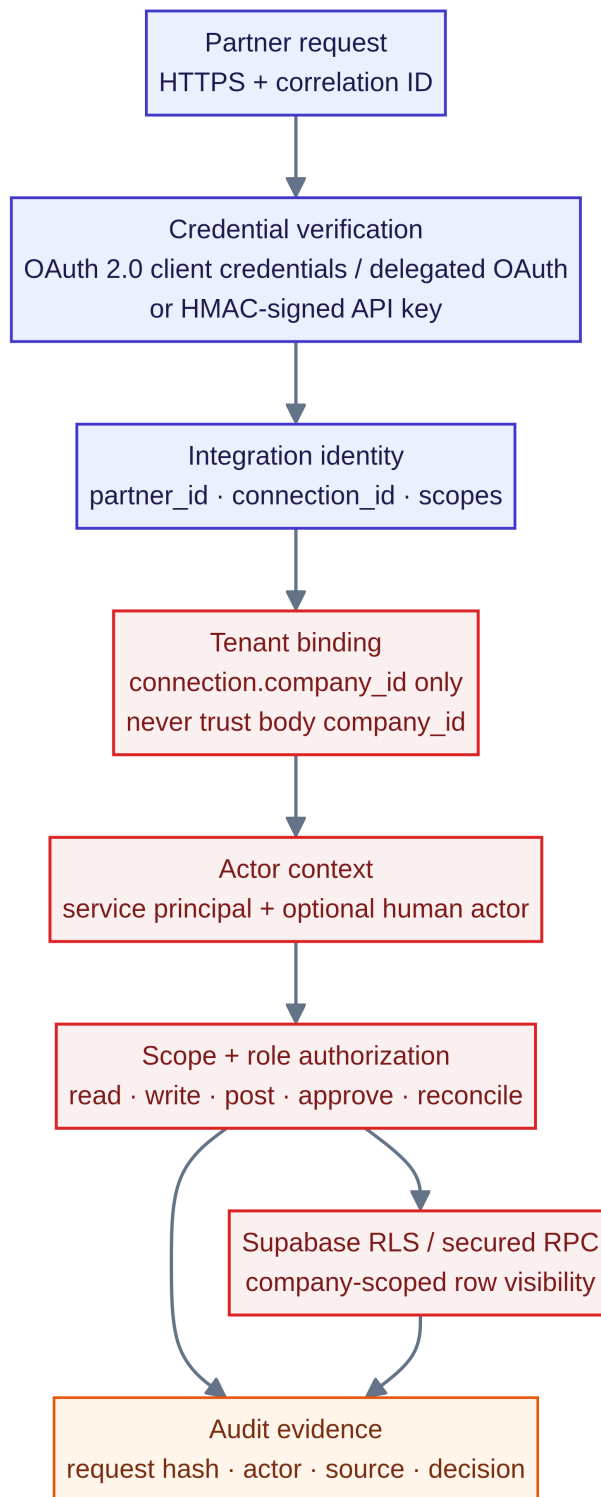


Figure 24: Documents and Secure Files evidence image from repository or captured application assets.

5.19 IV.19 Projects and Task Management

PARTIALLY IMPLEMENTED

5.19.1 Module overview

Projects provide tasks, milestones, expenses, and progress-oriented persistence boundaries so work can be coordinated with financial and operational context. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Projects and Task Management is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `project\{005F}expenses`, `project\{005F}milestones`, `project\{005F}tasks`, `projects`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Create project → add milestone/task → assign owner → update progress → record expense → review completion. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. The repository contains the persistence boundary, but complete project operations still depend on authenticated data and role-specific acceptance.

5.19.2 Purpose, users, navigation, and features

Primary roles: Project Manager; team member; executive. UI evidence: `client/src/components/PeopleCommandCenters.jsx`. Projects provide tasks, milestones, expenses, and progress-oriented persistence boundaries so work can be coordinated with financial and operational context.

5.19.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in- interpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: <code>client/src/components/PeopleCommandCenters.jsx</code> ; server: <code>server/routers.ts</code> ; <code>server/projectsPersistence.ts</code>
Purpose	Projects and Task Management	Primary users	Project Manager; team member; executive
Navigation	Authenticated module registry or dedicated workspace	Workflow	Create project → add milestone/task → assign owner → update progress → record expense → review completion.
Database	<code>project_expenses</code> , <code>project_milestones</code> , <code>project_tasks</code> , <code>projects</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	The repository contains the persistence boundary, but com-	Future improvement	Complete controlled authenticated CRUD and provider accep-

Area	Verified reference	Operational interpretation	in- Limitation
	plete project operations still depend on authenticated data and role-specific acceptance.		tance where the status is not PASSED.

5.19.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.19.5 Known limitations and future improvements

The repository contains the persistence boundary, but complete project operations still depend on authenticated data and role-specific acceptance. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

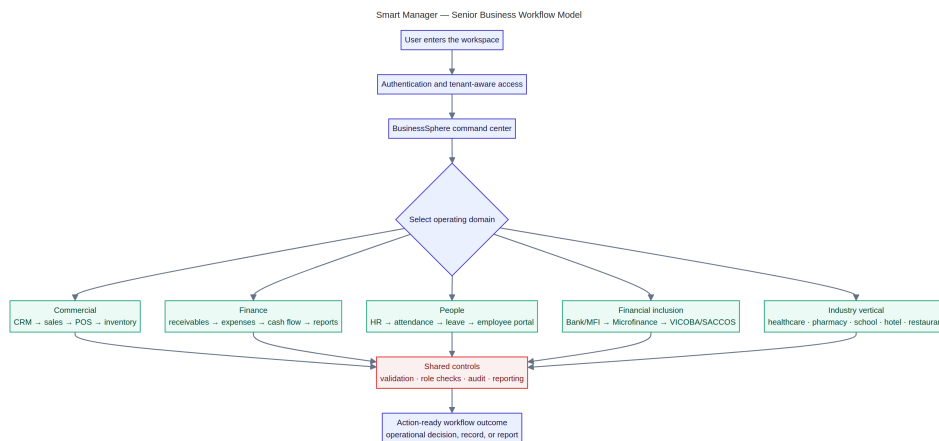


Figure 25: Projects and Task Management evidence image from repository or captured application assets.

5.20 IV.20 Customer Support and Helpdesk

TESTING

5.20.1 Module overview

Support includes tickets, timelines, internal notes, inbox/chat evidence, calls, SLA policies, workflow rules, search, and support metrics. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Customer Support and Helpdesk is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `support\agents`, `support\call\log`, `support\chat\conversations`, `support\chat\messages`, `support\message\templates`, `support\sla\policies`, `support\team\members`, `support\teams`, `support\ticket\activity`, `support\ticket\messages`, `support\ticket\notes`, `support\tickets`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Receive issue → create ticket → classify/SLA → assign → communicate → resolve → audit and measure. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Automated external support execution is intentionally not claimed without configured channels; internal ticket persistence remains the source of record.

5.20.2 Purpose, users, navigation, and features

Primary roles: Support agent; manager; customer; administrator. UI evidence: `client/src/components/IntelligenceCommandCenters.jsx`; `client/src/components/PeopleCommandCenters.jsx`. Support includes tickets, timelines, internal notes, inbox/chat evidence, calls, SLA policies, workflow rules, search, and support metrics.

5.20.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/IntelligenceCommandCenters.jsx</code> ; <code>client/src/components/PeopleCommandCenters.jsx</code> ; server: <code>server/supportOperations.ts</code> ; <code>server/supportMetrics.ts</code>
Purpose	Customer Support and Helpdesk	Primary users	Support agent; manager; customer; administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Receive issue → create ticket → classify/SLA → assign → com-

Area	Verified reference	Operational interpretation	in- Limitation
			municate → resolve → audit and measure.
Database	support_agents, support_call_log, support_chat_conversations, support_chat_messages, support_message_templates, support_sla_policies, support_team_members, support_teams, support_ticket_activity, support_ticket_messages, support_ticket_notes, support_tickets	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Automated external support execution is intentionally not claimed without configured channels; internal ticket persistence remains the source of record.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.20.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.20.5 Known limitations and future improvements

Automated external support execution is intentionally not claimed without configured channels; internal ticket persistence remains the source of record. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

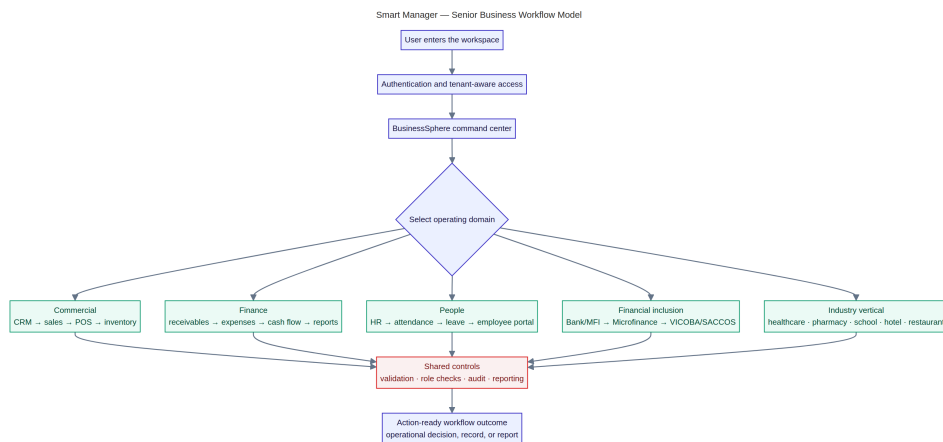


Figure 26: Customer Support and Helpdesk evidence image from repository or captured application assets.

5.21 IV.21 Enterprise Analytics and BI

PARTIALLY IMPLEMENTED

5.21.1 Module overview

Analytics combines KPI, benchmark, market, trend, and command-center views. It is a decision-support layer over recorded operations, not a replacement for source ledgers. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Enterprise Analytics and BI is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes competitors, custom_kpis, financial_benchmarks. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Select scope → read source metrics → compare or trend → investigate source record → document action. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Analytical quality depends on complete, correctly scoped input data and configured market services.

5.21.2 Purpose, users, navigation, and features

Primary roles: Executive; analyst; manager. UI evidence: client/src/components/IntelligenceCommandCenters.jsx; client/src/components/PredictiveAnalyticsWorkspace.jsx. Analytics combines KPI, benchmark, market, trend, and command-center views. It is a decision-support layer over recorded operations, not a replacement for source ledgers.

5.21.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: client/src/components/IntelligenceCommandCenters.jsx; client/src/components/PredictiveAnalyticsWorkspace.jsx server: server/dashboardReports.ts; server/marketIntelligence.ts
Purpose	Enterprise Analytics and BI	Primary users	Executive; analyst; manager
Navigation	Authenticated module registry or dedicated workspace	Workflow	Select scope → read source metrics → compare or trend → investigate source record → document action.
Database	competitors, custom_kpis, financial_benchmarks	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
Known limitation	Analytical quality depends on complete, correctly scoped input data and configured market services.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.21.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.21.5 Known limitations and future improvements

Analytical quality depends on complete, correctly scoped input data and configured market services. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.



Figure 27: Enterprise Analytics and BI evidence image from repository or captured application assets.

5.22 IV.22 Notifications and Alerting

TESTING

5.22.1 Module overview

Notifications capture reminders, delivery history, unread/read state, scheduled signals, and module-specific alerts while distinguishing a persisted notification from successful external delivery. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Notifications and Alerting is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `emails`, `notification\channels`, `notification\log`, `notification\rules`, `sms\group\members`, `sms\groups`, `sms\templates`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Detect event → create notification record → deliver through configured channel → mark read or retry → audit result. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. SMS, email, push, and WhatsApp delivery depend on external provider configuration and should show explicit unavailable states.

5.22.2 Purpose, users, navigation, and features

Primary roles: Authenticated users; managers; administrators. UI evidence: `client/src/BusinessSphereDashboard.jsx`; module workspaces. Notifications capture reminders, delivery history, unread/read state, scheduled signals, and module-specific alerts while distinguishing a persisted notification from successful external delivery.

5.22.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-terpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/BusinessSphereDashboard.jsx</code> ; module workspaces; server: <code>server/notificationHistory.ts</code> ; <code>server/transactionalEmail.ts</code>
Purpose	Notifications and Alerting	Primary users	Authenticated users; managers; administrators
Navigation	Authenticated module registry or dedicated workspace	Workflow	Detect event → create notification record → deliver through configured channel → mark read or retry → audit result.
Database	<code>emails</code> , <code>notification_channels</code> , <code>notification_log</code> ,	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
	notification_rules, sms_group_members, sms_groups, sms_templates		
Known limitation	SMS, email, push, and WhatsApp delivery depend on external provider configuration and should show explicit unavailable states.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.22.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.22.5 Known limitations and future improvements

SMS, email, push, and WhatsApp delivery depend on external provider configuration and should show explicit unavailable states. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

Smart Manager — Live Deployment Observation

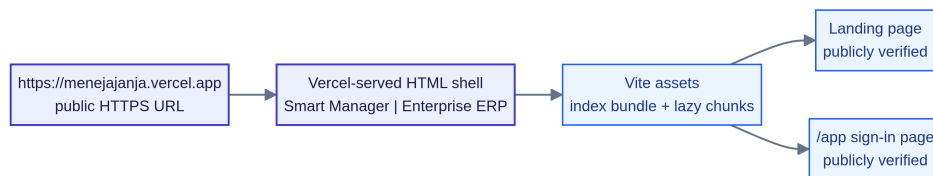


Figure 28: Notifications and Alerting evidence image from repository or captured application assets.

5.23 IV.23 Activity Stream and Audit Evidence

TESTING

5.23.1 Module overview

Audit evidence records who performed an action, where applicable, in which company scope, and with what result. Sensitive global administration records have a separate control boundary. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Activity Stream and Audit Evidence is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `audit\log`, `audit\log\entries`, `bank\audit\events`, `billing\plan\audit\log`, `community\group\audit\log`, `hospitality\audit\log`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Authenticated action → protected mutation → audit record → scoped search/export → review or incident follow-up. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. An audit record increases traceability but does not correct an unsafe write; mutation confirmation and database constraints remain primary controls.

5.23.2 Purpose, users, navigation, and features

Primary roles: Auditor; administrator; company manager with scope. UI evidence: `client/src/components/ComplianceAuditLogView.tsx`; dashboard views. Audit evidence records who performed an action, where applicable, in which company scope, and with what result. Sensitive global administration records have a separate control boundary.

5.23.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/ComplianceAuditLogView.tsx</code> ; dashboard views; server: <code>server/auditLogs.ts</code> ; <code>server/tenantAuditViewer.ts</code>
Purpose	Activity Stream and Audit Evidence	Primary users	Auditor; administrator; company manager with scope
Navigation	Authenticated module registry or dedicated workspace	Workflow	Authenticated action → protected mutation → audit record → scoped search/export → review or incident follow-up.

Area	Verified reference	Operational in- interpretation	Limitation
Database	audit_log, audit_log_entries, bank_audit_events, billing_plan_audit_log, community_group_audit_log, hospitality_audit_log	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	An audit record increases traceability but does not correct an unsafe write; mutation confirmation and database constraints remain primary controls.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.23.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.23.5 Known limitations and future improvements

An audit record increases traceability but does not correct an unsafe write; mutation confirmation and database constraints remain primary controls. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

SMART MANAGER — Integration Authentication and Tenant Isolation

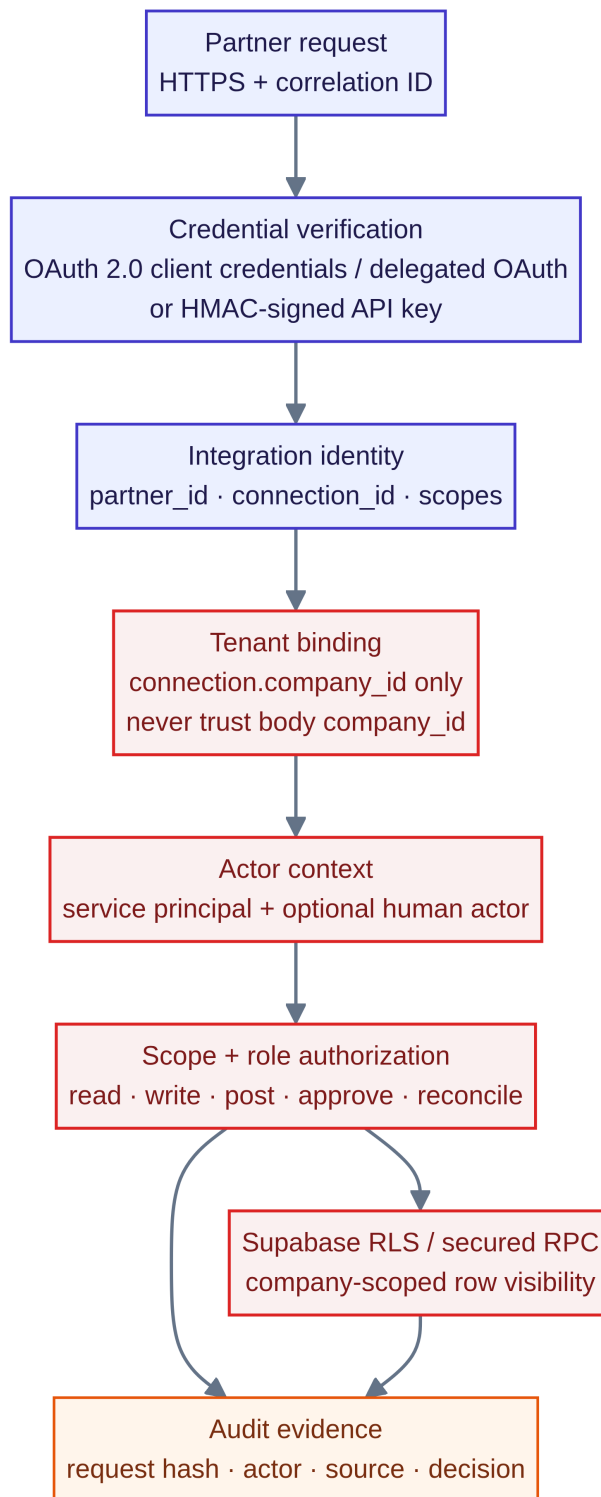


Figure 29: Activity Stream and Audit Evidence evidence image from repository or captured application assets.

5.24 IV.24 Integration Hub

CONFIGURATION REQUIRED

5.24.1 Module overview

The hub represents integration metadata, webhook configuration, provider readiness, delivery history, and safe test boundaries. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Integration Hub is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `integration\connections`, `notification\channels`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Define connection → validate configuration → test safely → enable dispatch → monitor delivery → retry or disable. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. No provider credential, endpoint, or delivery success is assumed when configuration is missing; controls must remain disabled or clearly unavailable.

5.24.2 Purpose, users, navigation, and features

Primary roles: Administrator; integration manager; finance manager. UI evidence: `client/src/components/FinanceCommandCenters.jsx`; `client/src/components/PeopleCommandCenters.jsx`. The hub represents integration metadata, webhook configuration, provider readiness, delivery history, and safe test boundaries.

5.24.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	CONFIGURATION REQUIRED	Evidence	UI: <code>client/src/components/FinanceCommandCenters.jsx</code> ; <code>client/src/components/PeopleCommandCenters.jsx</code> ; server: <code>server/webhooks.ts</code> ; <code>server/transactionalEmail.ts</code> ; <code>server/integration connections</code>
Purpose	Integration Hub	Primary users	Administrator; integration manager; finance manager
Navigation	Authenticated module registry or dedicated workspace	Workflow	Define connection → validate configuration → test safely → enable dispatch → monitor delivery → retry or disable.
Database	<code>integration_connections</code> , <code>notification_channels</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
Known limitation	No provider credential, endpoint, or delivery success is assumed when configuration is missing; controls must remain disabled or clearly unavailable.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.24.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.24.5 Known limitations and future improvements

No provider credential, endpoint, or delivery success is assumed when configuration is missing; controls must remain disabled or clearly unavailable. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

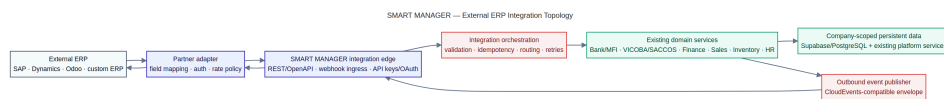


Figure 30: Integration Hub evidence image from repository or captured application assets.

5.25 IV.25 Workflow Studio and Marketplace

PARTIALLY IMPLEMENTED

5.25.1 Module overview

Workflow Studio represents reusable templates, steps, approvals, tasks, notifications, and audit actions. Official templates state what they do and what they do not do. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Workflow Studio and Marketplace is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `workflow\templates`, `workflows`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Choose template → inspect steps → configure target → approve or run → record task/result → review audit. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. A workflow step that drafts an email or notification is not proof of external dispatch; complete orchestration requires channel configuration.

5.25.2 Purpose, users, navigation, and features

Primary roles: Workflow administrator; approver; manager. UI evidence: `client/src/components/PeopleCommandCenters.jsx`; dashboard registry. Workflow Studio represents reusable templates, steps, approvals, tasks, notifications, and audit actions. Official templates state what they do and what they do not do.

5.25.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: <code>client/src/components/PeopleCommandCenters.jsx</code> ; dashboard registry; server: <code>server/workflows.ts</code> ; <code>server/aiApprovals.ts</code>
Purpose	Workflow Studio and Marketplace	Primary users	Workflow administrator; approver; manager
Navigation	Authenticated module registry or dedicated workspace	Workflow	Choose template → inspect steps → configure target → approve or run → record task/result → review audit.
Database	<code>workflow_marketplace_templates</code> , <code>workflows</code>	Templates, workflows	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
Known limitation	A workflow step that drafts an email or notification is not proof of external dispatch; complete orchestration requires channel configuration.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.25.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.25.5 Known limitations and future improvements

A workflow step that drafts an email or notification is not proof of external dispatch; complete orchestration requires channel configuration. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

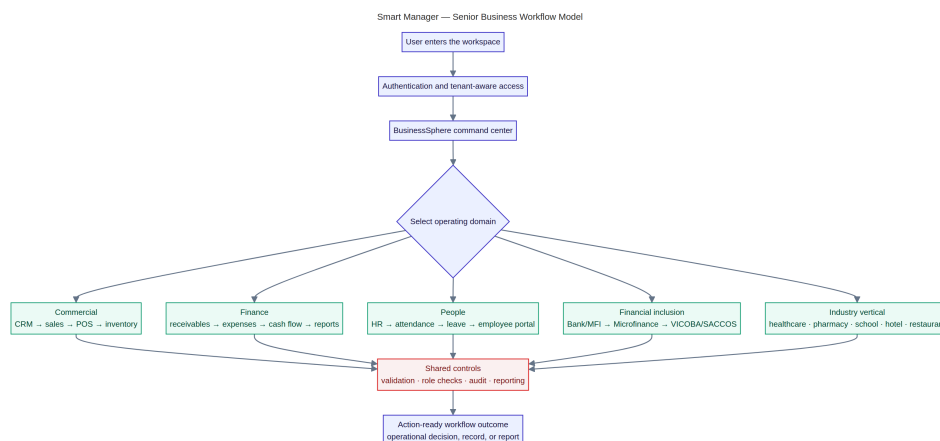


Figure 31: Workflow Studio and Marketplace evidence image from repository or captured application assets.

5.26 IV.26 Collaboration Hub

PARTIALLY IMPLEMENTED

5.26.1 Module overview

Collaboration connects messages, channels, presence, calendar, read receipts, and workflow signals where configured. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Collaboration Hub is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `calendar\events`, `collab\channels`, `collab\messages`, `emails`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Create channel/event → post message → notify participants → track read/presence → retain evidence. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. External collaboration delivery and realtime behavior depend on configuration and provider availability; local state is not a durable message authority.

5.26.2 Purpose, users, navigation, and features

Primary roles: Team member; manager; executive. UI evidence: `client/src/components/PeopleCommandCenters.jsx`. Collaboration connects messages, channels, presence, calendar, read receipts, and workflow signals where configured.

5.26.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: <code>client/src/components/PeopleCommandCenters.jsx</code> ; server: <code>server/collaborationEmailLinkCheck.ts</code> ; server: <code>server/collaborationPersistence.ts</code>
Purpose	Collaboration Hub	Primary users	Team member; manager; executive
Navigation	Authenticated module registry or dedicated workspace	Workflow	Create channel/event → post message → notify participants → track read/presence → retain evidence.
Database	<code>calendar_events</code> , <code>collab_channels</code> , <code>collab_messages</code> , <code>emails</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
Known limitation	External collaboration delivery and real-time behavior depend on configuration and provider availability; local state is not a durable message authority.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.26.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.26.5 Known limitations and future improvements

External collaboration delivery and realtime behavior depend on configuration and provider availability; local state is not a durable message authority. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

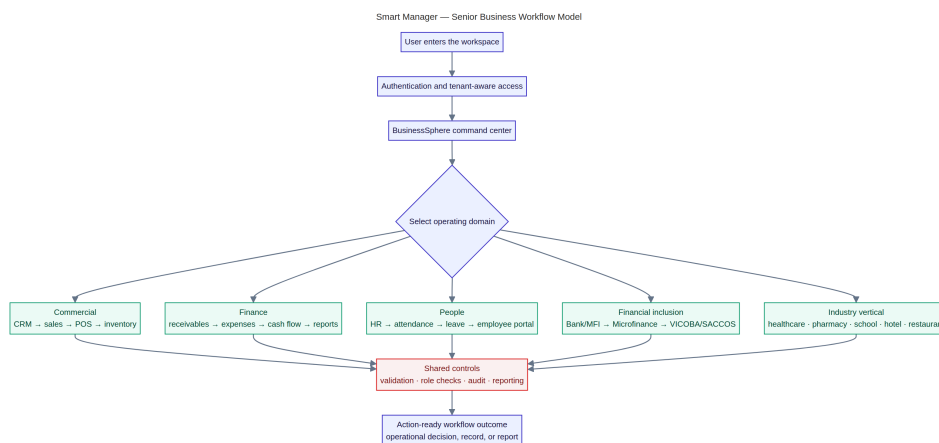


Figure 32: Collaboration Hub evidence image from repository or captured application assets.

5.27 IV.27 TRA, VFD, and Tanzania Fiscalization

CONFIGURATION REQUIRED

5.27.1 Module overview

The repository contains fiscal profiles, receipts, retry queues, Z-report archives, tax configuration, VAT anomaly analysis, and idempotent submission boundaries. It distinguishes configuration and provider readiness from an actual TRA submission. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of TRA, VFD, and Tanzania Fiscalization is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes shared company-scoped persistence tables. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Configure TIN/VRN and branch → validate readiness → submit idempotent transaction → receive provider result → store receipt/retry/audit evidence. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Production fiscalization requires approved provider credentials, device/branch configuration, and provider availability; a local receipt screen is not a TRA acceptance claim.

5.27.2 Purpose, users, navigation, and features

Primary roles: CFO; Finance Manager; administrator; fiscal operator. UI evidence: client/src/components/TraPortalModule.jsx. The repository contains fiscal profiles, receipts, retry queues, Z-report archives, tax configuration, VAT anomaly analysis, and idempotent submission boundaries. It distinguishes configuration and provider readiness from an actual TRA submission.

5.27.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	CONFIGURATION REQUIRED	Evidence	UI: client/src/components/TraPortalModule.jsx; server: server/traFiscalRouter.ts; server/traFiscal.ts; server/traZReportArchive.ts
Purpose	TRA, VFD, and Tanzania Fiscalization	Primary users	CFO; Finance Manager; administrator; fiscal operator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Configure TIN/VRN and branch → validate readiness → submit idempotent transaction → receive provider result → store receipt/retry/audit evidence.

Area	Verified reference	Operational in- interpretation	Limitation
Database	Shared company-scoped tables	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Production fiscalization requires approved provider credentials, device/branch configuration, and provider availability; a local receipt screen is not a TRA acceptance claim.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.27.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.27.5 Known limitations and future improvements

Production fiscalization requires approved provider credentials, device/branch configuration, and provider availability; a local receipt screen is not a TRA acceptance claim. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

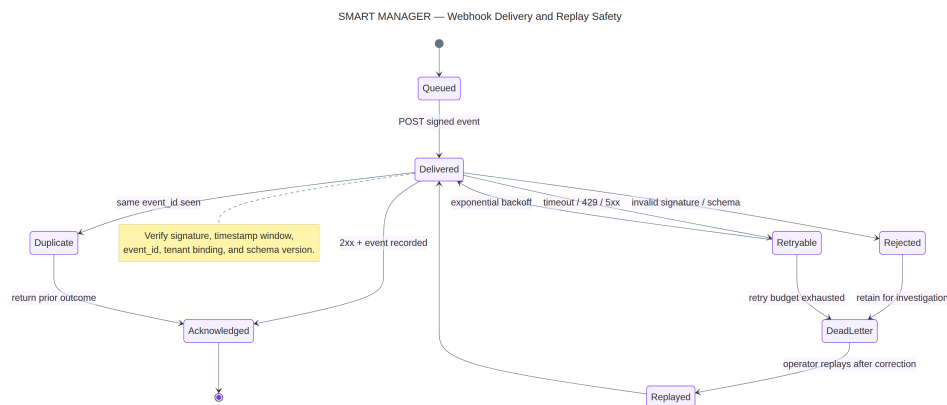


Figure 33: TRA, VFD, and Tanzania Fiscalization evidence image from repository or captured application assets.

5.28 IV.28 AI Assistant and Smart Intelligence

EXTERNAL SERVICE REQUIRED

5.28.1 Module overview

The AI assistant uses a built-in model boundary, limits history and context, treats user/business JSON as untrusted data, returns structured responses, restricts navigation targets, and produces proposals rather than silently mutating business records. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of AI Assistant and Smart Intelligence is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `custom\u{005F}kpis`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Verified user → scoped context → structured assistant response → optional navigation/proposal → independent role approval → confirmed business action. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. AI model availability and response quality depend on configured service access and supplied evidence; the assistant must not claim a mutation it did not execute.

5.28.2 Purpose, users, navigation, and features

Primary roles: Authenticated user; approver for proposed actions. UI evidence: `client/src/components/AIChatBox.tsx`; `client/src/components/IntelligenceCommandCenters.jsx`. The AI assistant uses a built-in model boundary, limits history and context, treats user/business JSON as untrusted data, returns structured responses, restricts navigation targets, and produces proposals rather than silently mutating business records.

5.28.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	EXTERNAL SERVICE REQUIRED	Evidence	UI: <code>client/src/components/AIChatBox.tsx</code> ; <code>client/src/components/IntelligenceCommandCenters.jsx</code> ; server: <code>server/smartAssistant.ts</code> ; <code>server/aiApprovals.ts</code>
Purpose	AI Assistant and Smart Intelligence	Primary users	Authenticated user; approver for proposed actions
Navigation	Authenticated module registry or dedicated workspace	Workflow	Verified user → scoped context → structured assistant response → optional navigation/proposal → independent role approval → confirmed business action.

Area	Verified reference	Operational interpretation	in-	Limitation
Database	custom_kpis	Security		Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	AI model availability and response quality depend on configured service access and supplied evidence; the assistant must not claim a mutation it did not execute.	Future improvement		Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.28.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.28.5 Known limitations and future improvements

AI model availability and response quality depend on configured service access and supplied evidence; the assistant must not claim a mutation it did not execute. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.



Figure 34: AI Assistant and Smart Intelligence evidence image from repository or captured application assets.

5.29 IV.29 WhatsApp Web Integration

CONFIGURATION REQUIRED

5.29.1 Module overview

WhatsApp surfaces support account/contact/message/conversation metadata and provider readiness boundaries. Credentials and provider access must remain server-side. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of WhatsApp Web Integration is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `whatsapp\account\links`, `whatsapp\accounts`, `whatsapp\contacts`, `whatsapp\conversations`, `whatsapp\message\events`, `whatsapp\messages`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Configure provider → link account/contact → draft message → validate content → dispatch through provider → record status. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Provider configuration and message delivery are not assumed; credential persistence in browser storage is a security risk and must remain removed or blocked.

5.29.2 Purpose, users, navigation, and features

Primary roles: Marketing manager; support agent; administrator. UI evidence: `client/src/BusinessSphereDashboard.jsx`; messaging surfaces. WhatsApp surfaces support account/contact/message/conversation metadata and provider readiness boundaries. Credentials and provider access must remain server-side.

5.29.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	CONFIGURATION REQUIRED	Evidence	UI: <code>client/src/BusinessSphereDashboard.jsx</code> ; messaging surfaces; server: <code>server/whatsappProvider.ts</code> ; <code>server/whatsappSecurity.ts</code>
Purpose	WhatsApp Web Integration	Primary users	Marketing manager; support agent; administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Configure provider → link account/contact → draft message → validate content → dispatch through provider → record status.

Area	Verified reference	Operational in- interpretation	Limitation
Database	whatsapp_account_links, whatsapp_accounts, whatsapp_contacts, whatsapp_conversations, whatsapp_message_events, whatsapp_messages	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Provider configuration and message delivery are not assumed; credential persistence in browser storage is a security risk and must remain removed or blocked.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.29.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.29.5 Known limitations and future improvements

Provider configuration and message delivery are not assumed; credential persistence in browser storage is a security risk and must remain removed or blocked. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

Smart Manager — Live Deployment Observation

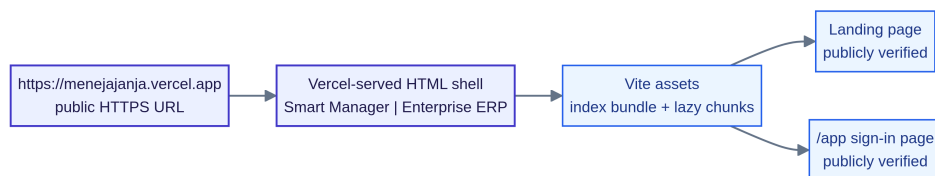


Figure 35: WhatsApp Web Integration evidence image from repository or captured application assets.

5.30 IV.30 Microfinance

TESTING

5.30.1 Module overview

Microfinance covers borrowers, groups, products, applications, approvals, disbursement, repayments, savings, cash sessions, collections, scoring, and escalation. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Microfinance is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `mfi\audit\logs`, `mfi\cash\sessions`, `mfi\cash\transactions`, `mfi\clients`, `mfi\collateral`, `mfi\collections`, `mfi\credit\scorecards`, `mfi\credit\scoring`, `mfi\settings`, `mfi\groups`, `mfi\guarantors`, `mfi\loan\applications`, `mfi\loan\products`, `mfi\loans`, `mfi\notifications`, `mfi\par\escalation\settings`, `mfi\repayment\schedules`, `mfi\repayments`, `mfi\savings`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Register borrower/group → choose product → submit application → approve → disburse → collect repayment/savings → monitor PAR/escalation. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Financial persistence and controlled data must be exercised against a real tenant schema; seeded UI examples are not production records.

5.30.2 Purpose, users, navigation, and features

Primary roles: MFI manager; loan officer; cashier; approver; auditor. UI evidence: `client/src/components/MicrofinanceWorkspace.jsx`; `client/src/components/MicrofinanceGovernanceDialogs.jsx`. Microfinance covers borrowers, groups, products, applications, approvals, disbursement, repayments, savings, cash sessions, collections, scoring, and escalation.

5.30.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/MicrofinanceWorkspace.jsx</code> ; <code>client/src/components/MicrofinanceGovernanceDialogs.jsx</code> ; server: <code>server/microfinanceOperations.ts</code>
Purpose	Microfinance	Primary users	MFI manager; loan officer; cashier; approver; auditor

Area	Verified reference	Operational in- interpretation	Limitation
Navigation	Authenticated module registry or dedicated workspace	Workflow	Register borrower/group → choose product → submit application → approve → disburse → collect repayment/savings → monitor PAR/escalation.
Database	mfi_audit_logs, mfi_cash_sessions, mfi_cash_transactions, mfi_clients, mfi_collateral, mfi_collections, mfi_credit_scorecards, mfi_credit_scoring_settings, mfi_groups, mfi_guarantors, mfi_loan_applications, mfi_loan_products, mfi_loans, mfi_notifications, mfi_par_escalation_settings, mfi_repayment_schedules	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Financial persistence and controlled data must be exercised against a real tenant schema; seeded UI examples are not production records.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.30.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.30.5 Known limitations and future improvements

Financial persistence and controlled data must be exercised against a real tenant schema; seeded UI examples are not production records. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

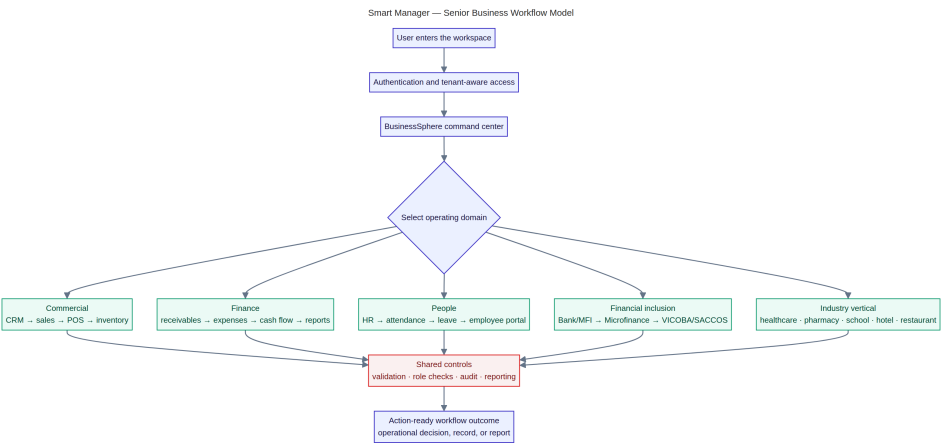


Figure 36: Microfinance evidence image from repository or captured application assets.

5.31 IV.31 Money Agent

IMPLEMENTED

5.31.1 Module overview

Money Agent covers agents, customers, wallets, cash actions, fees, commissions, ledger snapshots, approvals, reconciliation, and customer-facing summaries. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Money Agent is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `money\agent\agents`, `money\agent\alerts`, `money\agent\approvals`, `money\agent\audit\events`, `money\agent\branches`, `money\agent\commission\rules`, `money\agent\customers`, `money\agent\daily\summaries`, `money\agent\fee\rules`, `money\agent\ledger\entries`, `money\agent\limits`, `money\agent\notifications`, `money\agent\pin\credentials`, `money\agent\receipts`, `money\agent\reconciliations`, `money\agent\risk\events`, `money\agent\services`, `money\agent\settlements`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Register agent/customer → open cash/wallet context → initiate action → validate balance/role → record ledger → reconcile and audit. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. The live schema includes the migration history and tables, but real wallet/provider movement still requires controlled credentials and operational acceptance.

5.31.2 Purpose, users, navigation, and features

Primary roles: Money agent; supervisor; cashier; auditor. UI evidence: `client/src/components/MoneyAgentWorkspace.jsx`; `client/src/components/SectorCommandCenters.jsx`. Money Agent covers agents, customers, wallets, cash actions, fees, commissions, ledger snapshots, approvals, reconciliation, and customer-facing summaries.

5.31.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in- interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/MoneyAgentWorkspace.jsx</code> ; <code>client/src/components/SectorCommandCenters.jsx</code> ; server: <code>server/moneyAgentOperations.ts</code>

Area	Verified reference	Operational interpretation	in-	Limitation
Purpose	Money Agent	Primary users		Money agent; supervisor; cashier; auditor
Navigation	Authenticated module registry or dedicated workspace	Workflow		Register agent/customer → open cash/wallet context → initiate action → validate balance/role → record ledger → reconcile and audit.
Database	money_agent_agents, money_agent_alerts, money_agent_audit_events, money_agent_branches, money_agent_commission_rules, money_agent_customers, money_agent_daily_summaries, money_agent_fee_rules, money_agent_ledger_entries, money_agent_limits, money_agent_notifications, money_agent_pin_credentials, money_agent_receipts, money_agent_reconciliations, money_agent_risk_events	Security		Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	The live schema includes the migration history and tables, but real wallet/provider movement still requires controlled credentials and operational acceptance.	Future improvement		Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.31.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.31.5 Known limitations and future improvements

The live schema includes the migration history and tables, but real wallet/provider movement still requires controlled credentials and operational acceptance. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

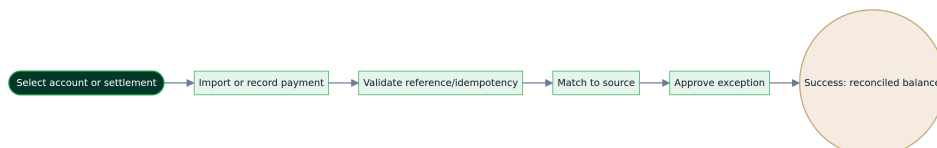


Figure 37: Money Agent evidence image from repository or captured application assets.

5.32 IV.32 VICوبا, SACCOS, and Community Groups

TESTING

5.32.1 Module overview

Community-group features cover groups, members, contributions, savings, loans, meetings, attendance, budgets, projects, welfare, votes, messages, notifications, approvals, and audit. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of VICوبا, SACCOS, and Community Groups is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `community\{005F}contributions`, `community\{005F}group\{005F}announcements`, `community\{005F}group\{005F}approvals`, `community\{005F}group\{005F}assets`, `community\{005F}group\{005F}attendance`, `community\{005F}group\{005F}audit\{005F}log`, `community\{005F}group\{005F}budgets`, `community\{005F}group\{005F}committee\{005F}members`, `community\{005F}group\{005F}committees`, `community\{005F}group\{005F}contributions`, `community\{005F}group\{005F}documents`, `community\{005F}group\{005F}events`, `community\{005F}group\{005F}expenses`, `community\{005F}group\{005F}fundraising`, `community\{005F}group\{005F}income`, `community\{005F}group\{005F}loan\{005F}guarantors`, `community\{005F}group\{005F}loan\{005F}penalties`, `community\{005F}group\{005F}loan\{005F}repayments`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Create group → add member → record contribution/meeting → approve loan or welfare action → update ledger/report. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Relationship guards and approvals are present, but broader group-accounting walkthroughs should still be completed with controlled tenant data.

5.32.2 Purpose, users, navigation, and features

Primary roles: Group chairperson; treasurer; secretary; member; auditor. UI evidence: `client/src/components/SectorCommandCenters.jsx`; community group surfaces. Community-group features cover groups, members, contributions, savings, loans, meetings, attendance, budgets, projects, welfare, votes, messages, notifications, approvals, and audit.

5.32.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-terpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/SectorCommandCenters.jsx</code> ; community group sur- faces; server: server/

Area	Verified reference	Operational interpretation	in- Limitation
			communityGroups.ts; server/routers.ts
Purpose	VICOBA, SACCOS, and Community Groups	Primary users	Group chairperson; treasurer; secretary; member; auditor
Navigation	Authenticated module registry or dedicated workspace	Workflow	Create group → add member → record contribution/meeting → approve loan or welfare action → update ledger/report.
Database	community_contributions, security, community_group_announcements, community_group_approvals, community_group_assets, community_group_attendance, community_group_audit_log, community_group_budgets, community_group_committee_members, community_group_committees, community_group_contributions, community_group_documents, community_group_events, community_group_expenses, community_group_fundraising, community_group_income, community_group_loan_guarantors		Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Relationship guards and approvals are present, but broader group-accounting walkthroughs should still be completed with controlled tenant data.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.32.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.32.5 Known limitations and future improvements

Relationship guards and approvals are present, but broader group-accounting walkthroughs should still be completed with controlled tenant data. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

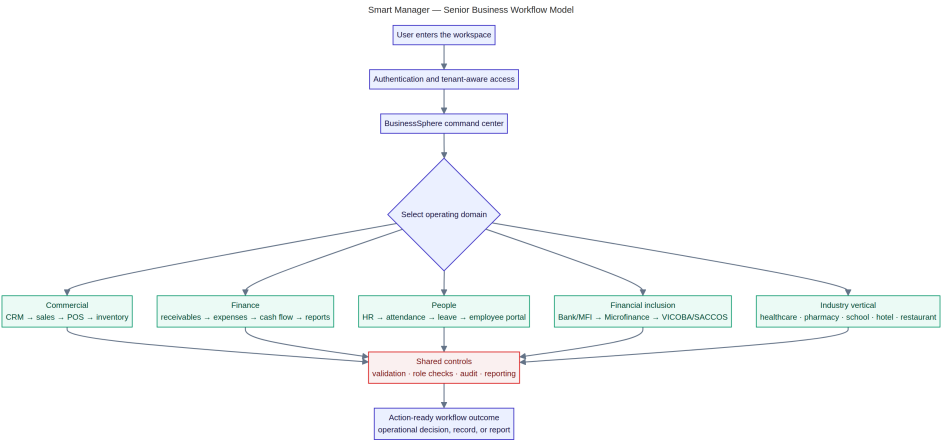


Figure 38: VICOBA, SACCOS, and Community Groups evidence image from repository or captured application assets.

5.33 IV.33 Healthcare and Clinic

PARTIALLY IMPLEMENTED

5.33.1 Module overview

Healthcare covers patients, appointments, visits, vitals, prescriptions, lab/radiology, claims, reports, reminders, patient SMS consent, portal references, and interoperability exports. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Healthcare and Clinic is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `hc\appointments`, `hc\doctors`, `hc\insurance\claims`, `hc\invoices`, `hc\lab\orders`, `hc\notifications`, `hc\patients`, `hc\portal\reference\approvals`, `hc\portal\reference\imports`, `hc\portal\reference\summary\settings`, `hc\prescriptions`, `hc\radiology`, `hc\reminder\deliveries`, `hc\reminder\settings`, `hc\reports`, `hc\visits`, `hc\vitals`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Register patient → book appointment → record visit/vitals → prescribe/order → issue report or claim → notify with consent → audit. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. SMS and interoperability depend on provider configuration; clinical data requires strict least-privilege review and must not be exposed through generic tenant screens.

5.33.2 Purpose, users, navigation, and features

Primary roles: Clinician; nurse; receptionist; finance user; patient portal user. UI evidence: `client/src/components/HealthcareClinicWorkspace.jsx`. Healthcare covers patients, appointments, visits, vitals, prescriptions, lab/radiology, claims, reports, reminders, patient SMS consent, portal references, and interoperability exports.

5.33.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-terpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: <code>client/src/components/HealthcareClinicWorkspace.jsx</code> ; server: <code>server/healthcareOperations.ts</code> ; <code>server/healthcareInteroperability.ts</code> ; <code>server/healthcareReminders.ts</code>
Purpose	Healthcare and Clinic	Primary users	Clinician; nurse; receptionist; finance user; patient portal user

Area	Verified reference	Operational interpretation	in- Limitation
Navigation	Authenticated module registry or dedicated workspace	Workflow	Register patient → book appointment → record visit/vitals → prescribe/order → issue report or claim → notify with consent → audit.
Database	hc_appointments, hc_doctors, hc_insurance_claims, hc_invoices, hc_lab_orders, hc_notifications, hc_patients, hc_portal_reference_approvals, hc_portal_reference_imports, hc_portal_reference_summary_settings, hc_prescriptions, hc_radiology, hc_reminder_deliveries, hc_reminder_settings, hc_reports, hc_visits	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	SMS and interoperability depend on provider configuration; clinical data requires strict least-privilege review and must not be exposed through generic tenant screens.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.33.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.33.5 Known limitations and future improvements

SMS and interoperability depend on provider configuration; clinical data requires strict least-privilege review and must not be exposed through generic tenant screens. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

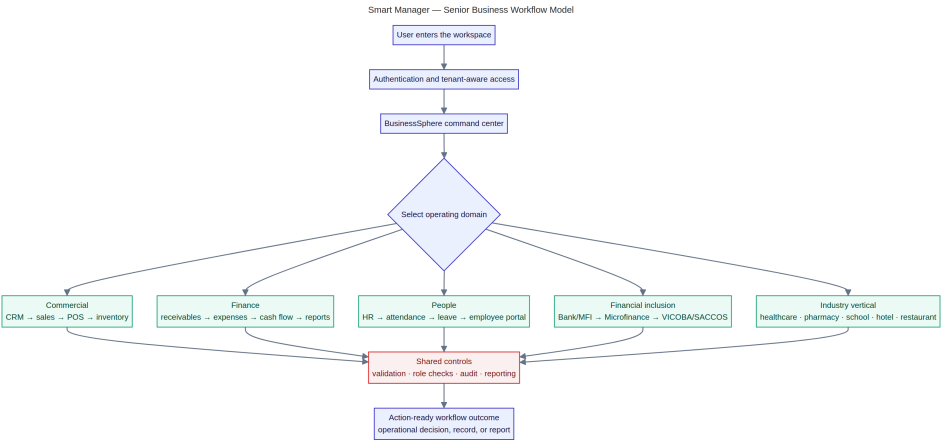


Figure 39: Healthcare and Clinic evidence image from repository or captured application assets.

5.34 IV.34 School Management

PASSED

5.34.1 Module overview

School Management includes academic years, terms, departments, subjects, classes, streams, teachers, admissions, attendance, assessments, report cards, assignments, fees, scholarships, transport, library, discipline, announcements, and portal links. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of School Management is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `sch\academic\years`, `sch\admissions`, `sch\announcements`, `sch\approval\requests`, `sch\assessment\scores`, `sch\assessments`, `sch\assignment\submissions`, `sch\assignments`, `sch\attendance\records`, `sch\attendance\sessions`, `sch\audit\logs`, `sch\books`, `sch\classes`, `sch\departments`, `sch\disciplinary\records`, `sch\documents`, `sch\enrollments`, `sch\exams`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Set academic structure → admit learner → assign teacher/class → record attendance/assessment → publish report → invoice/receive fees → communicate. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Automated browser coverage is strong for the surface, while full per-school CRUD remains a data-backed acceptance task.

5.34.2 Purpose, users, navigation, and features

Primary roles: School administrator; teacher; student/guardian portal user; finance user. UI evidence: `client/src/components/SchoolWorkspace.jsx`. School Management includes academic years, terms, departments, subjects, classes, streams, teachers, admissions, attendance, assessments, report cards, assignments, fees, scholarships, transport, library, discipline, announcements, and portal links.

5.34.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	in- Limitation
Status	PASSED	Evidence	UI: <code>client/src/components/SchoolWorkspace.jsx</code> ; server: <code>server/schoolOperations.ts</code>
Purpose	School Management	Primary users	School administrator; teacher; student/guardian portal user; finance user

Area	Verified reference	Operational in- interpretation	Limitation
Navigation	Authenticated module registry or dedicated workspace	Workflow	Set academic structure → admit learner → assign teacher/class → record attendance/assessment → publish report → invoice/receive fees → communicate.
Database	sch_academic_years, sch_admissions, sch_announcements, sch_approval_requests, sch_assessment_scores, sch_assessments, sch_assignment_submissions, sch_assignments, sch_attendance_records, sch_attendance_sessions, sch_audit_logs, sch_books, sch_classes, sch_departments, sch_disciplinary_records, sch_documents	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Automated browser coverage is strong for the surface, while full per-school CRUD remains a data-backed acceptance task.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.34.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.34.5 Known limitations and future improvements

Automated browser coverage is strong for the surface, while full per-school CRUD remains a data-backed acceptance task. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

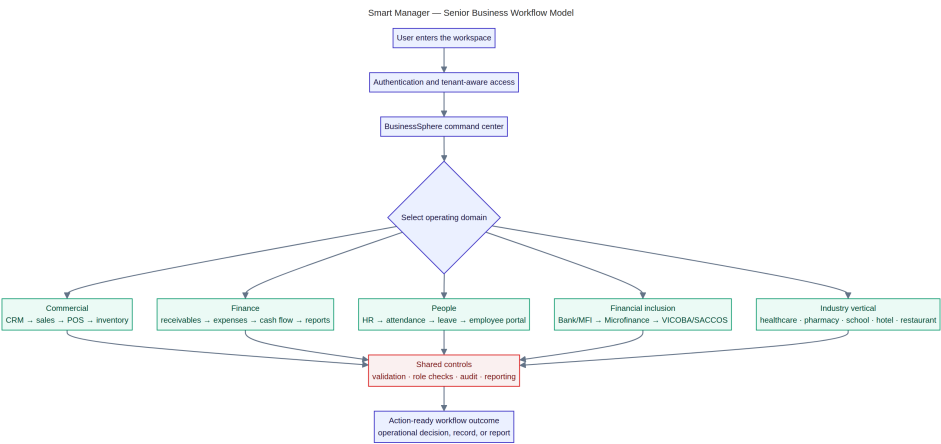


Figure 40: School Management evidence image from repository or captured application assets.

5.35 IV.35 Pharmacy Management

TESTING

5.35.1 Module overview

Pharmacy covers drug categories, brands, medicines, suppliers, purchase orders, stock receipt/adjustment/transfer, dispensing, sales, payments, claims, returns, reports, notifications, and audit. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Pharmacy Management is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `phm\audit\logs`, `phm\batches`, `phm\brands`, `phm\categories`, `phm\controlled\medicine\register`, `phm\dispense`, `phm\dispense\items`, `phm\drugs`, `phm\insurance\claims`, `phm\notifications`, `phm\payments`, `phm\purchase\order\items`, `phm\purchase\orders`, `phm\return\items`, `phm\returns`, `phm\sale\items`, `phm\sales`, `phm\stock`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Create medicine → receive stock → adjust/transfer → dispense or sell → collect payment → return/claim → reconcile. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Live dispensing and inventory walkthroughs remain dependent on controlled pharmacy records and appropriate clinical/financial permissions.

5.35.2 Purpose, users, navigation, and features

Primary roles: Pharmacist; storekeeper; cashier; clinician; administrator. UI evidence: `client/src/components/PharmacyWorkspace.jsx`. Pharmacy covers drug categories, brands, medicines, suppliers, purchase orders, stock receipt/adjustment/transfer, dispensing, sales, payments, claims, returns, reports, notifications, and audit.

5.35.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/PharmacyWorkspace.jsx</code> ; server: <code>server/pharmacyOperations.ts</code>
Purpose	Pharmacy Management	Primary users	Pharmacist; storekeeper; cashier; clinician; administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Create medicine → receive stock → adjust/transfer → dispense or sell → collect payment → return/claim → reconcile.

Area	Verified reference	Operational in- interpretation	Limitation
Database	phm_audit_logs, phm_batches, phm_brands, phm_categories, phm_controlled_medicine_register, phm_dispense, phm_dispense_items, phm_drugs, phm_insurance_claims, phm_notifications, phm_payments, phm_purchase_order_items, phm_purchase_orders, phm_return_items, phm_returns, phm_sale_items	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Live dispensing and inventory walk-throughs remain dependent on controlled pharmacy records and appropriate clinical/financial permissions.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.35.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.35.5 Known limitations and future improvements

Live dispensing and inventory walkthroughs remain dependent on controlled pharmacy records and appropriate clinical/financial permissions. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

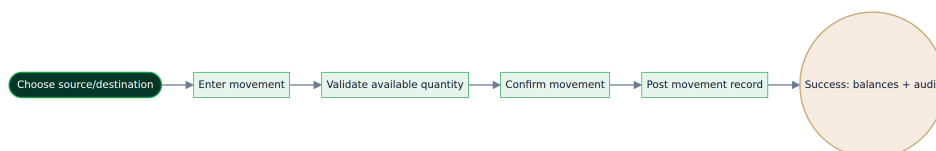


Figure 41: Pharmacy Management evidence image from repository or captured application assets.

5.36 IV.36 Hotel and Hospitality

TESTING

5.36.1 Module overview

Hospitality covers properties, room types, rooms, rate plans, reservations, guests, guest KYC, folios, payments, housekeeping, laundry, minibar, events, complaints, amenities, notifications, and reconciliation. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Hotel and Hospitality is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `hospitality\amenities`, `hospitality\audit\log`, `hospitality\complaints`, `hospitality\event\venues`, `hospitality\events`, `hospitality\finance\reconciliations`, `hospitality\folio\lines`, `hospitality\folios`, `hospitality\guest\kyc`, `hospitality\guest\requests`, `hospitality\guests`, `hospitality\housekeeping\tasks`, `hospitality\laundry\orders`, `hospitality\loyalty\accounts`, `hospitality\maintenance\requests`, `hospitality\menu\items`, `hospitality\menus`, `hospitality\minibar\postings`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Configure property/rooms → create reservation → check in → post folio/service → receive payment → housekeeping/check out → reconcile. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. External booking-channel and payment connections remain provider-dependent; local reservation state is not proof of channel synchronization.

5.36.2 Purpose, users, navigation, and features

Primary roles: Hotel manager; front desk; housekeeping; finance; guest. UI evidence: `client/src/components/HospitalityWorkspace.jsx`; `client/src/components/VerticalCommandCenters.jsx`. Hospitality covers properties, room types, rooms, rate plans, reservations, guests, guest KYC, folios, payments, housekeeping, laundry, minibar, events, complaints, amenities, notifications, and reconciliation.

5.36.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in- interpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/HospitalityWorkspace.jsx</code> ; <code>client/src/components/VerticalCommandCenters.jsx</code> ; server: <code>server/hospitalityOperations.ts</code>

Area	Verified reference	Operational in- interpretation	Limitation
Purpose	Hotel and Hospitality	Primary users	Hotel manager; front desk; house-keeping; finance; guest
Navigation	Authenticated module registry or dedicated workspace	Workflow	Configure property/rooms → create reservation → check in → post folio/service → receive payment → housekeeping/check out → reconcile.
Database	hospitality_amenities, hospitality_audit_log, hospitality_complaints, hospitality_event_venues, hospitality_events, hospitality_finance_reconciliations, hospitality_folio_lines, hospitality_folios, hospitality_guest_kyc, hospitality_guest_requests, hospitality_guests, hospitality_housekeeping_tasks, hospitality_laundry_orders, hospitality_loyalty_accounts, hospitality_maintenance_requests, hospitality_menu_items	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	External booking-channel and payment connections remain provider-dependent; local reservation state is not proof of channel synchronization.	Future improve- ment	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.36.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.36.5 Known limitations and future improvements

External booking-channel and payment connections remain provider-dependent; local reservation state is not proof of channel synchronization. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

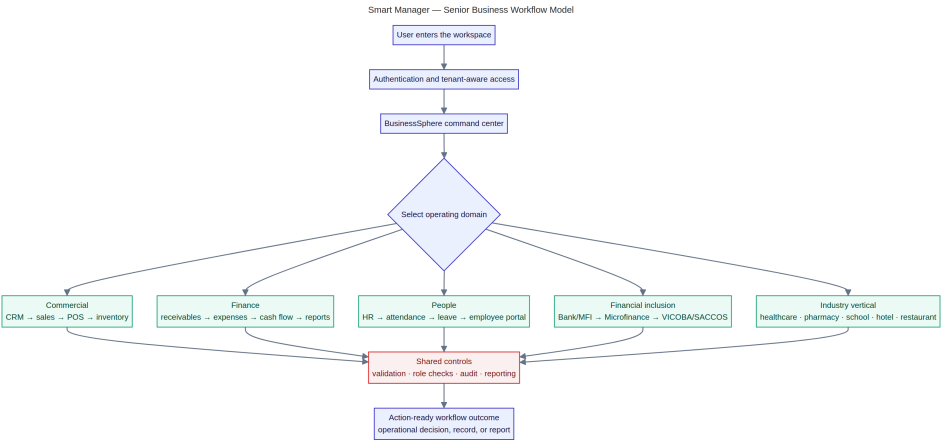


Figure 42: Hotel and Hospitality evidence image from repository or captured application assets.

5.37 IV.37 Restaurant and Food & Beverage

TESTING

5.37.1 Module overview

Restaurant operations cover tax/fiscal profiles, menus, menu items, tables, orders, reservations, waiters, payments, receipts, refunds, and Tanzania fiscal configuration. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Restaurant and Food & Beverage is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `restaurant\{alerts, restaurant\{audit\{events, restaurant\{bill\{splits, restaurant\{combo\{items, restaurant\{customers, restaurant\{dining\{areas, restaurant\{fiscal\{profiles, restaurant\{fiscal\{receipts, restaurant\{kitchen\{tickets, restaurant\{menu\{categories, restaurant\{menu\{items, restaurant\{mobile\{money\{intents, restaurant\{mobile\{money\{profiles, restaurant\{modifier\{groups, restaurant\{modifier\{options, restaurant\{order\{lines, restaurant\{orders, restaurant\{outlets`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Configure menu/table → open order → add items → prepare/serve → collect payment → issue receipt/fiscal evidence → reconcile. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Delivery and external payment channels require configuration; fiscal status is a readiness state until a provider confirms submission.

5.37.2 Purpose, users, navigation, and features

Primary roles: Restaurant manager; waiter; cashier; kitchen; finance. UI evidence: `client/src/components/RestaurantWorkspace.jsx`; `client/src/components/RestaurantTanzaniaFiscalPanel.jsx`. Restaurant operations cover tax/fiscal profiles, menus, menu items, tables, orders, reservations, waiters, payments, receipts, refunds, and Tanzania fiscal configuration.

5.37.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-terpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/RestaurantWorkspace.jsx</code> ; <code>client/src/components/RestaurantTanzaniaFiscalPanel.jsx</code> ; server: <code>server/restaurantManagement.ts</code> ; <code>server/traFiscal.ts</code>

Area	Verified reference	Operational interpretation	in- Limitation
Purpose	Restaurant and Food & Beverage	Primary users	Restaurant manager; waiter; cashier; kitchen; finance
Navigation	Authenticated module registry or dedicated workspace	Workflow	Configure menu/table → open order → add items → prepare/serve → collect payment → issue receipt/fiscal evidence → reconcile.
Database	restaurant_alerts, restaurant_bill_splits, restaurant_combo_items, restaurant_customers, restaurant_dining_areas, restaurant_fiscal_profiles, restaurant_fiscal_receipts, restaurant_kitchen_tickets, restaurant_menu_categories, restaurant_menu_items, restaurant_mobile_money_intents, restaurant_mobile_money_profiles, restaurant_modifier_groups, restaurant_modifier_options, restaurant_order_lines	Security audit events, restaurant_audit_events	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Delivery and external payment channels require configuration; fiscal status is a readiness state until a provider confirms submission.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.37.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.37.5 Known limitations and future improvements

Delivery and external payment channels require configuration; fiscal status is a readiness state until a provider confirms submission. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

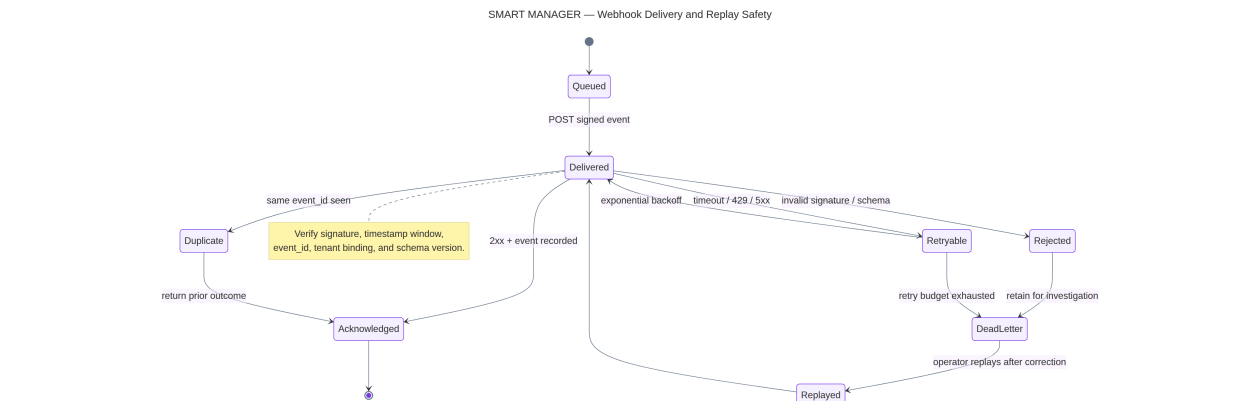


Figure 43: Restaurant and Food & Beverage evidence image from repository or captured application assets.

5.38 IV.38 Banking and MFI

TESTING

5.38.1 Module overview

Bank/MFI covers institutions, branches, customers, KYC documents, beneficial owners, accounts, beneficiaries, tellers, cash movements, wallets, loans, schedules, repayments, AML alerts, reconciliation, and audit events. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Banking and MFI is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `bank\u{005F}account\u{005F}beneficiaries`, `bank\u{005F}account\u{005F}types`, `bank\u{005F}accounts`, `bank\u{005F}agents`, `bank\u{005F}aml\u{005F}alerts`, `bank\u{005F}audit\u{005F}events`, `bank\u{005F}beneficial\u{005F}owners`, `bank\u{005F}branches`, `bank\u{005F}cash\u{005F}movements`, `bank\u{005F}collateral`, `bank\u{005F}customer\u{005F}documents`, `bank\u{005F}customers`, `bank\u{005F}fixed\u{005F}deposits`, `bank\u{005F}group\u{005F}members`, `bank\u{005F}groups`, `bank\u{005F}guarantors`, `bank\u{005F}idempotency\u{005F}keys`, `bank\u{005F}institutions`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Set institution → register customer/KYC → open account → post transaction → originate/approve loan → schedule/collect → reconcile/resolve AML. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. External banking rails are not connected without approved adapters; SECURITY DEFINER routines require endpoint-by-endpoint privilege review.

5.38.2 Purpose, users, navigation, and features

Primary roles: Bank administrator; teller; loan officer; compliance officer; auditor. UI evidence: `client/src/components/BankMfiWorkspace.jsx`; `client/src/components/SectorCommandCenters.jsx`. Bank/MFI covers institutions, branches, customers, KYC documents, beneficial owners, accounts, beneficiaries, tellers, cash movements, wallets, loans, schedules, repayments, AML alerts, reconciliation, and audit events.

5.38.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational in-terpretation	Limitation
Status	TESTING	Evidence	UI: <code>client/src/components/BankMfiWorkspace.jsx</code> ; <code>client/src/components/SectorCommandCenters.jsx</code> ; server: <code>server/bankMfiOperations.ts</code>

Area	Verified reference	Operational interpretation	in-	Limitation
Purpose	Banking and MFI	Primary users		Bank administrator; teller; loan officer; compliance officer; auditor
Navigation	Authenticated module registry or dedicated workspace	Workflow		Set institution → register customer/KYC → open account → post transaction → originate/approve loan → schedule/collect → reconcile/resolve AML.
Database	bank_account_beneficial_owners, bank_account_types, bank_accounts, bank_agents, bank_aml_alerts, bank_audit_events, bank_beneficial_owners, bank_branches, bank_cash_movements, bank_collateral, bank_customer_documents, bank_customers, bank_fixed_deposits, bank_group_members, bank_groups, bank_guarantors	Security		Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	External banking rails are not connected without approved adapters; SECURITY DEFINER routines require endpoint-by-endpoint privilege review.	Future improvement		Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.38.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.38.5 Known limitations and future improvements

External banking rails are not connected without approved adapters; SECURITY DEFINER routines require endpoint-by-endpoint privilege review. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

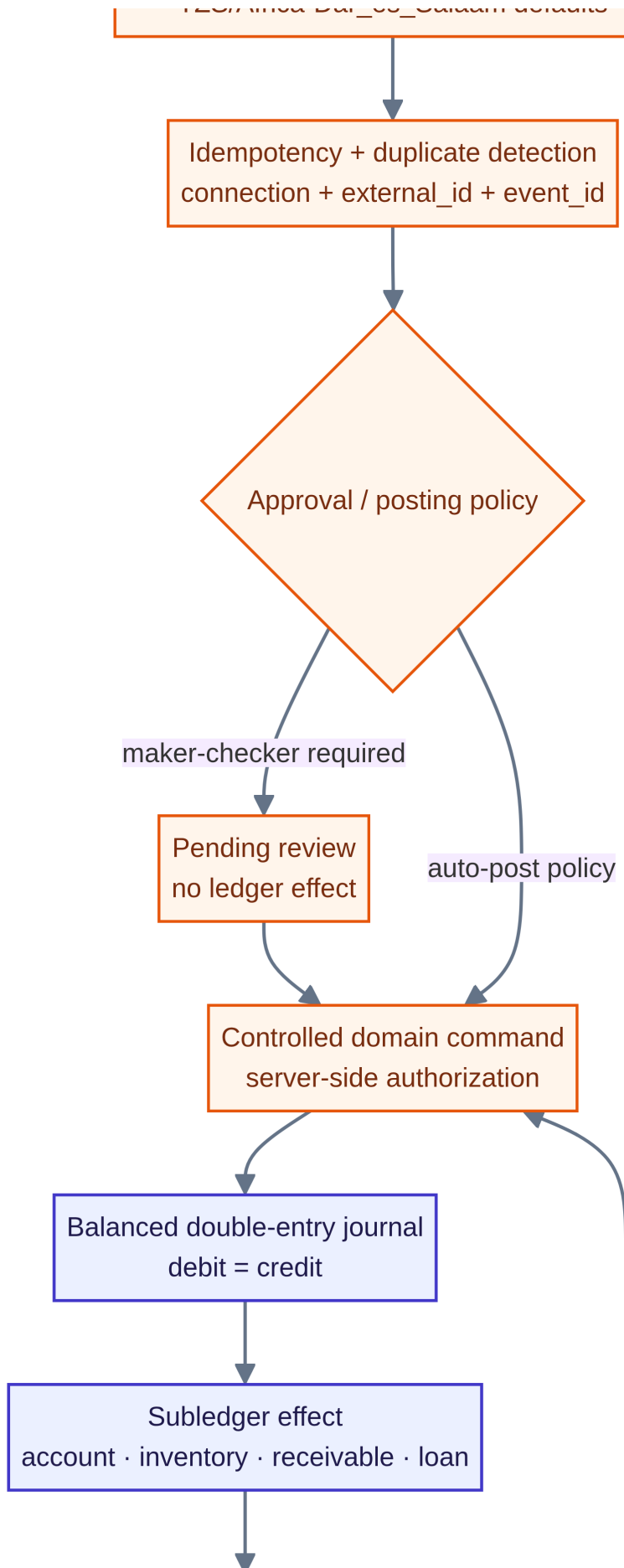


Figure 44: Banking and MFI evidence image from repository or captured application assets.

5.39 IV.39 Employee Portal

IMPLEMENTED

5.39.1 Module overview

The portal provides self-service views for profile/work context, attendance, leave, announcements, benefits, documents, and approved workforce actions while retaining company scope. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Employee Portal is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `hr\announcement\reads`, `hr\announcements`, `hr\approval\requests`, `hr\approval\steps`, `hr\attendance`, `hr\benefit\enrollments`, `hr\benefit\plans`, `hr\benefits`, `hr\candidates`, `hr\duties`, `hr\employee\documents`, `hr\employees`, `hr\expense\claims`, `hr\goal\updates`, `hr\goals`, `hr\holidays`, `hr\invite\codes`, `hr\leave\balances`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Accept secure invitation → verify profile → view or request employee action → manager/HR decision → record outcome. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. An employee view does not imply access to payroll or HR administration; role boundaries must be enforced by server and RLS.

5.39.2 Purpose, users, navigation, and features

Primary roles: Employee; manager; HR administrator. UI evidence: `client/src/components/EmployeePortalWorkspace.jsx`. The portal provides self-service views for profile/work context, attendance, leave, announcements, benefits, documents, and approved workforce actions while retaining company scope.

5.39.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/EmployeePortalWorkspace.jsx</code> ; server: <code>server/teamWorkforce.ts</code> ; <code>server/employeePortal.ts</code>
Purpose	Employee Portal	Primary users	Employee; manager; HR administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Accept secure invitation → verify profile → view or request

Area	Verified reference	Operational interpretation	Limitation
			employee action → manager/HR decision → record outcome.
Database	hr_announcement_read, hr_announcements, hr_approval_requests, hr_approval_steps, hr_attendance, hr_benefit_enrollments, hr_benefit_plans, hr_benefits, hr_candidates, hr_duties, hr_employee_documents, hr_employees, hr_expense_claims, hr_goal_updates, hr_goals, hr_holidays	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	An employee view does not imply access to payroll or HR administration; role boundaries must be enforced by server and RLS.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.39.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.39.5 Known limitations and future improvements

An employee view does not imply access to payroll or HR administration; role boundaries must be enforced by server and RLS. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

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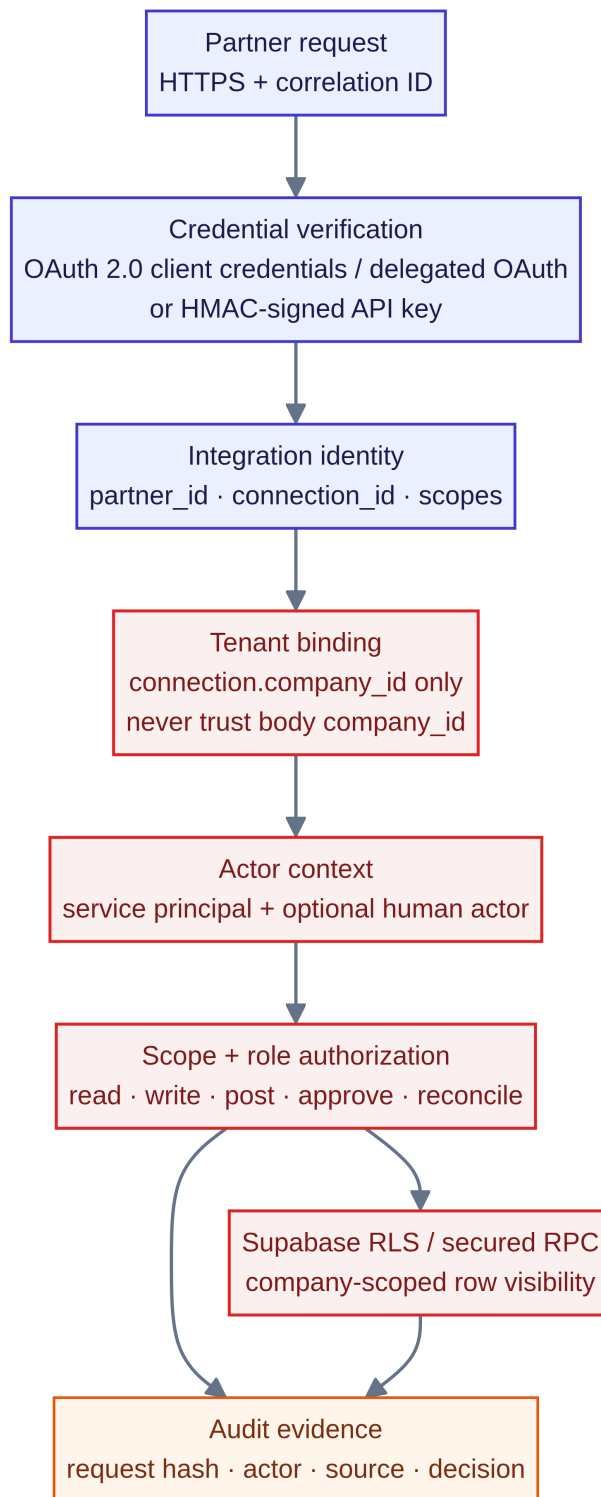


Figure 45: Employee Portal evidence image from repository or captured application assets.

5.40 IV.40 Property Management IMPLEMENTED

5.40.1 Module overview

Property Management covers portfolios, owners, buildings, plots, units, listings, agents, tenants, KYC documents, applications, leases, inspections, handover, rent schedules, tax/fee rules, service charges, utilities, invoices, payments, receipts, contractors, maintenance, work orders, expenses, budgets, insurance, notices, approvals, documents, ledgers, reconciliation, notifications, integrations, and audit. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Property Management is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `property\agents`, `property\applications`, `property\approvals`, `property\audit\log`, `property\budgets`, `property\buildings`, `property\contractors`, `property\documents`, `property\expenses`, `property\handover\records`, `property\inspection\items`, `property\inspections`, `property\insurances`, `property\integration\events`, `property\invoice\lines`, `property\invoices`, `property\leases`, `property\ledger\entries`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Portfolio → building/unit → owner/tenant/application → lease → inspection/handover → rent/utility invoice → payment/receipt → maintenance/expense → reconciliation. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. The live audit shows the property migration and tables present; external payment, map, messaging, and contractor channels still require configuration and controlled acceptance.

5.40.2 Purpose, users, navigation, and features

Primary roles: Property manager; landlord/owner; agent; tenant; finance manager; maintenance user. UI evidence: `client/src/components/PropertyManagementWorkspace.jsx`. Property Management covers portfolios, owners, buildings, plots, units, listings, agents, tenants, KYC documents, applications, leases, inspections, handover, rent schedules, tax/fee rules, service charges, utilities, invoices, payments, receipts, contractors, maintenance, work orders, expenses, budgets, insurance, notices, approvals, documents, ledgers, reconciliation, notifications, integrations, and audit.

5.40.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/PropertyManagementWorkspace.jsx</code> server: <code>server/propertyManagementOperations.ts</code> server: <code>propertyManagement.ts</code>

Area	Verified reference	Operational interpretation	in- Limitation
Purpose	Property Management	Primary users	Property manager; landlord/owner; agent; tenant; finance manager; maintenance user
Navigation	Authenticated module registry or dedicated workspace	Workflow	Portfolio → building/unit → owner/tenant/application → lease → inspection/handover → rent/utility invoice → payment/receipt → maintenance/expense → reconciliation.
Database	property_agents, property_approvals, property_audit_log, property_budgets, property_buildings, property_contractors, property_documents, property_expenses, property_handover_records, property_inspection_items, property_inspections, property_insurances, property_integration_events, property_invoice_lines, property_invoices	Security applications,	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	The live audit shows the property migration and tables present; external payment, map, messaging, and contractor channels still require configuration and controlled acceptance.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.40.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.40.5 Known limitations and future improvements

The live audit shows the property migration and tables present; external payment, map, messaging, and contractor channels still require configuration and controlled acceptance. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

5.41 IV.41 Subscription and Billing

IMPLEMENTED

5.41.1 Module overview

The live commercial contract contains FREE_15 at TZS 0 for 15 days and six paid monthly packages. Paid activation is released only after verified provider status; the server and database own amount, entitlement, idempotency, and access state. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Subscription and Billing is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `billing\plan\audit\log`, `billing\plans`, `billing\profiles`, `subscription\events`, `subscription\invoices`, `subscription\notifications`, `subscription\payments`, `subscription\usage`, `tenant\subscriptions`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Read catalog → start Free or create paid intent → send/verify provider request → reconcile payment → activate subscription/invoice → refresh access snapshot. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. HarakaPay account authorization and provider credentials remain deployment prerequisites; failed or cancelled requests must not be shown as active entitlement.

5.41.2 Purpose, users, navigation, and features

Primary roles: Billing manager; owner; CEO; CFO; finance manager; administrator. UI evidence: `client/src/components/SubscriptionBillingWorkspace.jsx`; `client/src/lib/subscriptionAccess.js`. The live commercial contract contains FREE_15 at TZS 0 for 15 days and six paid monthly packages. Paid activation is released only after verified provider status; the server and database own amount, entitlement, idempotency, and access state.

5.41.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/SubscriptionBillingWorkspace.jsx</code> ; <code>client/src/lib/subscriptionAccess.js</code> ; server: <code>server/subscriptionBilling.ts</code>
Purpose	Subscription and Billing	Primary users	Billing manager; owner; CEO; CFO; finance manager; administrator

Area	Verified reference	Operational interpretation	in- Limitation
Navigation	Authenticated module registry or dedicated workspace	Workflow	Read catalog → start Free or create paid intent → send/verify provider request → reconcile payment → activate subscription/invoice → refresh access snapshot.
Database	billing_plan_audit_log, billing_plans, billing_profiles, subscription_events, subscription_invoices, subscription_notifications, subscription_payments, subscription_usage, tenant_subscriptions	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	HarakaPay account authorization and provider credentials remain deployment prerequisites; failed or cancelled requests must not be shown as active entitlement.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.41.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.41.5 Known limitations and future improvements

HarakaPay account authorization and provider credentials remain deployment prerequisites; failed or cancelled requests must not be shown as active entitlement. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

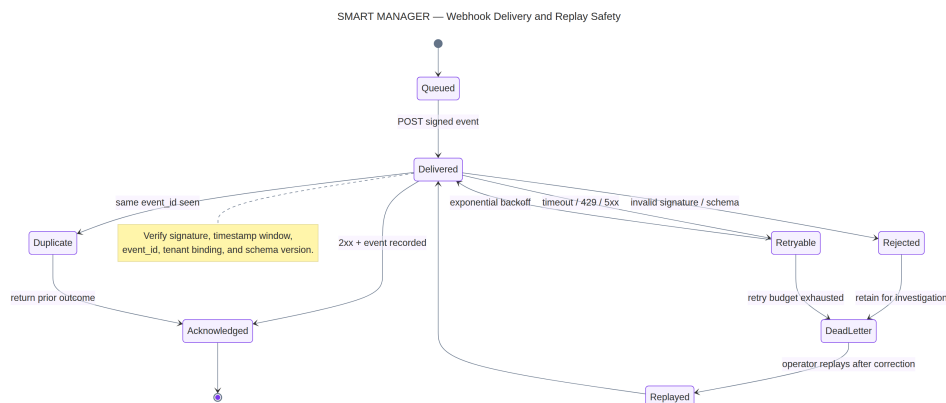


Figure 46: Subscription and Billing evidence image from repository or captured application assets.

5.42 IV.42 Global Admin Control Center

IMPLEMENTED

5.42.1 Module overview

Global administration is a separate control plane for platform-wide company, user, subscription, security, integration, health, and audit actions. It is not a broad tenant policy. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Global Admin Control Center is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes `billing\plan\audit\log`, `companies`, `platform\admin\actions`, `profiles`. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Verify platform-admin authority → inspect scoped snapshot → perform explicit action → record platform audit → review outcome. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. The `platform_admin_actions` table is intentionally protected from a broad authenticated policy; access uses the protected platform-admin RPC boundary.

5.42.2 Purpose, users, navigation, and features

Primary roles: Super Administrator; platform administrator. UI evidence: `client/src/components/GlobalAdminControlCenter.tsx`. Global administration is a separate control plane for platform-wide company, user, subscription, security, integration, health, and audit actions. It is not a broad tenant policy.

5.42.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: <code>client/src/components/GlobalAdminControlCenter.tsx</code> ; server: <code>server/globalAdmin.ts</code> ; <code>server/routers.ts</code>
Purpose	Global Admin Control Center	Primary users	Super Administrator; platform administrator
Navigation	Authenticated module registry or dedicated workspace	Workflow	Verify platform-admin authority → inspect scoped snapshot → perform explicit action → record platform audit → review outcome.
Database	<code>billing_plan_audit_log</code> , <code>companies</code> , <code>platform_admin_actions</code> , <code>profiles</code>	Security	Verified profile, company scope, role checks, RLS/policies where applicable

Area	Verified reference	Operational in- interpretation	Limitation
Known limitation	The platform_admin_actions table is intentionally protected from a broad authenticated policy; access uses the protected platform-admin RPC boundary.	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

5.42.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.42.5 Known limitations and future improvements

The platform_admin_actions table is intentionally protected from a broad authenticated policy; access uses the protected platform-admin RPC boundary. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

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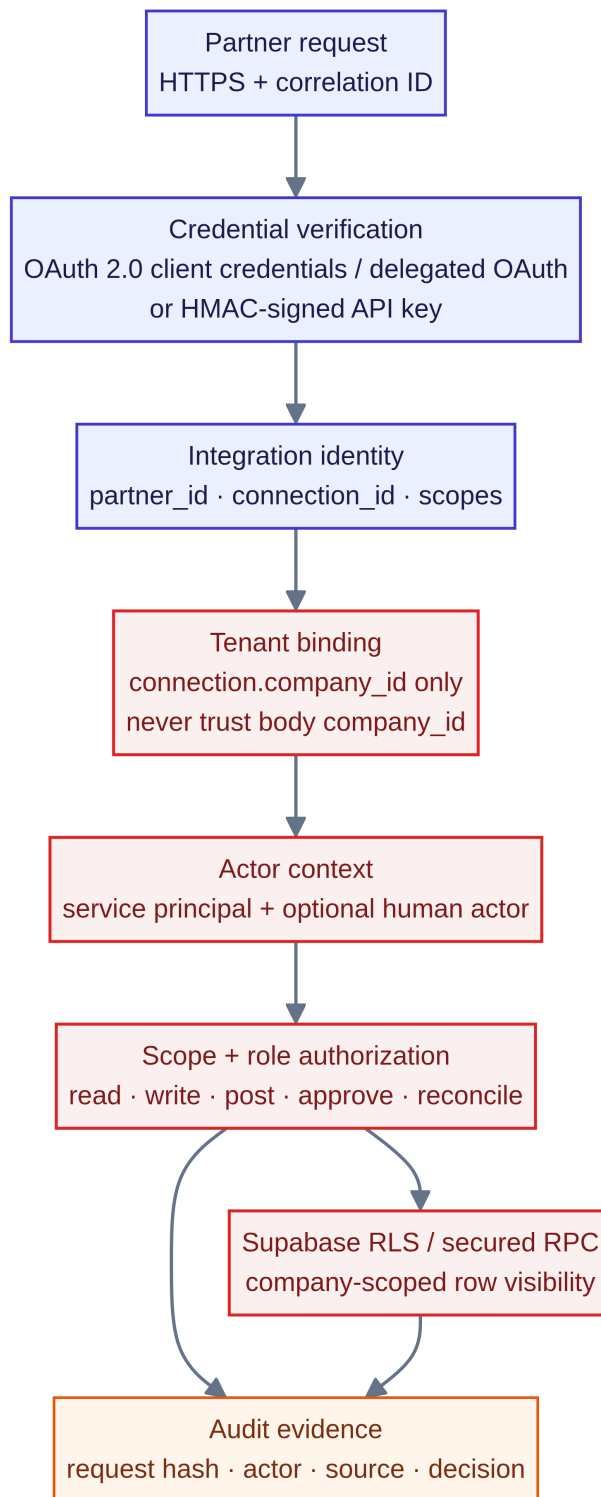


Figure 47: Global Admin Control Center evidence image from repository or captured application assets.

5.43 IV.43 Enterprise Settings and Security Control Center

IMPLEMENTED

5.43.1 Module overview

Settings cover company profile, branding, language, timezone/currency, receipts, idle timeout, modules, branches, departments, preferences, passkeys, and approval-aware role changes. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Enterprise Settings and Security Control Center is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes branches, company\{modules, company\{profile\{settings, departments, user\{table\{preferences. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Open settings → read confirmed configuration → edit permitted field → server validates and persists → refresh UI → audit sensitive change. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Provider-specific email, SMS, API-key, and storage configuration remains environment-dependent; secret values must be redacted.

5.43.2 Purpose, users, navigation, and features

Primary roles: Organization owner; administrator; manager; user with permitted settings. UI evidence: client/src/components/DashboardPreferencesDrawer.jsx; client/src/components/ProfileIdentityCenter.jsx. Settings cover company profile, branding, language, timezone/currency, receipts, idle timeout, modules, branches, departments, preferences, passkeys, and approval-aware role changes.

5.43.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	IMPLEMENTED	Evidence	UI: client/src/components/DashboardPreferencesDrawer.jsx; client/src/components/ProfileIdentityCenter.jsx; server: server/workspaceSettings.ts; server/workspaceBranding.ts; server/roleChangeApprovals.ts
Purpose	Enterprise Settings and Security Control Center	Primary users	Organization owner; administrator; manager; user with permitted settings
Navigation	Authenticated module registry or dedicated workspace	Workflow	Open settings → read confirmed configuration → edit permitted field → server validates and persists → refresh UI → audit sensitive change.

Area	Verified reference	Operational in- interpretation	Limitation
Database	branches, company_modules, company_profile_settings, departments, user_table_preferences	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Provider-specific email, SMS, API-key, and storage configura- tion remains environ- ment-dependent; se- cret values must be redacted.	Future improve- ment	Complete controlled authenti- cated CRUD and provider accep- tance where the status is not PASSED.

5.43.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.43.5 Known limitations and future improvements

Provider-specific email, SMS, API-key, and storage configuration remains environment-dependent; secret values must be redacted. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.

SMART MANAGER — Integration Authentication and Tenant Isolation

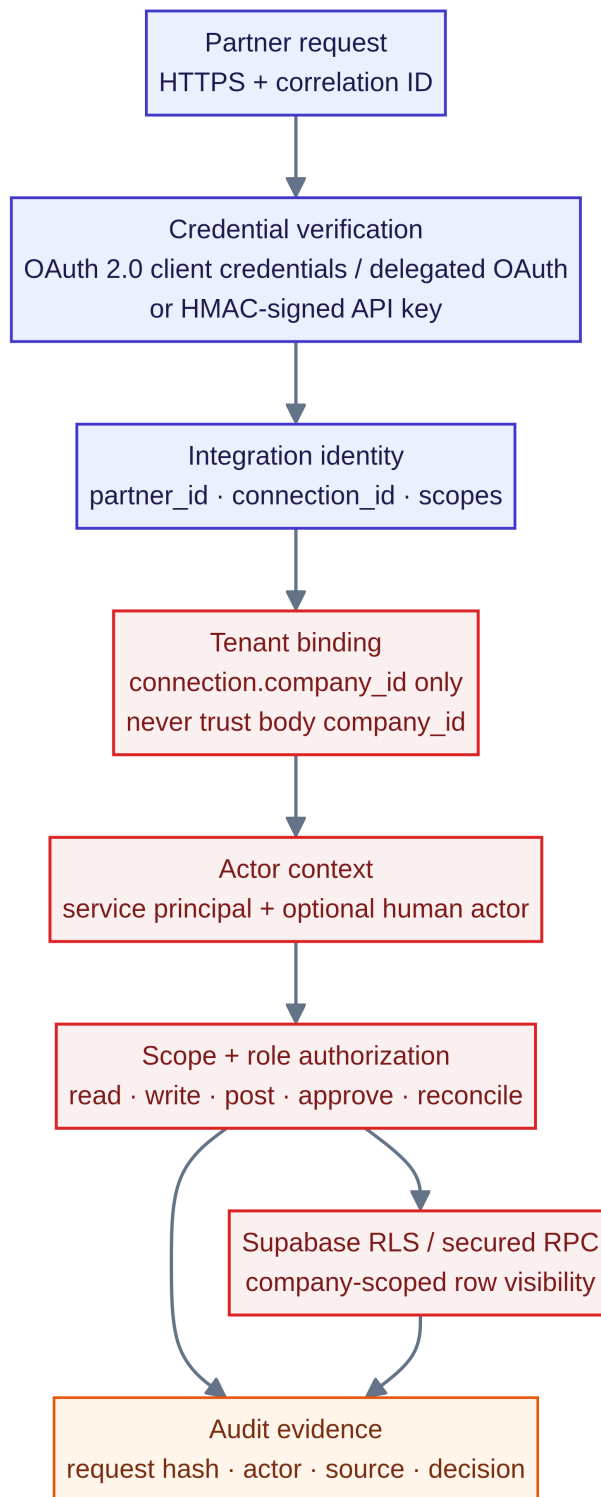


Figure 48: Enterprise Settings and Security Control Center evidence image from repository or captured application assets.

5.44 IV.44 Predictive Analytics

PARTIALLY IMPLEMENTED

5.44.1 Module overview

Predictive surfaces present trends, forecasts, benchmarks, and next-best-action ideas when sufficient data and configured intelligence services exist. This chapter is written from the repository UI, server boundary, migration history, and the read-only live Supabase inventory. The strongest evidence is named; anything that depends on configuration, connected data, or an external provider remains qualified. The purpose of Predictive Analytics is to make a repeatable business operation visible and controlled. The primary business problem is not the absence of a screen; it is the risk that the same fact is entered in several places, that a role sees too much, or that a failed write is presented as success. The module therefore belongs to a path from input to validation, persistence, evidence, and management review. The representative persistence surface observed for this module includes competitors, custom_kpis, financial_benchmarks. These names are evidence from the current live inventory or the repository contract; they are not an assertion that every table is used by every deployment or that every listed record contains production data. The verified workflow is: Select evidence set → compute signal → present confidence/limitation → inspect source metrics → decide manually. Operators should complete the step in sequence, confirm the returned state, and use the module's error or empty state rather than substituting a local guess. Predictions are decision support, not guarantees. Sparse or stale data, configuration, and model availability constrain their reliability.

5.44.2 Purpose, users, navigation, and features

Primary roles: Executive; analyst; manager. UI evidence: client/src/components/PredictiveAnalyticsWorkspace.jsx. Predictive surfaces present trends, forecasts, benchmarks, and next-best-action ideas when sufficient data and configured intelligence services exist.

5.44.3 Database, permissions, workflow, and reports

Area	Verified reference	Operational interpretation	Limitation
Status	PARTIALLY IMPLEMENTED	Evidence	UI: client/src/components/PredictiveAnalyticsWorkspace.jsx server: server/marketIntelligence.ts; server/smartAssistant.ts
Purpose	Predictive Analytics	Primary users	Executive; analyst; manager
Navigation	Authenticated module registry or dedicated workspace	Workflow	Select evidence set → compute signal → present confidence/limitation → inspect source metrics → decide manually.
Database	competitors, custom_kpis, financial_benchmarks	Security	Verified profile, company scope, role checks, RLS/policies where applicable
Known limitation	Predictions are decision support, not guarantees. Sparse or stale data, configura-	Future improvement	Complete controlled authenticated CRUD and provider acceptance where the status is not PASSED.

Area	Verified reference	Operational interpretation	in-	Limitation
	tion, and model availability constrain their reliability.			

5.44.4 Security and mobile experience

Use verified profile resolution, company scope, role checks, database constraints, RLS/policies, and confirmed server responses. Responsive UI changes layout, not authorization. A failed or unavailable write remains failed or unavailable.

5.44.5 Known limitations and future improvements

Predictions are decision support, not guarantees. Sparse or stale data, configuration, and model availability constrain their reliability. The next improvement is evidence-driven: complete controlled authenticated CRUD, provider readiness, role-specific regression, or safe code decomposition.



Figure 49: Predictive Analytics evidence image from repository or captured application assets.

6 Part V — Business Workflows

6.1 V.1 Authentication and onboarding

Public entry → Supabase configuration → password/OAuth/passkey → session persistence → verified profile → company context → protected workspace.

6.1.1 Kiswahili

Ukurasa wa umma → configuration ya Supabase → nenosiri/OAuth/passkey → kikao → wasifu uliothibitishwa → kampuni → workspace salama.

6.1.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.2 V.2 Sales to receipt

Customer → quotation/order → invoice → confirmed payment → receipt/ledger → management report and audit.

6.2.1 Kiswahili

Mteja → quotation/order → ankara → malipo yaliyothibitishwa → receipt/ledger → ripoti na audit.

6.2.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.3 V.3 Procurement to stock

Supplier → request/order → approval → receipt → stock movement → payable/finance evidence.

6.3.1 Kiswahili

Msambazaji → ombi/order → approval → mapokezi → stock movement → ushahidi wa deni/finance.

6.3.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.4 V.4 POS to reconciliation

Register/shift → item/customer → price/stock validation → tender → confirmed sale → receipt → return/reconciliation.

6.4.1 Kiswahili

Register/shift → item/mteja → uthibitishaji wa bei na stock → tender → sale → receipt → return/reconciliation.

6.4.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.5 V.5 Finance close

Transaction → journal or subledger → approval → bank/tax reconciliation → period close → report.

6.5.1 Kiswahili

Muamala → journal/subledger → approval → reconciliation ya benki/kodi → kufunga kipindi → ripoti.

6.5.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.6 V.6 HR and payroll

Secure invite → employee profile → attendance/leave → approval → payroll calculation → report.

6.6.1 Kiswahili

Invitation salama → wasifu wa mfanyakazi → attendance/leave → approval → payroll → ripoti.

6.6.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.7 V.7 Property rental

Portfolio → unit → owner/tenant/application → lease → inspection → invoice → payment → maintenance/expense → reconciliation.

6.7.1 Kiswahili

Portfolio → unit → owner/tenant/application → lease → inspection → invoice → malipo → maintenance/expense → reconciliation.

6.7.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.8 V.8 Healthcare care path

Patient → appointment → visit/vitals → prescription/lab → report/claim → consent-based reminder → audit.

6.8.1 Kiswahili

Mgonjwa → appointment → visit/vitals → prescription/lab → report/claim → reminder yenye consent → audit.

6.8.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.9 V.9 Subscription lifecycle

Catalog → Free activation or paid intent → provider status → verified settlement → subscription/invoice → access snapshot → expiry/renewal decision.

6.9.1 Kiswahili

Catalog → Free au paid intent → status ya provider → settlement iliyothibitishwa → subscription/invoice → access snapshot → expiry/renewal.

6.9.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

6.10 V.10 AI proposal path

Verified user → bounded context → structured response → navigation/proposal → independent approval → confirmed module mutation.

6.10.1 Kiswahili

Mtumiaji aliyehakikiwa → context yenye mipaka → response iliyopangwa → proposal → approval huru → mutation iliyothibitishwa.

6.10.2 Control points

At each transition, confirm identity, company scope, role, input validation, database response, and audit/result state. A failed step must remain failed; a local optimistic row is not a durable business record.

7 Part VI — User Manual

7.1 VI.1 First-day onboarding

Sign in through the supported gateway, verify the company context, review your role, open one permitted module, and create only a record that the server confirms.

7.1.1 Kiswahili

Siku ya kwanza ingia kupitia gateway, hakikisha company context, soma role yako, fungua module inayoruhusiwa na tengeneza record ambayo server imethibitisha.

7.1.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

7.2 VI.2 Daily operating rhythm

Start with the dashboard, inspect exceptions, execute the module workflow, verify returned state, and leave approvals or reconciliation items visible to the responsible role.

7.2.1 Kiswahili

Anza na dashibodi, kagua exceptions, tekeleza workflow, thibitisha hali iliyorejeshwa na acha approvals/reconciliation zionekane kwa role husika.

7.2.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

7.3 VI.3 Safe data entry

Use required fields, valid dates and amounts, company-scoped references, and clear descriptions. Do not paste secrets into business notes.

7.3.1 Kiswahili

Tumia fields zinazohitajika, tarehe na kiasi sahihi, references za kampuni na maelezo wazi. Usiweke siri ndani ya notes za biashara.

7.3.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

7.4 VI.4 Errors and retries

Read the error state, preserve the form, correct the input or permission, and retry only after the cause is understood. Do not click repeatedly on payment or approval actions without an idempotency path.

7.4.1 Kiswahili

Soma error, hifadhi form, rekebisha input au permission na retry baada ya kuelewa sababu. Usibonyeze payment au approval mara nyingi bila idempotency.

7.4.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

7.5 VI.5 Reports and review

Use reports to ask a question of recorded data. Follow a surprising number back to the source document, payment, movement, or journal entry.

7.5.1 Kiswahili

Tumia ripoti kuuliza swali la data iliyorekodiwa. Namba ya kushangaza ifuatilie hadi document, payment, movement au journal entry ya chanzo.

7.5.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

7.6 VI.6 Subscription access

Read the server access state. FREE_15 is free for 15 days; paid access requires verified payment. Do not use local browser state as entitlement authority.

7.6.1 Kiswahili

Soma access state ya server. FREE_15 ni bure kwa siku 15; paid access inahitaji payment iliyothibitishwa. Browser si mamlaka ya entitlement.

7.6.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

7.7 VI.7 Working on mobile

Use responsive views for review and entry, maintain stable connectivity for writes, and wait for confirmed response before navigating away from a critical action.

7.7.1 Kiswahili

Tumia responsive views kwa review na entry, hakikisha connection kwa writes na subiri response iliyothibitishwa kabla ya kuondoka kwenye action muhimu.

7.7.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

7.8 VI.8 Support handoff

Record company, module, time, user role, exact message, and safe reproduction steps without including passwords or tokens.

7.8.1 Kiswahili

Andika kampuni, module, muda, role, ujumbe kamili na hatua salama za kurudia bila kuingiza password au token.

7.8.2 Practical control

Treat the server-confirmed response as the completion signal. If it is unavailable, keep the operation visible as pending, failed, or unavailable rather than completed.

8 Part VII — Administrator Manual

8.1 VII.1 Provision a company

Confirm the authenticated owner, create or join the company through the supported path, verify membership and role, and review module entitlements before inviting staff.

8.1.1 Kiswahili

Thibitisha owner, unda au jiunge na kampuni, hakikisha membership na role, kagua entitlements kabla ya kualika wafanyakazi.

8.1.2 Administrator evidence

An administrator should be able to show the source migration, the live ledger, the policy boundary, the successful verification result, and the remaining limitation. A green UI alone is not administration evidence.

8.2 VII.2 Manage roles

Use role-change approval where required. Review the target user, old role, new role, approver, and audit record.

8.2.1 Kiswahili

Tumia role-change approval inapohitajika. Kagua mtumiaji, role ya zamani/mpya, approver na audit record.

8.2.2 Administrator evidence

An administrator should be able to show the source migration, the live ledger, the policy boundary, the successful verification result, and the remaining limitation. A green UI alone is not administration evidence.

8.3 VII.3 Configure providers

Keep provider keys server-side, test readiness, expose disabled states, and document which environment is sandbox or production.

8.3.1 Kiswahili

Hifadhi keys server-side, pima readiness, onyesha disabled state na andika mazingira ya sandbox au production.

8.3.2 Administrator evidence

An administrator should be able to show the source migration, the live ledger, the policy boundary, the successful verification result, and the remaining limitation. A green UI alone is not administration evidence.

8.4 VII.4 Review RLS

For each table, identify the company boundary, intended commands, membership helper, and policy. Avoid broad authenticated policies that bypass tenant scope.

8.4.1 Kiswahili

Kwa kila table tambua company boundary, commands, helper ya membership na policy. Epuka policy pana ya authenticated inayovunja tenant scope.

8.4.2 Administrator evidence

An administrator should be able to show the source migration, the live ledger, the policy boundary, the successful verification result, and the remaining limitation. A green UI alone is not administration evidence.

8.5 VII.5 Run migrations

Use source-versioned migrations, review prerequisites and compatibility guards, back up the target environment, apply through the controlled Supabase migration path, then rerun schema and RLS checks.

8.5.1 Kiswahili

Tumia migrations zenye version, kagua prerequisites na guards, hifadhi backup, apply kupitia njia iliyodhibitiwa ya Supabase, kisha rudia schema/RLS checks.

8.5.2 Administrator evidence

An administrator should be able to show the source migration, the live ledger, the policy boundary, the successful verification result, and the remaining limitation. A green UI alone is not administration evidence.

8.6 VII.6 Respond to incidents

Preserve audit evidence, fail closed, rotate exposed credentials, isolate the affected company or provider, and record the recovery decision.

8.6.1 Kiswahili

Hifadhi audit, kata access pale inavyohitajika, badilisha credentials zilizoonekana, tenga kampuni/provider iliyoathirika na andika uamuzi wa recovery.

8.6.2 Administrator evidence

An administrator should be able to show the source migration, the live ledger, the policy boundary, the successful verification result, and the remaining limitation. A green UI alone is not administration evidence.

9 Part VIII — Developer Manual

9.1 VIII.1 Source navigation

Start with `client/src/App.tsx`, `BusinessSphereDashboard.jsx`, dashboard registries, `server/_core/apiApp.ts`, `server/routers.ts`, feature operations, and `supabase/migrations`. Follow an action from UI to handler/RPC to table and back to confirmed state.

9.1.1 Kiswahili

Anza na `App.tsx`, `BusinessSphereDashboard.jsx`, registries, `apiApp.ts`, `routers.ts`, operations na migrations. Fuatilia action kutoka UI hadi handler/RPC, table na response iliyohibitishwa.

9.1.2 Developer control

Source, database, API, UI, RLS, tests, build, and deployment evidence must agree before a feature is described as complete.

9.2 VIII.2 Persistence contract

Critical writes should validate input, resolve verified profile, enforce company scope, call the server/database boundary, require returned row/result, and retain retryable input on failure.

9.2.1 Kiswahili

Writes muhimu zithibitishe input, profile, company scope, server/database, response ya row/result na zibaki na form kwa retry ikishindikana.

9.2.2 Developer control

Source, database, API, UI, RLS, tests, build, and deployment evidence must agree before a feature is described as complete.

9.3 VIII.3 Migration discipline

Use timestamped source migrations. Prefer additive and guarded changes. Never silently coerce incompatible production columns or create a duplicate architecture around `auth.users/profiles`.

9.3.1 Kiswahili

Tumia migrations zenye timestamp. Pendelea additive guards. Usibadilishe column za production kimya kimya wala kuunda `users/profiles` duplicate.

9.3.2 Developer control

Source, database, API, UI, RLS, tests, build, and deployment evidence must agree before a feature is described as complete.

9.4 VIII.4 Testing

Use contract tests for source promises, Vitest for server logic, Playwright for browser journeys, and controlled Supabase staging for authenticated CRUD and provider acceptance.

9.4.1 Kiswahili

Tumia contract tests, Vitest, Playwright na staging ya Supabase kwa CRUD ya authenticated na acceptance ya providers.

9.4.2 Developer control

Source, database, API, UI, RLS, tests, build, and deployment evidence must agree before a feature is described as complete.

9.5 VIII.5 Observability

Log safe error categories, request correlation where available, audit outcomes, provider order IDs, and schema-verification status. Never log credentials or raw private payloads.

9.5.1 Kiswahili

Log categories salama za errors, correlation, audit outcomes, provider order IDs na schema status. Usi-log credentials au payload za faragha.

9.5.2 Developer control

Source, database, API, UI, RLS, tests, build, and deployment evidence must agree before a feature is described as complete.

9.6 VIII.6 Extension strategy

Add new modules through a route/registry decision, server boundary, schema contract, RLS policy, migration, tests, and documentation status. Keep the module's limitation visible until all evidence exists.

9.6.1 Kiswahili

Ongeza module kupitia route/registry, server boundary, schema contract, RLS, migration, tests na status ya docs. Usifiche limitation kabla ya ushahidi kamili.

9.6.2 Developer control

Source, database, API, UI, RLS, tests, build, and deployment evidence must agree before a feature is described as complete.

10 Part IX — Security Book

The security model is layered and the risk register below records actual findings from the repository and live advisors.

10.1 IX.1 Authenticated SECURITY DEFINER routines

WARN Evidence: The live security advisor reported 118 WARN and 1 INFO lint at the audit timestamp; many WARNs identify signed-in execution of SECURITY DEFINER routines. Recommended fix: Review each signature, keep only intentionally callable endpoints, pin search paths, apply narrow grants, and move internal helpers out of the exposed API surface where possible.

10.1.1 Kiswahili

Ushahidi: The live security advisor reported 118 WARN and 1 INFO lint at the audit timestamp; many WARNs identify signed-in execution of SECURITY DEFINER routines. Hatua inayopendekezwa: Review each signature, keep only intentionally callable endpoints, pin search paths, apply narrow grants, and move internal helpers out of the exposed API surface where possible. Mfumo unapaswa kutumia utambulisho ulio wazi, role yenye mipaka, company scope, confirmation ya server, enforcement ya database na audit.

Control	Expected behavior	Review question	Evidence
Identity	Verified profile and supported token	Who is the caller?	auth/profile boundary
Tenant	Company-scoped record access	Does the row belong to the company?	company_id/RLS
Role	Specific permitted action	Is this role allowed?	server helper/policy
Mutation	Confirmed server/database result	Did the write really complete?	returned row/status
Audit	Safe activity evidence	Can the outcome be traced?	audit table/log

10.2 IX.2 Multiple permissive RLS policies

WARN Evidence: The live performance advisor reported 851 lints, including multiple permissive policies on the same table/action. Recommended fix: Consolidate overlapping policies by command and role after verifying semantics; do not blindly drop production policies.

10.2.1 Kiswahili

Ushahidi: The live performance advisor reported 851 lints, including multiple permissive policies on the same table/action. **Hatua inayopendekezwa:** Consolidate overlapping policies by command and role after verifying semantics; do not blindly drop production policies. **Mfumo unapaswa kutumia utambulisho ulio wazi,** role yenye mipaka, company scope, confirmation ya server, enforcement ya database na audit.

Control	Expected behavior	Review question	Evidence
Identity	Verified profile and supported token	Who is the caller?	auth/profile boundary
Tenant	Company-scoped record access	Does the row belong to the company?	company_id/RLS
Role	Specific permitted action	Is this role allowed?	server helper/policy
Mutation	Confirmed server/database result	Did the write really complete?	returned row/status
Audit	Safe activity evidence	Can the outcome be traced?	audit table/log

10.3 IX.3 Legacy non-atomic invoice payment path

P0 historical finding Evidence: The audit report records separate sales_payments insertion and invoice balance update without a proven atomic idempotency RPC. Recommended fix: Add a reviewed tenant-scoped atomic RPC and durable idempotency key before claiming concurrent-safe posting.

10.3.1 Kiswahili

Ushahidi: The audit report records separate sales_payments insertion and invoice balance update without a proven atomic idempotency RPC. Hatua inayopendekezwa: Add a reviewed tenant-scoped atomic RPC and durable idempotency key before claiming concurrent-safe posting. Mfumo unapaswa kutumia utambulisho ulio wazi, role yenye mipaka, company scope, confirmation ya server, enforcement ya database na audit.

Control	Expected behavior	Review question	Evidence
Identity	Verified profile and supported token	Who is the caller?	auth/profile boundary
Tenant	Company-scoped record access	Does the row belong to the company?	company_id/RLS
Role	Specific permitted action	Is this role allowed?	server helper/policy
Mutation	Confirmed server/database result	Did the write really complete?	returned row/status
Audit	Safe activity evidence	Can the outcome be traced?	audit table/log

10.4 IX.4 Large dashboard boundary

P2 Evidence: BusinessSphereDashboard.jsx remains a very large monolithic component and the build reports a non-fatal large-chunk warning. Recommended fix: Decompose incrementally after persistence and live-environment blockers are addressed; avoid cosmetic rewrites that increase risk.

10.4.1 Kiswahili

Ushahidi: BusinessSphereDashboard.jsx remains a very large monolithic component and the build reports a non-fatal large-chunk warning. Hatua inayopendekezwa: Decompose incrementally after persistence and live-environment blockers are addressed; avoid cosmetic rewrites that increase risk. Mfumo unapaswa kutumia utambulisho ulio wazi, role yenye mipaka, company scope, confirmation ya server, enforcement ya database na audit.

Control	Expected behavior	Review question	Evidence
Identity	Verified profile and supported token	Who is the caller?	auth/profile boundary
Tenant	Company-scoped record access	Does the row belong to the company?	company_id/RLS
Role	Specific permitted action	Is this role allowed?	server helper/policy
Mutation	Confirmed server/database result	Did the write really complete?	returned row/status
Audit	Safe activity evidence	Can the outcome be traced?	audit table/log

10.5 IX.5 External provider readiness

Configuration boundary Evidence: HarakaPay, TRA/VFD, WhatsApp, email/SMS, storage, and AI services depend on deployment configuration and approved credentials. Recommended fix: Keep provider secrets server-side, expose readiness states, and test only controlled sandbox or authorized production paths.

10.5.1 Kiswahili

Ushahidi: HarakaPay, TRA/VFD, WhatsApp, email/SMS, storage, and AI services depend on deployment configuration and approved credentials. Hatua inayopendekezwa: Keep provider secrets server-side, expose readiness states, and test only controlled sandbox or authorized production paths. Mfumo unapaswa kutumia utambulisho ulio wazi, role yenye mipaka, company scope, confirmation ya server, enforcement ya database na audit.

Control	Expected behavior	Review question	Evidence
Identity	Verified profile and supported token	Who is the caller?	auth/profile boundary
Tenant	Company-scoped record access	Does the row belong to the company?	company_id/RLS
Role	Specific permitted action	Is this role allowed?	server helper/policy
Mutation	Confirmed server/database result	Did the write really complete?	returned row/status
Audit	Safe activity evidence	Can the outcome be traced?	audit table/log

10.6 IX.6 Demo fallback risk

Medium Evidence: The client has an explicit seed-data fallback when Supabase is not configured. Recommended fix: Production deployments must fail closed with a clear configuration message; demo mode must remain explicit and non-operational.

10.6.1 Kiswahili

Ushahidi: The client has an explicit seed-data fallback when Supabase is not configured. Hatua inayopendekezwa: Production deployments must fail closed with a clear configuration message; demo mode must remain explicit and non-operational. Mfumo unapaswa kutumia utambulisho ulio wazi, role yenye mipaka, company scope, confirmation ya server, enforcement ya database na audit.

Control	Expected behavior	Review question	Evidence
Identity	Verified profile and supported token	Who is the caller?	auth/profile boundary
Tenant	Company-scoped record access	Does the row belong to the company?	company_id/RLS
Role	Specific permitted action	Is this role allowed?	server helper/policy
Mutation	Confirmed server/database result	Did the write really complete?	returned row/status
Audit	Safe activity evidence	Can the outcome be traced?	audit table/log

11 Part X — Database Dictionary

The live read-only Supabase inventory returned 542 tables (519 public and 23 auth), with 535 reporting RLS enabled and 7 not enabled in the returned inventory. The following table index is evidence metadata only; it does not expose row contents.

11.1 X.1 Public table segment 1-24

Table	Observed columns	Primary key	RLS
approval_signatures	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
audit_log	id, company_id, action, module, actor, details, subject, detail, created_at, updated_at	id	Enabled
bank_account_beneficiaries	id, company_id, customer_id, account_id, beneficiary_name, beneficiary_account_number, bank_name, phone, status, verified_at, created_by, created_at, updated_at	id	Enabled
bank_account_types	id, company_id, code, name, product_kind, currency, minimum_opening_balance, minimum_operating_balance, annual_interest_rate, withdrawal_fee, status, data, created_at, updated_at	id	Enabled
bank_accounts	id, company_id, name, status, amount, notes, data, created_at, updated_at, account_number, customer_id, account_type_id, branch_id, currency, ledger_balance, available_balance, hold_amount, opened_at, closed_at, version, created_by	id	Enabled
bank_agents	id, company_id, agent_code, name, phone, national_id, branch_id, status, float_balance, commission_rate, data, created_at, updated_at	id	Enabled
bank_aml_alerts	id, company_id, alert_number, customer_id, transaction_id, rule_code, risk_level, status, rationale, assigned_to, mlro_decision, closed_at, data, created_at, updated_at	id	Enabled
bank_audit_events	id, company_id, actor_id, operation, entity_type, entity_id, outcome, request_id, redacted_payload, created_at	id	Enabled
bank_beneficial_owners	id, company_id, customer_id, full_name, national_id, ownership_percent, verification_status, data, created_at, updated_at	id	Enabled
bank_branches	id, company_id, institution_id, code, name, region, district, address, phone, status, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
bank_cash_movements	id, company_id, teller_id, branch_id, movement_type, amount, currency, status, transaction_id, approved_by, approved_at, idempotency_key, narration, created_by, created_at	id	Enabled
bank_collateral	id, company_id, application_id, collateral_type, description, ownership_document, estimated_value, valuation_date, verification_status, data, created_at	id	Enabled
bank_customer_documents	id, company_id, customer_id, document_type, document_number, file_url, issued_at, expires_at, verification_status, verified_by, verified_at, data, created_at, updated_at	id	Enabled
bank_customers	id, company_id, customer_number, customer_kind, full_name, phone, email, date_of_birth, gender, occupation, address, national_id, tin, risk_rating, pep_status, source_of_funds, relationship_purpose, kyc_status, kyc_verified_at, kyc_verified_by, kyc_expires_at, status, branch_id, data, created_by, cr	id	Enabled
bank_fixed_deposits	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
bank_group_members	id, company_id, group_id, customer_id, role, shares_count, joined_at, status	id	Enabled
bank_groups	id, company_id, group_number, name, group_type, meeting_frequency, status, branch_id, data, created_at, updated_at	id	Enabled
bank_guarantors	id, company_id, application_id, customer_id, guarantee_amount, consent_status, consented_at, data, created_at	id	Enabled
bank_idempotency_keys	id, company_id, idempotency_key, operation, request_hash, status, result, created_by, created_at, completed_at	id	Enabled
bank_institutions	id, company_id, legal_name, trading_name, institution_type, licence_number, licence_status, country_code, currency, currency_exponent, timezone, fiscal_year_start_month, data, created_by, created_at, updated_at	id	Enabled
bank_journal_batches	id, company_id, batch_number, currency, total_debit, total_credit, status, source_type,	id	Enabled

Table	Observed columns	Primary key	RLS
	source_id, idempotency_key, posted_by, posted_at, created_at		
bank_journal_lines	id, company_id, batch_id, account_id, gl_code, line_description, debit, credit, created_at	id	Enabled
bank_loan_application	id, company_id, application_number, customer_id, product_id, amount, term_months, purpose, status, credit_score, score_inputs, submitted_by, submitted_at, decision_by, decision_at, decision_note, branch_id, data, created_at, updated_at, disbursement_account_id	id	Enabled
bank_loan_approvals	id, company_id, application_id, step_number, approver_id, decision, note, decided_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.2 X.2 Public table segment 25–48

Table	Observed columns	Primary key	RLS
bank_loan_products	id, company_id, code, name, product_kind, currency, minimum_amount, maximum_amount, minimum_term_months, maximum_term_months, annual_interest_rate, interest_method, processing_fee_rate, late_penalty_rate, collateral_required, guarantors_required, approval_threshold, status, data, created_at, updated	id	Enabled
bank_loan_repayments	id, company_id, repayment_number, loan_id, account_id, amount, principal_amount, interest_amount, fee_amount, penalty_amount, channel, status, idempotency_key, transaction_id, posted_by, posted_at	id	Enabled
bank_loan_schedules	id, company_id, loan_id, installment_number, due_date, principal_due, interest_due, fee_due, penalty_due, principal_paid, interest_paid, fee_paid, penalty_paid, status, paid_at	id	Enabled
bank_loans	id, company_id, name, status, amount, notes, data, created_at, updated_at, loan_number, application_id, customer_id, product_id, principal, outstanding_principal, outstanding_interest, outstanding_fees, outstanding_penalties, annual_interest_rate, term_months, interest_method, disbursed_at, maturity	id	Enabled
bank_notifications	id, company_id, profile_id, customer_id, notification_type, title, body, channel, status, sent_at, data, created_at	id	Enabled
bank_payment_instructions	id, company_id, instruction_number, payment_type, channel, source_account_id, destination_account_id, amount, currency, provider, msisdn, provider_reference, status, requested_at, confirmed_at, failure_reason, idempotency_key, data, created_by	id	Enabled
bank_reconciliations	id, company_id, reconciliation_number, account_id, period_start, period_end, statement_balance, ledger_balance, difference, status, reviewed_by, reviewed_at, notes, created_by, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
bank_shares	id, company_id, group_id, customer_id, shares_count, price_per_share, transaction_id, status, created_at	id	Enabled
bank_standing_orders	id, company_id, name, status, amount, notes, data, created_at, updated_at, order_number, source_account_id, destination_account_id, destination_msisdn, frequency, next_run_date, end_date, last_run_at, last_result, created_by	id	Enabled
bank_tellers	id, company_id, profile_id, branch_id, teller_code, name, status, opening_balance, closing_balance, opened_at, closed_at, version, data, created_at, updated_at	id	Enabled
bank_transactions	id, company_id, name, status, amount, notes, data, created_at, updated_at, transaction_number, transaction_type, channel, source_account_id, destination_account_id, customer_id, fee_amount, currency, idempotency_key, provider, provider_reference, narration, journal_batch_id, teller_id, initiated_by,	id	Enabled
bank_wallets	id, company_id, wallet_number, customer_id, provider, msisd, balance, status, provider_customer_ref, data, created_at, updated_at	id	Enabled
billing_plan_audit_log	id, plan_id, company_id, action, changed_by, source, previous_values, new_values, created_at	id	Enabled
billing_plans	id, company_id, code, name, description, status, currency, monthly_price, annual_price, annual_savings_label, included_users, included_branches, included_storage_mb, included_transactions, features, module_entitlements, sort_order, recommended, created_by, created_at, updated_at, plan_category, badg	id	Enabled
billing_profiles	id, company_id, legal_name, contact_name, email, phone, tax_identifier, address, notes, created_at, updated_at	id	Enabled
bnk_accounts	id, company_id, member_id, name, type, balance, status, open_date, branch, acct_no, interest, data, created_at, updated_at	id	Enabled
bnk_applications	id, company_id, member_id, member, product, amount, term, purpose, collateral,	id	Enabled

Table	Observed columns	Primary key	RLS
	submitted_date, status, officer, score, data, created_at, updated_at		
bnk_loans	id, company_id, member_id, member, product, principal, rate, term, disbursed, maturity, balance, status, collateral, emi, paid, dpd, data, created_at, updated_at	id	Enabled
bnk_members	id, company_id, name, dob, national_id, phone, email, gender, occupation, kyc_status, join_date, branch, data, created_at, updated_at	id	Enabled
bnk_transactions	id, company_id, acct_no, member, type, amount, balance, date, channel, narration, ref, data, created_at, updated_at	id	Enabled
branches	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
business_loans	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
calendar_events	id, company_id, attendees, description, end_time, event_date, event_type, meeting_link, start_time, title, created_at, updated_at	id	Enabled
collab_channels	id, company_id, description, name, scope, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.3 X.3 Public table segment 49–72

Table	Observed columns	Primary key	RLS
collab_messages	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
community_contributions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
community_group_announcements	id, company_id, group_id, title, body, audience, status, published_at, expires_at, created_by, created_at, updated_at	id	Enabled
community_group_applications	id, company_id, group_id, entity_type, entity_id, action, status, requested_by, decided_by, decision_notes, requested_at, decided_at, created_at, updated_at	id	Enabled
community_group_assets	id, company_id, group_id, asset_code, name, category, acquisition_date, acquisition_cost, current_value, location, custodian, status, created_at, updated_at	id	Enabled
community_group_attachments	id, company_id, meeting_id, member_id, status, notes, created_at, updated_at	id	Enabled
community_group_audit_logs	id, company_id, group_id, actor_id, actor_name, action, entity_type, entity_id, details, created_at, updated_at	id	Enabled
community_group_budgets	id, company_id, group_id, project_id, category, budget_amount, fiscal_year, status, approved_by, approved_at, created_at, updated_at	id	Enabled
community_group_committees	id, company_id, group_id, committee_id, member_id, committee_role, start_date, end_date, created_at, updated_at	id	Enabled
community_group_committees	id, company_id, group_id, name, committee_type, status, created_at, updated_at	id	Enabled
community_group_contributions	id, company_id, group_id, member_id, contribution_number, contribution_type, amount, currency, contribution_date, due_date, payment_method, mobile_money_provider, payment_reference, status, receipt_number, notes, created_by, created_at, updated_at	id	Enabled
community_group_documents	id, company_id, group_id, document_type, title, file_url, document_date, expires_at, status, uploaded_by, created_at, updated_at	id	Enabled
community_group_events	id, company_id, group_id, title, event_type, event_date, start_time, venue, description, reminder_sent_at, status, created_by, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
community_group_expenses	id, company_id, group_id, project_id, category, description, amount, expense_date, payment_method, payment_reference, status, approved_by, approved_at, created_by, created_at, updated_at	id	Enabled
community_group_fundraising	id, company_id, group_id, project_id, donor_name, amount, donation_date, payment_method, payment_reference, status, notes, created_at, updated_at	id	Enabled
community_group_income	id, company_id, group_id, income_type, description, amount, income_date, payment_method, payment_reference, status, created_at, updated_at	id	Enabled
community_group_loan_guaranty	id, guarantor_id, loan_id, guarantor_member_id, guaranteed_amount, consent_status, consented_at, created_at, updated_at	id	Enabled
community_group_loan_penalty	id, penalty_id, loan_id, penalty_date, reason, amount, status, created_by, created_at, updated_at	id	Enabled
community_group_loan_repayment	id, repayment_id, loan_id, repayment_number, repayment_date, amount, principal_amount, interest_amount, penalty_amount, payment_method, mobile_money_provider, payment_reference, status, receipt_number, notes, created_by, created_at, updated_at	id	Enabled
community_group_loans	id, company_id, group_id, member_id, loan_number, purpose, principal, interest_rate, interest_method, term_months, application_date, approval_status, status, approved_by, approved_at, disbursed_at, first_due_date, total_interest, total_repayable, outstanding_principal, outstanding_interest, currency	id	Enabled
community_group_meetings	id, company_id, group_id, meeting_number, meeting_date, start_time, venue, agenda, minutes, chairperson_id, status, reminder_sent_at, created_by, created_at, updated_at	id	Enabled
community_group_members	id, company_id, group_id, member_number, full_name, phone, email, national_id, id_type, gender, date_of_birth, address, occupation, next_of_kin, next_of_kin_phone, join_date, exit_date, role, kyc_status, membership_status,	id	Enabled

Table	Observed columns	Primary key	RLS
	kyc_data, data, created_by, created_at, updated_at		
community_group_messages	company_id, group_id, sender_member_id, subject, body, channel, status, created_at, updated_at	id	Enabled
community_group_notifications	company_id, group_id, member_id, notification_type, title, body, channel, status, scheduled_for, sent_at, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.4 X.4 Public table segment 73–96

Table	Observed columns	Primary key	RLS
community_group_projects	company_id, group_id, project_number, name, description, start_date, end_date, target_amount, status, created_by, created_at, updated_at	id	Enabled
community_group_savings	company_id, group_id, member_id, transaction_type, amount, transaction_date, payment_method, reference, status, notes, created_by, created_at, updated_at	id	Enabled
community_group_votes_ballots	company_id, vote_id, option_id, member_id, cast_at	id	Enabled
community_group_votes_options	company_id, vote_id, label, candidate_member_id, vote_count, created_at, updated_at	id	Enabled
community_group_votes_polls	company_id, group_id, title, description, vote_type, opens_at, closes_at, status, quorum_percent, created_by, created_at, updated_at	id	Enabled
community_group_welfare_claims	company_id, group_id, member_id, event_type, description, amount_requested, amount_approved, claim_date, status, payment_method, payment_reference, approved_by, approved_at, created_by, created_at, updated_at	id	Enabled
community_groups	id, company_id, name, status, amount, notes, data, created_at, updated_at, group_number, group_type, registration_number, description, country, region, district, ward, village, meeting_frequency, contribution_frequency, contribution_amount, currency, rules, created_by	id	Enabled
companies	id, name, category, tin, vrn, phone, email, address, city, country, currency, tax_rate, logo, join_code, created_at, updated_at, website, tax_id, business_scale, timezone, receipt_width, receipt_footer, receipt_show_logo, brand_primary_color, brand_accent_color	id	Enabled
company_memberships	user_id, company_id, role, created_at	user_id, company_id	Enabled
company_modules	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
company_profile_settings	company_id, profile_data, updated_at	company_id	Enabled
competitors	id, company_id, category, name, notes, threat_level, created_at, updated_at	id	Enabled
crm_contacts	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
crm_interactions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
crm_leads	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
custom_kpis	id, company_id, label, metric_id, target_value, created_at, updated_at	id	Enabled
customer_feedback	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
departments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
digital_signatures	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
documents	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
ecommerce_orders	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
ecommerce_products	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
emails	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
expense_budgets	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.5 X.5 Public table segment 97–120

Table	Observed columns	Primary key	RLS
fin_accounts	id, company_id, account_code, account_name, account_type, normal_side, parent_id, is_postable, is_cash, currency, status, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
fin_approval_requests	id, company_id, entity_type, entity_id, action, requested_by, status, required_approvals, decided_by, decided_at, decision_note, maker_checker_key, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
fin_idempotency_key	id, company_id, scope, idempotency_key, request_hash, response, status, expires_at, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
fin_journal_batches	id, company_id, batch_number, source_module, source_type, source_id, business_date, currency, status, debit_total, credit_total, posted_at, posted_by, reversal_of_batch_id, narration, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
fin_journal_lines	id, company_id, journal_batch_id, line_no, business_date, account_id, debit, credit, currency, branch_id, member_id, customer_id, description, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
fin_periods	id, company_id, period_start, period_end, status, timezone, closed_by, closed_at, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
fin_posting_links	id, company_id, journal_batch_id, source_table, source_id, link_role, created_by, created_at, updated_at, version, metadata	id	Enabled
fin_reconciliation_batches	id, company_id, account_scope, external_source, statement_date, opening_balance, closing_balance, status, file_reference, import_hash, approved_by, approved_at, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
fin_reconciliation_items	id, company_id, batch_id, external_reference, external_date, amount, direction, provider, provider_status,	id	Enabled

Table	Observed columns	Primary key	RLS
	matched_source_table, matched_source_id, match_status, exception_reason, resolved_by, resolved_at, created_by, created_at, updated_by, updated_at, version, metadata		
finance_assets	id, company_id, acquisition_date, category, cost, name, useful_life_years, created_at, updated_at	id	Enabled
finance_expenses	id, company_id, amount, category, due_date, expense_date, method, status, vendor, created_at, updated_at	id	Enabled
financial_benchmarks	id, company_id, benchmark_value, label, metric_id, created_at, updated_at	id	Enabled
fleet_alerts	id, company_id, alert_key, severity, alert_type, entity_type, entity_id, title, body, due_on, status, acknowledged_by, acknowledged_at, created_at	id	Enabled
fleet_audit_events	id, company_id, actor_profile_id, action, entity_type, entity_id, before_data, after_data, metadata, created_at	id	Enabled
fleet_driver_assignments	id, company_id, vehicle_id, driver_id, starts_at, ends_at, status, notes, assigned_by, created_at	id	Enabled
fleet_drivers	id, company_id, employee_id, profile_id, full_name, mobile_number, licence_number, licence_class, licence_expires_on, status, safety_score, notes, created_by, created_at, updated_at	id	Enabled
fleet_fuel_cards	id, company_id, card_number_masked, issuer, vehicle_id, status, daily_limit, monthly_limit, expires_on, created_at	id	Enabled
fleet_fuel_transactions	id, company_id, vehicle_id, trip_id, fuel_card_id, transaction_at, station_name, litres, unit_price, total_cost, odometer_km, receipt_url, payment_reference, notes, created_by, created_at	id	Enabled
fleet_incidents	id, company_id, vehicle_id, driver_id, trip_id, incident_type, occurred_at, location, description, cost, status, evidence_url, created_by, created_at	id	Enabled
fleet_maintenance_jobs	id, company_id, job_number, vehicle_id, plan_id, workshop_id, maintenance_type, status, priority, requested_on, due_on, odometer_km, estimated_cost, approved_by, completed_on, notes, created_by, created_at, updated_at	id	Enabled
fleet_maintenance_plans	id, company_id, vehicle_id, name, maintenance_type, interval_km, interval_days, last_completed_odometer_km,	id	Enabled

Table	Observed columns	Primary key	RLS
	last_completed_on, next_due_odometer_km, next_due_on, active, notes, created_by, created_at, updated_at		
fleet_routes	id, company_id, name, origin, destination, planned_distance_km, expected_duration_minutes, toll_budget, active, notes, created_by, created_at, updated_at	id	Enabled
fleet_service_records	id, company_id, maintenance_job_id, vehicle_id, service_date, odometer_km, labour_cost, parts_cost, total_cost, invoice_reference, notes, created_by, created_at	id	Enabled
fleet_spare_parts	id, company_id, inventory_item_id, part_number, name, quantity_on_hand, reorder_level, average_cost, location, created_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.6 X.6 Public table segment 121–144

Table	Observed columns	Primary key	RLS
fleet_telematics_events	id, company_id, vehicle_id, provider, external_event_id, captured_at, latitude, longitude, speed_kph, odometer_km, ignition_on, payload, created_at	id	Enabled
fleet_trips	id, company_id, trip_number, vehicle_id, driver_id, route_id, purpose, customer_reference, dispatch_status, planned_departure_at, dispatched_at, completed_at, origin, destination, start_odometer_km, end_odometer_km, distance_km, toll_cost, parking_cost, other_cost, approved_by, created_by, notes, cr	id	Enabled
fleet_tyres	id, company_id, vehicle_id, position, brand, size, serial_number, installed_on, installed_odometer_km, expected_life_km, status, notes, created_at	id	Enabled
fleet_vehicle_categories	id, company_id, name, description, created_at	id	Enabled
fleet_vehicle_documents	id, company_id, vehicle_id, document_type, document_number, issuer, issued_on, expires_on, document_url, status, notes, created_by, created_at, updated_at	id	Enabled
fleet_vehicles	id, company_id, registration_number, ownership_type, status, category_id, make, model, model_year, vin, engine_number, fuel_type, odometer_km, seats, acquisition_type, acquisition_date, acquisition_cost, lease_end_date, home_branch, cost_center, notes, metadata, created_by, created_at, updated_at	id	Enabled
fleet_workshops	id, company_id, name, contact_name, phone, email, address, supplier_id, status, created_at	id	Enabled
flt_maintenance	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
flt_trips	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
flt_vehicles	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_appointments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_doctors	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
hc_insurance_claims	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_invoices	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_lab_orders	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_notifications	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_patients	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_portal_reference_approvals	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_portal_reference_imports	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_portal_reference_signments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_prescriptions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_radiology	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_reminder_deliveries	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_reminder_settings	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.7 X.7 Public table segment 145–168

Table	Observed columns	Primary key	RLS
hc_reports	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_visits	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hc_vitals	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hospitality_amenities	id, company_id, name, category, status, data	id	Enabled
hospitality_audit_log	id, company_id, actor_profile_id, action, subject, detail, created_at	id	Enabled
hospitality_complaints	id, company_id, reservation_id, guest_id, category, description, status, resolution, data, created_at, updated_at	id	Enabled
hospitality_event_venues	id, company_id, property_id, name, capacity, base_rate, currency, status, data	id	Enabled
hospitality_events	id, company_id, property_id, venue_id, guest_id, name, start_at, end_at, status, amount, currency, data	id	Enabled
hospitality_finance_reconciliation	id, company_id, property_id, business_date, status, currency, gross_revenue, tax_total, payment_total, refund_total, variance, pos_transaction_id, journal_entry_id, finance_reference, data, created_by, created_at, updated_at	id	Enabled
hospitality_folio_lines	id, company_id, folio_id, line_type, description, quantity, unit_amount, amount, tax_amount, currency, source_table, source_record_id, posted_by, posted_at, data	id	Enabled
hospitality_folios	id, company_id, property_id, reservation_id, guest_id, folio_number, status, currency, finance_reference, data, created_at, updated_at	id	Enabled
hospitality_guest_kyc	id, company_id, guest_id, document_id, id_type, id_number, issuing_country, expires_at, verification_status, data, created_at	id	Enabled
hospitality_guest_requests	id, company_id, reservation_id, guest_id, request_type, description, priority, status, assigned_employee_id, data, created_at, updated_at	id	Enabled
hospitality_guests	id, company_id, profile_id, first_name, last_name, email, phone, nationality, date_of_birth, loyalty_number, status, data, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
hospitality_housekeeping_tasks	id, company_id, property_id, room_id, assigned_employee_id, task_type, status, due_at, completed_at, notes, data, created_at, updated_at	id	Enabled
hospitality_laundry_orders	id, company_id, reservation_id, guest_id, status, amount, currency, data, created_at, updated_at	id	Enabled
hospitality_loyalty_accounts	id, company_id, guest_id, tier, points, status, data	id	Enabled
hospitality_maintenance_requests	id, company_id, property_id, room_id, category, priority, status, assigned_employee_id, notes, data, created_at, updated_at	id	Enabled
hospitality_menu_items	id, company_id, menu_id, inventory_item_id, name, price, currency, status, data	id	Enabled
hospitality_menus	id, company_id, property_id, name, meal_period, status, data	id	Enabled
hospitality_minibar_purchases	id, company_id, reservation_id, inventory_item_id, quantity, amount, status, data, created_at	id	Enabled
hospitality_notifications	id, company_id, profile_id, employee_id, title, body, type, module, record_id, read_at, created_at	id	Enabled
hospitality_order_lines	id, company_id, order_id, menu_item_id, name, quantity, unit_price, status, data	id	Enabled
hospitality_orders	id, company_id, property_id, table_id, reservation_id, folio_id, order_number, status, currency, data, created_by, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.8 X.8 Public table segment 169–192

Table	Observed columns	Primary key	RLS
hospitality_payments	id, company_id, folio_id, payment_method, amount, currency, status, reference, finance_payment_id, received_by, received_at, data	id	Enabled
hospitality_properties	id, company_id, branch_id, name, code, address, timezone, currency, status, data, created_at, updated_at	id	Enabled
hospitality_rate_plans	id, company_id, property_id, room_type_id, name, currency, nightly_rate, effective_from, effective_to, status, data	id	Enabled
hospitality_reservations	id, company_id, property_id, guest_id, room_type_id, room_id, confirmation_code, arrival_date, departure_date, adults, children, status, nightly_rate, currency, source, special_requests, checked_in_at, checked_out_at, data, created_at, updated_at	id	Enabled
hospitality_restaurant_tables	id, company_id, property_id, table_number, capacity, zone, status, data	id	Enabled
hospitality_room_types	id, company_id, property_id, name, code, capacity_adults, capacity_children, base_rate, currency, amenities, status, data	id	Enabled
hospitality_rooms	id, company_id, property_id, room_type_id, room_number, floor, status, housekeeping_status, maintenance_status, data, created_at, updated_at	id	Enabled
hospitality_taxes	id, company_id, property_id, name, code, rate, applies_to, effective_from, effective_to, status, data	id	Enabled
hr_announcement_reads	announcement_id, profile_id, read_at	announcement_id, profile_id	Enabled
hr_announcements	id, company_id, title, body, audience_type, department_id, status, published_at, expires_at, created_by, created_at, updated_at	id	Enabled
hr_approval_requests	id, company_id, request_type, source_table, source_record_id, requester_employee_id, subject_employee_id, status, current_step, data, created_at, updated_at	id	Enabled
hr_approval_steps	id, company_id, approval_request_id, step_number, approver_profile_id, approver_role, status, decision_note, decided_at, created_at	id	Enabled

Table	Observed columns	Primary key	RLS
hr_attendance	id, company_id, name, status, amount, notes, data, created_at, updated_at, employee_id, attendance_date, shift_id, clock_in_at, clock_out_at, worked_minutes, source	id	Enabled
hr_benefit_enrollments	id, company_id, employee_id, benefit_plan_id, status, effective_from, effective_to, data, created_at, updated_at	id	Enabled
hr_benefit_plans	id, company_id, name, provider, benefit_type, employee_contribution, employer_contribution, currency, status, data, created_at, updated_at	id	Enabled
hr_benefits	id, company_id, name, status, amount, notes, data, created_at, updated_at, employee_id, benefit_plan_id, effective_from, effective_to	id	Enabled
hr_candidates	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hr_duties	id, company_id, name, status, amount, notes, data, created_at, updated_at, employee_id, assignee_profile_id, duty_date, started_at, completed_at, approved_by, approved_at	id	Enabled
hr_employee_documents	id, company_id, employee_id, document_id, title, document_type, file_url, status, expires_at, data, created_by, created_at, updated_at	id	Enabled
hr_employees	id, company_id, name, status, amount, notes, data, created_at, updated_at, profile_id, department_id, position_id, manager_employee_id, employee_number, employment_start_date, employment_end_date, timezone	id	Enabled
hr_expense_claims	id, company_id, employee_id, expense_date, category, amount, currency, merchant, description, status, finance_expense_id, document_id, decided_by, decided_at, decision_note, data, created_at, updated_at	id	Enabled
hr_goal_updates	id, company_id, goal_id, employee_id, current_value, note, created_at	id	Enabled
hr_goals	id, company_id, employee_id, title, description, metric_name, target_value, current_value, unit, status, due_date, owner_employee_id, data, created_at, updated_at	id	Enabled
hr_holidays	id, company_id, holiday_date, name, holiday_type, paid, branch_id, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.9 X.9 Public table segment 193–216

Table	Observed columns	Primary key	RLS
hr_invite_codes	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
hr_leave_balances	id, company_id, employee_id, leave_policy_id, period_year, opening_balance, accrued_days, used_days, adjustment_days, updated_at	id	Enabled
hr_leave_policies	id, company_id, name, leave_type, annual_entitlement, carry_forward_limit, requires_approval, status, data, created_at, updated_at	id	Enabled
hr_leave_requests	id, company_id, name, status, amount, notes, data, created_at, updated_at, employee_id, leave_policy_id, start_date, end_date, requested_days, decision_by, decided_at, decision_note	id	Enabled
hr_notifications	id, company_id, profile_id, employee_id, title, body, notification_type, link_module, link_record_id, read_at, created_at	id	Enabled
hr_offboarding_cases	id, company_id, employee_id, status, last_working_date, reason, owner_profile_id, data, created_at, updated_at	id	Enabled
hr_onboarding_cases	id, company_id, employee_id, status, start_date, due_date, owner_profile_id, data, created_at, updated_at	id	Enabled
hr_onboarding_tasks	id, company_id, onboarding_case_id, employee_id, title, owner_profile_id, due_date, status, completed_at, data, created_at, updated_at	id	Enabled
hr_payroll_items	id, company_id, payroll_run_id, employee_id, gross_pay, taxable_pay, deductions, net_pay, currency, status, data, created_at, updated_at, employer_contributions, employer_cost	id	Enabled
hr_payroll_runs	id, company_id, name, status, amount, notes, data, created_at, updated_at, period_start, period_end, currency, timezone, approved_by, approved_at, posted_at, finance_reference	id	Enabled
hr_payslips	id, company_id, payroll_item_id, employee_id, pay_period, status, issued_at, document_id, data, created_at, updated_at	id	Enabled
hr_performance_reviews	id, company_id, name, status, amount, notes, data, created_at, updated_at, employee_id, reviewer_employee_id, review_period_start,	id	Enabled

Table	Observed columns	Primary key	RLS
	review_period_end, due_date, submitted_at, completed_at		
hr_positions	id, company_id, department_id, title, code, grade, status, description, created_at, updated_at	id	Enabled
hr_service_requests	id, company_id, employee_id, request_type, subject, description, status, assigned_to, decided_by, decided_at, decision_note, data, created_at, updated_at	id	Enabled
hr_shift_assignments	id, company_id, employee_id, shift_id, assignment_date, status, notes, created_by, created_at, updated_at	id	Enabled
hr_shifts	id, company_id, name, start_time, end_time, unpaid_break_minutes, timezone, status, data, created_at, updated_at	id	Enabled
hr_statutory_rules	id, company_id, name, rule_code, effective_from, effective_to, applies_to, calculation_type, rate, fixed_amount, threshold_amount, currency, status, data, created_at, updated_at	id	Enabled
hr_timesheet_entries	id, company_id, timesheet_id, work_date, minutes, project_reference, work_note, created_at, updated_at	id	Enabled
hr_timesheets	id, company_id, employee_id, period_start, period_end, total_minutes, status, submitted_at, decided_by, decided_at, decision_note, data, created_at, updated_at	id	Enabled
hr_training	id, company_id, completion_date, course, due_date, employee_name, hr_employees, is_compliance, is_mandatory, status, video_url, created_at, updated_at	id	Enabled
hr_training_assignments	id, company_id, employee_id, course_id, assigned_by, due_date, status, completed_at, data, created_at, updated_at	id	Enabled
hr_training_courses	id, company_id, title, provider, duration_minutes, mandatory, status, content_url, data, created_at, updated_at	id	Enabled
htl_bookings	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
htl_rooms	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.10 X.10 Public table segment 217–240

Table	Observed columns	Primary key	RLS
integration_connections	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
inventory_batches	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
inventory_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
inventory_stock_movements	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
inventory_suppliers	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
inventory_transfers	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
inventory_warehouses	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
journal_entries	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
kb_articles	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
loan_repayments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
manufacturing_bom_components	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
manufacturing_boms	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
manufacturing_machines	id, company_id, machine_type, name, purchase_date, status, warehouse_id, created_at, updated_at	id	Enabled
manufacturing_maintenance	id, company_id, cost, machine_name, maintenance_date, maintenance_type, next_due_date, notes, technician, created_at, updated_at	id	Enabled
manufacturing_qc_inspections	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
manufacturing_work_orders	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
marketing_campaigns	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_audit_logs	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_cash_sessions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
mfi_cash_transactions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_clients	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_collateral	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_collections	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_credit_scorecards	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.11 X.11 Public table segment 241-264

Table	Observed columns	Primary key	RLS
mfi_credit_scoring_settings	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_groups	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_guarantors	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_loan_applications	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_loan_products	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_loans	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_notifications	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_par_escalation_settings	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_repayment_schedules	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_repayments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_savings	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
mfi_staff_commissions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
money_agent_agents	id, company_id, profile_id, branch_id, supervisor_id, agent_code, full_name, phone, national_id, kyc_status, kyb_status, status, daily_limit, monthly_limit, notes, metadata, created_by, created_at, updated_at	id	Enabled
money_agent_alerts	id, company_id, agent_id, alert_type, severity, title, body, status, created_at, acknowledged_by, acknowledged_at	id	Enabled
money_agent_approvals	id, company_id, transaction_id, status, requested_by, decided_by, note, requested_at, decided_at	id	Enabled
money_agent_audit_events	id, company_id, actor_profile_id, action, entity_type, entity_id, before_data, after_data, metadata, created_at	id	Enabled
money_agent_branches	id, company_id, branch_code, name, region, district, ward, address, phone, status, created_by, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
money_agent_commission	id, company_id, service_code, commission_type, commission_value, active, created_by, created_at, updated_at	id	Enabled
money_agent_customers	id, company_id, profile_id, full_name, phone, national_id, kyc_status, status, address, meta-data, created_by, created_at, updated_at	id	Enabled
money_agent_daily_summaries	id, company_id, agent_id, branch_id, business_date, transaction_count, successful_count, failed_count, cash_in_amount, cash_out_amount, fee_amount, commission_amount, updated_at	id	Enabled
money_agent_fee_rules	id, company_id, service_code, min_amount, max_amount, fee_type, fee_value, active, created_by, created_at, updated_at	id	Enabled
money_agent_ledger_entries	id, company_id, transaction_id, account_code, entry_type, amount, currency, posted_at, meta-data	id	Enabled
money_agent_limits	id, company_id, agent_id, transaction_type, max_single_amount, daily_amount, monthly_amount, velocity_window_minutes, velocity_count, active, created_by, created_at, updated_at	id	Enabled
money_agent_notifications	id, company_id, transaction_id, agent_id, channel, status, title, body, provider_reference, created_at, sent_at, read_at	id	Enabled

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11.12 X.12 Public table segment 265–288

Table	Observed columns	Primary key	RLS
money_agent_pin_credentials	id, company_id, agent_id, pin_hash, failed_attempts, locked_until, last_used_at, status, created_at, updated_at	id	Enabled
money_agent_receipts	id, company_id, transaction_id, receipt_number, channel, recipient_phone, issued_at, metadata	id	Enabled
money_agent_reconciliations	id, company_id, settlement_id, status, expected_amount, actual_amount, variance, reviewed_by, reviewed_at, notes, created_at, updated_at	id	Enabled
money_agent_risk_events	id, company_id, agent_id, transaction_id, risk_type, severity, status, score, reason, metadata, created_at, resolved_by, resolved_at	id	Enabled
money_agent_services	id, company_id, service_code, name, service_type, provider_code, requires_provider, active, metadata, created_by, created_at, updated_at	id	Enabled
money_agent_settlements	id, company_id, agent_id, branch_id, business_date, opening_float, closing_float, expected_float, variance, status, submitted_by, settled_by, notes, created_at, updated_at	id	Enabled
money_agent_transactions	id, company_id, transaction_ref, idempotency_key, agent_id, branch_id, customer_id, service_id, transaction_type, amount, fee, commission, currency, status, authorization_method, authorization_reference_hash, provider_code, provider_reference, failure_code, failure_reason, requested_at, authorized_at	id	Enabled
money_agent_wallets	id, company_id, owner_type, owner_id, wallet_type, currency, available_balance, status, created_at, updated_at	id	Enabled
network_profiles	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
network_rfqs	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
notebook_notes	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
notification_channels	id, company_id, business_number, channel_id, enabled, from_address, from_number, server_key, webhook_url, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
notification_log	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
notification_rules	id, company_id, alert_type, channels, created_at, updated_at	id	Enabled
other_debtors	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
other_income	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
period_closes	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_audit_logs	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_batches	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_brands	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_categories	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_controlled_medication_registry	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_dispense	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_dispense_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.13 X.13 Public table segment 289–312

Table	Observed columns	Primary key	RLS
phm_drugs	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_insurance_claims	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_notifications	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_payments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_purchase_order_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_purchase_orders	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_return_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_returns	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_sale_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_sales	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_stock	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_stock_adjustments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_stock_movements	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_stock_receipts	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_stock_transfers	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
phm_suppliers	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
platform_admin_actions	id, actor_user_id, actor_role, action, target_type, target_id, reason, confirmation_text, details, created_at	id	Enabled
pos_cash_movements	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
pos_discount_rules	id, company_id, discount_code, name, scope_type, inventory_item_id, discount_type, value, max_discount_amount, minimum_subtotal, stackable, requires_approval, contra_revenue_account_id, effective_from,	id	Enabled

Table	Observed columns	Primary key	RLS
	effective_to, status, approval_request_id, created_by, created_at, updated_by, updated_at, version		
pos_loyalty_ledger	id, company_id, member_id, entry_type, points_delta, points_balance_after, sale_id, redemption_id, idempotency_key, reference, status, occurred_at, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_loyalty_members	id, company_id, program_id, customer_id, member_number, status, points_balance, lifetime_earned, lifetime_redeemed, joined_at, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_loyalty_programs	id, company_id, program_code, name, earn_points_per_100_tzs, redemption_tzs_per_point, minimum_redeem_points, expiry_days, points_liability_account_id, status, approval_request_id, effective_from, effective_to, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_loyalty_redemptions	id, company_id, member_id, reward_id, sale_id, points_spent, cash_value, status, approval_request_id, journal_batch_id, idempotency_key, request_hash, applied_at, applied_by, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_loyalty_rewards	id, company_id, program_id, reward_code, name, points_cost, cash_value, inventory_item_id, status, approval_request_id, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.14 X.14 Public table segment 313–336

Table	Observed columns	Primary key	RLS
pos_promotion_items	id, company_id, promotion_id, inventory_item_id, item_role, required_quantity, reward_price, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_promotions	id, company_id, promotion_code, name, trigger_type, benefit_type, minimum_spend, minimum_quantity, benefit_value, reward_quantity, points_multiplier_bps, stackable, priority, customer_limit, daily_limit, effective_from, effective_to, status, approval_request_id, requires_approval, created_by, create	id	Enabled
pos_registers	id, company_id, register_code, name, branch_id, warehouse_id, default_currency, status, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_return_commits	id, company_id, return_id, idempotency_key, created_by, created_at, updated_at	id	Enabled
pos_return_headers	id, company_id, return_number, sale_id, register_id, terminal_id, shift_id, cashier_id, reason, refund_method, currency, refund_total, status, approval_request_id, journal_batch_id, idempotency_key, request_hash, legacy_pos_return_id, posted_at, posted_by, created_by, created_at, updated_by, updated	id	Enabled
pos_return_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
pos_return_lines	id, company_id, return_id, sale_line_id, line_no, quantity, unit_price, tax_amount, refund_amount, restock_quantity, condition, status, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_returns	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
pos_sale_adjustments	id, company_id, sale_id, sale_line_id, adjustment_no, adjustment_type, tax_rule_id, discount_rule_id, promotion_id, base_amount, rate_bps, amount, status, approval_request_id, journal_batch_id, idempotency_key,	id	Enabled

Table	Observed columns	Primary key	RLS
	request_hash, applied_at, applied_by, created_by, created_at, updated_by, updated_at, ver		
pos_sale_headers	id, company_id, sale_number, register_id, terminal_id, shift_id, cashier_id, customer_id, customer_name, business_date, source_channel, status, payment_status, currency, subtotal, discount_total, tax_total, total, paid_total, change_total, refunded_total, journal_batch_id, idempotency_key, request_h	id	Enabled
pos_sale_lines	id, company_id, sale_id, line_no, inventory_item_id, item_sku, item_name, quantity, unit_price, discount_amount, tax_amount, line_subtotal, line_total, cost_total, returned_quantity, status, legacy_pos_transaction_item_id, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_sale_tax_lines	id, company_id, sale_id, sale_line_id, tax_rule_id, taxable_amount, rate_bps, tax_amount, included_in_price, status, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_sale_tenders	id, company_id, sale_id, tender_no, method, currency, tendered_amount, applied_amount, change_amount, reference, provider_code, provider_reference, provider_status, status, journal_batch_id, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_shift_cash_movements	id, company_id, shift_id, movement_type, amount, reason, reference, approval_request_id, journal_batch_id, status, occurred_at, legacy_pos_cash_movement_id, created_by, created_at, updated_by, updated_at, version, metadata, idempotency_key, request_hash, posted_at, posted_by, reversal_of_movement_id	id	Enabled
pos_shift_sessions	id, company_id, shift_number, register_id, terminal_id, cashier_id, business_date, opened_at, opening_float, expected_cash, counted_cash, variance, status, closed_at, closed_by, close_reason, open_idempotency_key, close_idempotency_key, legacy_pos_shift_id, created_by, created_at, updated_by, update	id	Enabled

Table	Observed columns	Primary key	RLS
pos_shifts	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
pos_sync_devices	id, company_id, device_key, terminal_id, last_sequence, last_seen_at, status, created_by, created_at, updated_by, updated_at, version, metadata, last_accepted_hash	id	Enabled
pos_sync_events	id, company_id, idempotency_key, transaction_id, status, message, created_by, created_at, updated_at	id	Enabled
pos_tax_rules	id, company_id, tax_code, name, tax_type, scope_type, inventory_item_id, rate_bps, calculation_method, tax_account_id, effective_from, effective_to, status, approval_request_id, requires_approval, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_terminals	id, company_id, register_id, device_key, device_label, app_version, last_seen_at, status, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
pos_transaction_comments	id, company_id, transaction_id, idempotency_key, created_by, created_at, updated_at	id	Enabled
pos_transaction_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
pos_transactions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
procurement_contracts	id, company_id, contract_type, doc_number, end_date, notes, start_date, supplier, value, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.15 X.15 Public table segment 337–360

Table	Observed columns	Primary key	RLS
procurement_purchases	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
profiles	id, company_id, full_name, email, role, phone, avatar_url, is_active, created_at, updated_at, customer_ref, onboarding_tour_completed_at, onboarding_tour_role_track, preferred_name, first_name, middle_name, last_name, date_of_birth, gender, address, country, preferred_language, currency_display, pro	id	Enabled
project_expenses	id, company_id, amount, description, expense_date, project_ref, created_at, updated_at	id	Enabled
project_milestones	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
project_tasks	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
projects	id, company_id, budget, client, end_date, manager, name, start_date, status, created_at, updated_at	id	Enabled
property_agents	id, company_id, profile_id, agent_code, full_name, phone, email, licence_number, commission_rate, status, branch_label, metadata, created_by, created_at, updated_at	id	Enabled
property_applications	id, company_id, application_number, unit_id, tenant_id, agent_id, requested_start_date, proposed_rent, status, decision_note, decided_by, decided_at, idempotency_key, metadata, created_by, created_at, updated_at	id	Enabled
property_approvals	id, company_id, entity_type, entity_id, action, status, requested_by, decided_by, note, requested_at, decided_at	id	Enabled
property_audit_log	id, company_id, actor_id, action, entity_type, entity_id, details, created_at	id	Enabled
property_budgets	id, company_id, portfolio_id, fiscal_year, category, budget_amount, status, approved_by, approved_at, created_by, created_at	id	Enabled
property_buildings	id, company_id, portfolio_id, property_code, name, property_type, address, country, region, district, ward, village, latitude, longitude, year_built, floors, status, metadata, created_by, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
property_contractors	id, company_id, contractor_code, name, phone, email, trade, tax_number, status, created_by, created_at, updated_at	id	Enabled
property_documents	id, company_id, entity_type, entity_id, document_type, title, storage_key, file_url, document_date, expires_at, verification_status, uploaded_by, metadata, created_at	id	Enabled
property_expenses	id, company_id, expense_number, property_id, unit_id, work_order_id, category, description, amount, expense_date, status, payment_method, payment_reference, approved_by, approved_at, created_by, created_at, updated_at	id	Enabled
property_handover_records	id, company_id, lease_id, handover_type, handover_date, keys_count, meter_snapshot, signed_by_tenant, signed_by_manager, notes, created_by, created_at	id	Enabled
property_inspection_items	id, company_id, inspection_id, area_name, condition, notes, estimated_cost, created_at	id	Enabled
property_inspections	id, company_id, lease_id, inspection_type, inspection_date, condition_summary, status, inspector_id, metadata, created_at	id	Enabled
property_insurances	id, company_id, property_id, unit_id, insurer, policy_number, cover_type, premium, start_date, end_date, status, notes, created_by, created_at	id	Enabled
property_integration_events	id, company_id, target_module, entity_type, entity_id, event_type, status, payload, created_by, created_at, processed_at	id	Enabled
property_invoice_lines	id, company_id, invoice_id, line_type, description, quantity, unit_amount, line_total, account_code, created_at	id	Enabled
property_invoices	id, company_id, invoice_number, lease_id, tenant_id, unit_id, invoice_type, period_start, period_end, issue_date, due_date, subtotal, tax_amount, late_fee_amount, total_amount, amount_paid, status, currency, idempotency_key, notes, created_by, created_at, updated_at	id	Enabled
property_leases	id, company_id, lease_number, unit_id, tenant_id, owner_id, application_id, start_date, end_date, rent_amount, service_charge_amount, deposit_amount, rent_frequency, notice_days, status, terms, created_by, approved_by,	id	Enabled

Table	Observed columns	Primary key	RLS
	approved_at, terminated_at, created_at, updated_at		
property_ledger_entries	id, company_id, source_type, source_id, account_code, entry_type, amount, currency, metadata, created_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.16 X.16 Public table segment 361–384

Table	Observed columns	Primary key	RLS
property_listings	id, company_id, unit_id, agent_id, listing_type, asking_amount, commission_rate, status, available_from, published_at, expires_at, description, metadata, created_by, created_at, updated_at	id	Enabled
property_maintenance_requests	id, company_id, request_number, unit_id, lease_id, tenant_id, category, priority, title, description, status, requested_at, completed_at, created_by, created_at, updated_at	id	Enabled
property_meter_readings	id, company_id, meter_id, reading_date, reading_value, previous_value, consumption, captured_by, source, notes, created_at	id	Enabled
property_notices	id, company_id, lease_id, tenant_id, notice_type, title, body, notice_date, effective_date, status, created_by, created_at	id	Enabled
property_notifications	id, company_id, recipient_profile_id, tenant_id, notice_id, notification_type, title, body, channel, status, scheduled_for, sent_at, dedupe_key, created_at	id	Enabled
property_owners	id, company_id, profile_id, owner_type, legal_name, phone, email, national_id, tin, kyc_status, status, notes, metadata, created_by, created_at, updated_at	id	Enabled
property_payments	id, company_id, payment_number, invoice_id, tenant_id, amount, payment_method, provider_code, provider_reference, status, idempotency_key, paid_at, posted_by, metadata, created_at	id	Enabled
property_plots	id, company_id, portfolio_id, plot_code, title_number, land_use, area_sqm, address, region, district, ward, latitude, longitude, owner_id, status, metadata, created_by, created_at, updated_at	id	Enabled
property_portfolios	id, company_id, portfolio_code, name, description, currency, timezone, status, created_by, created_at, updated_at	id	Enabled
property_receipts	id, company_id, payment_id, receipt_number, channel, recipient_phone, issued_at, metadata	id	Enabled
property_reconciliations	id, company_id, payment_id, invoice_id, expected_amount, actual_amount, variance, sta-	id	Enabled

Table	Observed columns	Primary key	RLS
	tus, reviewed_by, reviewed_at, notes, created_by, created_at		
property_rent_schedules	id, company_id, lease_id, next_invoice_date, frequency, active, last_invoice_id, created_at, updated_at	id	Enabled
property_service_charges	id, company_id, unit_id, name, amount, frequency, status, created_by, created_at, updated_at	id	Enabled
property_tax_fee_rules	id, company_id, code, name, applies_to, rate, flat_amount, status, created_by, created_at, updated_at	id	Enabled
property_tenant_documents	id, company_id, tenant_id, document_type, document_number, storage_key, file_url, verification_status, expires_at, metadata, uploaded_by, created_at	id	Enabled
property_tenants	id, company_id, profile_id, tenant_code, full_name, phone, email, national_id, tin, kyc_status, status, address, emergency_contact, metadata, created_by, created_at, updated_at	id	Enabled
property_units	id, company_id, building_id, plot_id, owner_id, unit_code, unit_type, floor_label, bedrooms, bathrooms, area_sqm, rent_amount, service_charge_amount, deposit_amount, currency, status, furnishing, notes, metadata, created_by, created_at, updated_at	id	Enabled
property_utility_meters	id, company_id, unit_id, utility_type, meter_number, unit_of_measure, rate, status	id	Enabled
property_work_orders	id, company_id, work_order_number, request_id, contractor_id, assigned_profile_id, estimated_cost, actual_cost, status, due_date, completion_note, created_by, created_at, updated_at	id	Enabled
purchase_order_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
resource_bookings	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
restaurant_alerts	id, company_id, outlet_id, alert_key, alert_type, severity, title, body, status, due_at, created_at	id	Enabled
restaurant_audit_events	id, company_id, outlet_id, actor_id, action, subject_type, subject_id, detail, created_at	id	Enabled
restaurant_bill_splits	id, company_id, parent_order_id, split_order_id, split_number, amount, status	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.17 X.17 Public table segment 385–408

Table	Observed columns	Primary key	RLS
restaurant_combo_items	id, company_id, parent_menu_item_id, child_menu_item_id, quantity	id	Enabled
restaurant_customers	id, company_id, guest_id, name, phone, email, loyalty_points, status, data	id	Enabled
restaurant_dining_areas	id, company_id, outlet_id, name, area_type, layout, status	id	Enabled
restaurant_fiscal_profiles	id, company_id, outlet_id, tax_profile_id, tin, vrn, business_name, trading_name, physical_address, region, district, device_serial, provider_code, environment, status, receipt_prefix, fiscalized_at, data, created_by, created_at, updated_at	id	Enabled
restaurant_fiscal_receipts	id, company_id, outlet_id, fiscal_profile_id, order_id, internal_reference, official_receipt_number, fiscal_serial, verification_code, qr_payload, status, gross_amount, vat_amount, net_amount, currency, idempotency_key, provider_response, failure_reason, queued_at, submitted_at, verified_at, created	id	Enabled
restaurant_kitchen_tickets	id, company_id, outlet_id, order_id, ticket_number, station, status, opened_at, completed_at	id	Enabled
restaurant_menu_categories	id, company_id, outlet_id, name, sort_order, status	id	Enabled
restaurant_menu_items	id, company_id, outlet_id, category_id, inventory_item_id, sku, name, description, price, cost_price, preparation_minutes, station, tax_rate, status, data	id	Enabled
restaurant_mobile_order_intents	id, company_id, outlet_id, order_id, profile_id, provider_reference, phone_last_four, amount, currency, status, provider_payload, failure_reason, expires_at, paid_at, created_by, created_at, updated_at	id	Enabled
restaurant_mobile_order_profiles	id, company_id, outlet_id, provider, merchant_label, merchant_account_reference, collection_mode, status, webhook_configured, data, created_by, created_at, updated_at	id	Enabled
restaurant_modifier_groups	id, company_id, outlet_id, name, min_select, max_select, status	id	Enabled
restaurant_modifier_options	id, company_id, group_id, inventory_item_id, name, price_delta, status	id	Enabled

Table	Observed columns	Primary key	RLS
restaurant_order_lines	id, company_id, order_id, menu_item_id, name, quantity, unit_price, discount_amount, status, modifiers, notes, stock_consumed, created_at	id	Enabled
restaurant_orders	id, company_id, outlet_id, table_id, reservation_id, customer_id, hotel_folio_id, waiter_employee_id, order_number, order_type, status, subtotal, discount_amount, tax_amount, service_charge_amount, tip_amount, total_amount, currency, opened_at, closed_at, data	id	Enabled
restaurant_outlets	id, company_id, property_id, branch_id, name, code, timezone, currency, tax_rate, service_charge_rate, status, data, created_at, updated_at	id	Enabled
restaurant_payments	id, company_id, order_id, method, amount, reference, status, received_by, received_at, data	id	Enabled
restaurant_promotions	id, company_id, outlet_id, code, name, discount_type, discount_value, starts_at, ends_at, status	id	Enabled
restaurant_purchase_lines	id, company_id, request_id, inventory_item_id, quantity, unit_cost, received_quantity	id	Enabled
restaurant_purchase_requests	id, company_id, outlet_id, supplier_id, request_number, status, total_amount, currency, data, created_at	id	Enabled
restaurant_recipe_ingredients	id, company_id, menu_item_id, inventory_item_id, quantity, unit, waste_pct	id	Enabled
restaurant_refunds	id, company_id, order_id, payment_id, reason, amount, status, idempotency_key, created_by, created_at	id	Enabled
restaurant_reservations	id, company_id, outlet_id, table_id, customer_id, reference, reservation_at, duration_minutes, covers, status, notes, created_at	id	Enabled
restaurant_shifts	id, company_id, outlet_id, employee_id, role, starts_at, ends_at, status, opening_cash, closing_cash, data, created_at	id	Enabled
restaurant_staff_roles	id, company_id, outlet_id, employee_id, role, status	id	Enabled

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11.18 X.18 Public table segment 409–432

Table	Observed columns	Primary key	RLS
restaurant_suppliers	id, company_id, name, phone, email, address, status, data	id	Enabled
restaurant_tables	id, company_id, outlet_id, area_id, code, capacity, status, position, current_order_id, data, updated_at	id	Enabled
restaurant_tax_profiles	id, company_id, outlet_id, code, name, tax_type, rate_percent, is_inclusive, is_default, is_active, legal_basis, created_at, updated_at	id	Enabled
restaurant_wastage	id, company_id, outlet_id, inventory_item_id, quantity, reason, cost, created_by, created_at	id	Enabled
rst_menu	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
rst_orders	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
rst_reservations	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
rst_tables	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sales_invoice_items	id, company_id, name, status, amount, notes, data, created_at, updated_at, invoice_id, item_name, item_sku, qty, rate, sort_order	id	Enabled
sales_invoices	id, company_id, name, status, amount, notes, data, created_at, updated_at, doc_number, customer, issue_date, due_date, order_id, amount_paid	id	Enabled
sales_order_items	id, company_id, name, status, amount, notes, data, created_at, updated_at, order_id, item_name, item_sku, qty, rate, sort_order	id	Enabled
sales_order_return_items	id, company_id, name, status, amount, notes, data, created_at, updated_at, return_id, item_name, item_sku, qty, rate	id	Enabled
sales_order_returns	id, company_id, name, status, amount, notes, data, created_at, updated_at, order_id, reason	id	Enabled
sales_orders	id, company_id, customer, doc_number, order_date, owner_id, quotation_id, sales_order_items, sales_order_returns, status, created_at, updated_at, quotation_reference, owner_name	id	Enabled
sales_payments	id, company_id, name, status, amount, notes, data, created_at, updated_at, invoice_id, method, payment_date, reference	id	Enabled

Table	Observed columns	Primary key	RLS
sales_quotation_items	id, company_id, name, status, amount, notes, data, created_at, updated_at, quotation_id, item_name, item_sku, qty, rate, sort_order	id	Enabled
sales_quotations	id, company_id, name, status, amount, notes, data, created_at, updated_at, doc_number, customer, issue_date, valid_until, owner_id	id	Enabled
sales_subscriptions	id, company_id, name, status, amount, notes, data, created_at, updated_at, doc_number, customer, plan, cycle, start_date, next_billing_date	id	Enabled
sch_academic_years	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_admissions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_announcements	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_approval_requests	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_assessment_scores	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_assessments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

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11.19 X.19 Public table segment 433–456

Table	Observed columns	Primary key	RLS
sch_assignment_submissions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_assignments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_attendance_records	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_attendance_sessions	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_audit_logs	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_books	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_classes	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_departments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_disciplinary_records	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_documents	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_enrollments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_exams	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_fee_invoice_lines	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_fee_invoices	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_fee_structures	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_fees	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_grading_scales	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_guardians	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_hostel_allocations	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_hostels	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_inventory_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
sch_inventory_movements	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_library_loans	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_messages	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.20 X.20 Public table segment 457–480

Table	Observed columns	Primary key	RLS
sch_notifications	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_payments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_portal_links	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_report_cards	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_scholarships	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_streams	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_student_guardians	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_students	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_subjects	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_teacher_assignments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_teachers	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_terms	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_timetables	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_transport	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sch_transport_assignments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
scheduled_reports	id, company_id, format, frequency, last_run, recipient_email, report_type, status, created_at, updated_at	id	Enabled
scm_shipments	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
scm_vehicles	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
signatures	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sms_group_members	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
sms_groups	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
sms_templates	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
stock_audit_items	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
stock_audits	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.21 X.21 Public table segment 481–504

Table	Observed columns	Primary key	RLS
subscription_events	id, company_id, subscription_id, payment_id, event_type, previous_status, new_status, actor_profile_id, actor_type, details, created_at	id	Enabled
subscription_invoices	id, company_id, subscription_id, payment_id, invoice_number, status, currency, subtotal, tax_amount, total_amount, paid_amount, issued_at, due_at, paid_at, document_data, created_at, updated_at	id	Enabled
subscription_notifications	id, company_id, subscription_id, notification_key, notification_type, title, message, status, scheduled_for, delivered_at, read_at, metadata, created_at	id	Enabled
subscription_payments	id, company_id, subscription_id, plan_id, provider, internal_reference, idempotency_key, provider_order_id, amount, fee, net_amount, currency, phone, description, billing_cycle, status, initiated_by, provider_response, verified_at, paid_at, failure_reason, created_at, updated_at	id	Enabled
subscription_usage	id, company_id, usage_key, period_start, period_end, usage_value, limit_value, source, metadata, recorded_at	id	Enabled
support_agents	id, company_id, profile_id, team_id, availability, workload_limit, is_active, created_at, updated_at	id	Enabled
support_call_log	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
support_chat_conversations	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
support_chat_messages	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
support_message_templates	id, company_id, name, provider, channel, language, category, body, variables, approval_status, is_active, created_by, created_at, updated_at	id	Enabled
support_sla_policies	id, company_id, name, priority, first_response_minutes, resolution_minutes, warning_minutes, is_active, created_at, updated_at	id	Enabled
support_team_members	id, company_id, team_id, profile_id, role, created_at	id	Enabled
support_teams	id, company_id, name, department_name, is_active, created_by, created_at, updated_at	id	Enabled

Table	Observed columns	Primary key	RLS
support_ticket_activities	id, company_id, ticket_id, actor_profile_id, event_type, details, created_at	id	Enabled
support_ticket_messages	id, company_id, name, status, amount, notes, data, created_at, updated_at, ticket_id, body, sender_kind, sender_profile_id, channel, is_internal, delivery_status, provider_message_id, sent_at	id	Enabled
support_ticket_notes	id, company_id, ticket_id, author_profile_id, body, kind, created_at	id	Enabled
support_tickets	id, company_id, assignee, category, created_date, customer, doc_number, priority, status, subject, support_ticket_messages, created_at, updated_at, assigned_profile_id, team_id, source_channel, customer_reference, due_at, resolved_at, closed_at	id	Enabled
tenant_subscriptions	id, company_id, plan_id, status, billing_cycle, amount, currency, started_at, renewed_at, expires_at, grace_expires_at, cancelled_at, cancellation_reason, source_payment_id, meta-data, created_at, updated_at, offer_code, paid_months, bonus_months, total_months, duration_days	id	Enabled
user_table_preferences	company_id, user_id, preference_key, value, updated_at	company_id, user_id, preference_key	Enabled
vicoba_loans	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
vicoba_meetings	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
vicoba_members	id, company_id, name, status, amount, notes, data, created_at, updated_at	id	Enabled
whatsapp_account_links	id, company_id, profile_id, phone_e164, code_hash, expires_at, used_at, created_by, created_at	id	Enabled
whatsapp_accounts	id, company_id, provider, phone_number_id, display_phone_number, enabled, allowed_capabilities, created_by, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.22 X.22 Public table segment 505–519

Table	Observed columns	Primary key	RLS
whatsapp_contacts	id, company_id, profile_id, phone_e164, display_name, preferred_language, linked_at, last_seen_at, created_at, updated_at	id	Enabled
whatsapp_conversations	id, company_id, contact_id, status, last_message_at, context_expires_at, created_at, updated_at	id	Enabled
whatsapp_message_events	id, provider_message_id, company_id, phone_e164, event_type, status, payload_hash, error_category, received_at, processed_at	id	Enabled
whatsapp_messages	id, company_id, name, status, amount, notes, data, created_at, updated_at, provider_message_id, direction, phone_e164, body, message_type, provider_timestamp, conversation_id, contact_id, request_id, error_category, tool_name, ai_model	id	Enabled
workflow_marketplace	id, template_id, company_id, category, description, install_count, is_official, name, published_by_company_name, steps, trigger_type, created_at, updated_at	id	Enabled
workflows	id, company_id, condition, enabled, last_run, name, steps, trigger_type, created_at, updated_at	id	Enabled
workforce_approval_limits	id, company_id, target_profile_id, target_role_id, permission_id, currency, single_transaction_limit, daily_limit, requires_checker, status, effective_from, effective_to, approval_request_id, assigned_by, assigned_at, revoked_by, revoked_at, version, metadata	id	Enabled
workforce_data_scopes	id, company_id, target_profile_id, target_role_id, scope_type, scope_id, effect, status, effective_from, effective_to, approval_request_id, assigned_by, assigned_at, revoked_by, revoked_at, version, metadata	id	Enabled
workforce_member_roles	id, company_id, profile_id, employee_id, role_id, status, effective_from, effective_to, approval_request_id, assigned_by, assigned_at, revoked_by, revoked_at, version, metadata	id	Enabled
workforce_module_access	id, company_id, target_profile_id, target_role_id, module_id, permission_action, effect, status, effective_from, effective_to,	id	Enabled

Table	Observed columns	Primary key	RLS
	approval_request_id, assigned_by, assigned_at, revoked_by, revoked_at, version, metadata		
workforce_permissions	id, company_id, conflict_code, permission_a_id, permission_b_id, severity, resolution_policy, status, description, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
workforce_permissions	id, company_id, code, module_id, resource, permission_action, description, is_sensitive, status, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
workforce_role_permissions	id, company_id, role_id, permission_id, effect, status, effective_from, effective_to, approval_request_id, granted_by, granted_at, revoked_by, revoked_at, version, metadata	id	Enabled
workforce_roles	id, company_id, code, name, role_kind, description, hierarchy_level, is_assignable, status, created_by, created_at, updated_by, updated_at, version, metadata	id	Enabled
workspaces	id, company_id, channel_ref, department, description, members, name, created_at, updated_at	id	Enabled

The index is a documentation aid. It does not prescribe direct client writes; server boundaries, constraints, RPCs, and RLS remain the enforcement layers.

11.23 X.23 Canonical identity and subscription contracts

Table or surface	Purpose	Security boundary	Evidence
auth.users	Supabase Auth identity	Auth session/token	PublicAuthGateway and profile resolution
profiles	User/company/role context	Authenticated self-service and company scope	Profile Identity Center migration
companies	Organization identity	Tenant boundary	Core migrations and live inventory
company_memberships	Membership role/status	Company + user + role/status	Invitation and auth workflows
billing_plans	Plan catalog	Catalog/service/admin boundaries	Subscription migrations
tenant_subscriptions	Company subscription state	Server/database access snapshot	Subscription migrations
subscription_payments	Provider payment state	Server service RPC and idempotency	Subscription handlers
billing_access_snapshots	Authoritative access result	RPC, not a client cache	Subscription access adapter

12 Part XI — Integrations

Integration	Purpose	Current evidence	Boundary
Supabase	Auth, Postgres/REST, RPC, RLS	Live schema and migration evidence	Service keys server-side
HarakaPay	USSD collection/status/web-hook	Handlers present; readiness required	Order matching and verified settlement
TRA/VFD	Fiscal profile/receipt/retry/Z report	Router and provider boundary	TIN/VRN, branch, credentials
Storage/S3	Avatars, property files, exports	Storage helper and proxy	Scoped keys and validation
Email/SMS/WhatsApp	Notifications and outreach	Persistence/provider boundaries	No delivery claim without provider
AI model	Structured assistant proposals	Built-in model boundary	Context limits and approvals
Vercel	Deployment/runtime	Build and serverless contract	Environment variables

12.1 XI.1 Supabase

Purpose: Auth, Postgres/REST, RPC, RLS. Evidence: Live schema and migration evidence. Boundary: Service keys server-side. Kiswahili: Supabase hutumika kwa auth, postgres/rest, rpc, rls. Live schema and migration evidence. Service keys server-side.

12.2 XI.2 HarakaPay

Purpose: USSD collection/status/webhook. Evidence: Handlers present; readiness required. Boundary: Order matching and verified settlement. Kiswahili: HarakaPay hutumika kwa ussd collection/status/webhook. Handlers present; readiness required. Order matching and verified settlement.

12.3 XI.3 TRA/VFD

Purpose: Fiscal profile/receipt/retry/Z report. Evidence: Router and provider boundary. Boundary: TIN/VRN, branch, credentials. Kiswahili: TRA/VFD hutumika kwa fiscal profile/receipt/retry/z report. Router and provider boundary. TIN/VRN, branch, credentials.

12.4 XI.4 Storage/S3

Purpose: Avatars, property files, exports. Evidence: Storage helper and proxy. Boundary: Scoped keys and validation. Kiswahili: Storage/S3 hutumika kwa avatars, property files, exports. Storage helper and proxy. Scoped keys and validation.

12.5 XI.5 Email/SMS/WhatsApp

Purpose: Notifications and outreach. Evidence: Persistence/provider boundaries. Boundary: No delivery claim without provider. Kiswahili: Email/SMS/WhatsApp hutumika kwa notifications and outreach. Persistence/provider boundaries. No delivery claim without provider.

12.6 XI.6 AI model

Purpose: Structured assistant proposals. Evidence: Built-in model boundary. Boundary: Context limits and approvals. Kiswahili: AI model hutumika kwa structured assistant proposals. Built-in model boundary. Context limits and approvals.

12.7 XI.7 Vercel

Purpose: Deployment/runtime. Evidence: Build and serverless contract. Boundary: Environment variables.
Kiswahili: Vercel hutumika kwa deployment/runtime. Build and serverless contract. Environment variables.

13 Part XII — Troubleshooting

13.1 XII.1 Login fails

Possible cause: Missing public Supabase configuration, invalid credentials, expired recovery session, or provider rejection. Diagnosis: Check `/api/config/public`, browser network response, Auth status, and the exact user-facing error; do not log passwords or tokens. Solution: Restore approved configuration or repeat the supported recovery flow; keep error messages generic enough to avoid account enumeration.

13.1.1 Kiswahili

Sababu: Missing public Supabase configuration, invalid credentials, expired recovery session, or provider rejection. Utambuzi: Check `/api/config/public`, browser network response, Auth status, and the exact user-facing error; do not log passwords or tokens. Suluhisho: Restore approved configuration or repeat the supported recovery flow; keep error messages generic enough to avoid account enumeration.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.2 XII.2 Signup does not enter workspace

Possible cause: Email confirmation, profile provisioning, or company assignment is incomplete. Diagnosis: Inspect Auth confirmation state, verified profile, company_memberships, and server logs without exposing secrets. Solution: Complete verification and secure onboarding; do not create a duplicate users table.

13.2.1 Kiswahili

Sababu: Email confirmation, profile provisioning, or company assignment is incomplete. Utambuzi: Inspect Auth confirmation state, verified profile, company_memberships, and server logs without exposing secrets. Suluhisho: Complete verification and secure onboarding; do not create a duplicate users table.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.3 XII.3 RLS permission error

Possible cause: Session missing, wrong company scope, role not permitted, or policy/column contract mismatch. Diagnosis: Verify bearer extraction, profile company_id, role, table company_id, and policy command. Solution: Fail closed, fix the specific policy or caller contract, and add a regression test.

13.3.1 Kiswahili

Sababu: Session missing, wrong company scope, role not permitted, or policy/column contract mismatch.

Utambuzi: Verify bearer extraction, profile company_id, role, table company_id, and policy command.

Suluhisho: Fail closed, fix the specific policy or caller contract, and add a regression test.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.4 XII.4 Catalog returns 503

Possible cause: Server-side Supabase billing configuration missing or billing RPC unavailable. Diagnosis: Check /api/billing/catalog, server env presence, migration ledger, and billing_public_plan_catalog readiness. Solution: Restore server configuration or apply the intended migration in a controlled window; never send the service key to the browser.

13.4.1 Kiswahili

Sababu: Server-side Supabase billing configuration missing or billing RPC unavailable. Utambuzi: Check /api/billing/catalog, server env presence, migration ledger, and billing_public_plan_catalog readiness. Suluhisho: Restore server configuration or apply the intended migration in a controlled window; never send the service key to the browser.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.5 XII.5 Free activation rejected

Possible cause: No verified manager role, missing company profile, or FREE_15 catalog/routine not ready.

Diagnosis: Confirm authenticated profile, company scope, active catalog row, and billing_start_free_plan routine. Solution: Correct prerequisites and retry idempotently; never treat a failed call as active.

13.5.1 Kiswahili

Sababu: No verified manager role, missing company profile, or FREE_15 catalog/routine not ready.

Utambuzi: Confirm authenticated profile, company scope, active catalog row, and billing_start_free_plan routine. Suluhisho: Correct prerequisites and retry idempotently; never treat a failed call as active.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.6 XII.6 Paid payment remains pending

Possible cause: Provider has not verified the order, webhook/status polling has not settled, or provider configuration is unavailable. Diagnosis: Check payment record, provider order ID, server status call, and webhook receipt. Solution: Keep access inactive until verified; retry only through idempotent server/database flow.

13.6.1 Kiswahili

Sababu: Provider has not verified the order, webhook/status polling has not settled, or provider configuration is unavailable. Utambuzi: Check payment record, provider order ID, server status call, and webhook receipt. Suluhisho: Keep access inactive until verified; retry only through idempotent server/database flow.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.7 XII.7 Data appears missing

Possible cause: Wrong company scope, empty state, failed query, or UI cache not refreshed. Diagnosis: Compare authenticated company scope, raw server response, loading/error state, and table RLS. Solution: Show unavailable/empty/error explicitly; never replace a failed read with fake seed data in production.

13.7.1 Kiswahili

Sababu: Wrong company scope, empty state, failed query, or UI cache not refreshed. Utambuzi: Compare authenticated company scope, raw server response, loading/error state, and table RLS. Suluhisho: Show unavailable/empty/error explicitly; never replace a failed read with fake seed data in production.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.8 XII.8 File upload fails

Possible cause: Storage configuration, size/type validation, path scope, or signed URL problem. Diagnosis: Inspect server validation and storage response; verify only metadata after confirmed upload. Solution: Keep bytes in storage, metadata in database, and show a retryable failure.

13.8.1 Kiswahili

Sababu: Storage configuration, size/type validation, path scope, or signed URL problem. Utambuzi: Inspect server validation and storage response; verify only metadata after confirmed upload. Suluhisho: Keep bytes in storage, metadata in database, and show a retryable failure.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.9 XII.9 TRA/VFD submission blocked

Possible cause: Profile not configured, provider readiness false, missing credentials, or duplicate idempotency key. Diagnosis: Read fiscal profile, readiness, receipt/retry queue status, and audit event. Solution: Configure approved provider or retain a truthful blocked state; do not call a local receipt fiscalized.

13.9.1 Kiswahili

Sababu: Profile not configured, provider readiness false, missing credentials, or duplicate idempotency key. Utambuzi: Read fiscal profile, readiness, receipt/retry queue status, and audit event. Suluhisho: Configure approved provider or retain a truthful blocked state; do not call a local receipt fiscalized.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.10 XII.10 AI response cannot be used as an action

Possible cause: AI returned a proposal, not an approved mutation, or the request exceeded safe targets.

Diagnosis: Inspect structured response, proposal operation, approval state, and source context. Solution:

Require independent role approval and confirmed module write; explain missing evidence.

13.10.1 Kiswahili

Sababu: AI returned a proposal, not an approved mutation, or the request exceeded safe targets. Utambuzi:

Inspect structured response, proposal operation, approval state, and source context. Suluhisho: Require

independent role approval and confirmed module write; explain missing evidence.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.11 XII.11 Deployment build fails

Possible cause: Schema verification credentials, dependency resolution, TypeScript, or route/runtime incompatibility. Diagnosis: Run pnpm check, pnpm test, verify:supabase-schema with approved env, and inspect the exact build diagnostic. Solution: Fix the specific contract; do not bypass the schema guard or suppress audit failures.

13.11.1 Kiswahili

Sababu: Schema verification credentials, dependency resolution, TypeScript, or route/runtime incompatibility. Utambuzi: Run pnpm check, pnpm test, verify:supabase-schema with approved env, and inspect the exact build diagnostic. Suluhisho: Fix the specific contract; do not bypass the schema guard or suppress audit failures.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

13.12 XII.12 Webhook is not reconciled

Possible cause: Unknown provider order, failed status verification, invalid payload, or endpoint configuration. Diagnosis: Confirm order lookup, provider status request, response order match, and service RPC result. Solution: Return a safe error, preserve audit state, and retry through a controlled path.

13.12.1 Kiswahili

Sababu: Unknown provider order, failed status verification, invalid payload, or endpoint configuration.

Utambuzi: Confirm order lookup, provider status request, response order match, and service RPC result.

Suluhisho: Return a safe error, preserve audit state, and retry through a controlled path.

Step	Instruction	Evidence	
1	Preserve the visible error and input	Browser/UI state	2
Check authenticated company and role	Verified profile	3	Check server/database response
API/RPC status	4	Retry only through idempotent path	Request key or provider order
5	Record outcome for support/audit	Audit result	

14 Part XIII — Operations

14.1 XIII.1 Release readiness

Confirm source migration, tests, TypeScript, build, browser journeys, live schema, RLS, provider configuration, and rollback evidence.

14.1.1 Kiswahili

Thibitisha migration, tests, TypeScript, build, browser journeys, schema, RLS, provider na rollback.

14.2 XIII.2 Backup and recovery

Distinguish database reachability from managed PITR or snapshot configuration. Do not report backup controls that were not checked.

14.2.1 Kiswahili

Tenganisha database reachability na PITR/snapshot. Usiripoti backup ambayo haikuthibitishwa.

14.3 XIII.3 Monitoring

Watch API status, scheduled handlers, webhook deliveries, provider errors, failed mutations, security advisors, and schema drift.

14.3.1 Kiswahili

Fuatilia API, scheduled handlers, webhooks, provider errors, failed writes, security advisors na schema drift.

14.4 XIII.4 Support handover

Keep the repository, migration ledger, environment checklist, role matrix, error reference, and incident notes together.

14.4.1 Kiswahili

Hifadhi repository, migration ledger, env checklist, role matrix, error reference na incident notes pamoja.

14.5 XIII.5 Data governance

Define retention, privacy, least privilege, export, deletion, and incident escalation by business and regulatory context.

14.5.1 Kiswahili

Weka retention, privacy, least privilege, export, deletion na escalation kulingana na biashara na compliance.

14.6 XIII.6 Change control

Every schema, role, provider, and entitlement change needs a source commit, migration/version record, test evidence, and updated status.

14.6.1 Kiswahili

Kila mabadiliko ya schema, role, provider na entitlement yanahitaji commit, migration/version, tests na status mpya.

15 Part XIV — Future Roadmap

15.1 XIV.1 Atomic legacy invoice payment RPC

Replace the separate payment insert and invoice update with a reviewed atomic, idempotent database boundary.

15.1.1 Kiswahili

Hatua hii inapaswa kufanywa baada ya uthibitisho wa source, data, ruhusa na deployment. Lengo si kuongeza madai ya bidhaa, bali kuimarisha ushahidi wa kazi iliyopo: Replace the separate payment insert and invoice update with a reviewed atomic, idempotent database boundary.

15.2 XIV.2 Broader authenticated CRUD journeys

Expand role-by-role Playwright and controlled Supabase staging coverage across legacy modules.

15.2.1 Kiswahili

Hatua hii inapaswa kufanywa baada ya uthibitisho wa source, data, ruhusa na deployment. Lengo si kuongeza madai ya bidhaa, bali kuimarisha ushahidi wa kazi iliyopo: Expand role-by-role Playwright and controlled Supabase staging coverage across legacy modules.

15.3 XIV.3 Signature-specific SECURITY DEFINER hardening

Review each exposed signature, preserve intentional endpoints, and revoke or relocate internal helpers.

15.3.1 Kiswahili

Hatua hii inapaswa kufanywa baada ya uthibitisho wa source, data, ruhusa na deployment. Lengo si kuongeza madai ya bidhaa, bali kuimarisha ushahidi wa kazi iliyopo: Review each exposed signature, preserve intentional endpoints, and revoke or relocate internal helpers.

15.4 XIV.4 RLS policy consolidation

Reduce overlapping permissive policies after proving equivalent tenant semantics.

15.4.1 Kiswahili

Hatua hii inapaswa kufanywa baada ya uthibitisho wa source, data, ruhusa na deployment. Lengo si kuongeza madai ya bidhaa, bali kuimarisha ushahidi wa kazi iliyopo: Reduce overlapping permissive policies after proving equivalent tenant semantics.

15.5 XIV.5 Provider readiness

Complete approved HarakaPay, TRA/VFD, storage, email/SMS/WhatsApp, maps, and AI configuration and acceptance.

15.5.1 Kiswahili

Hatua hii inapaswa kufanywa baada ya uthibitisho wa source, data, ruhusa na deployment. Lengo si kuongeza madai ya bidhaa, bali kuimarisha ushahidi wa kazi iliyopo: Complete approved HarakaPay, TRA/VFD, storage, email/SMS/WhatsApp, maps, and AI configuration and acceptance.

15.6 XIV.6 Dashboard decomposition

Extract safe, high-risk boundaries and code-split the large dashboard only after persistence and live blockers are addressed.

15.6.1 Kiswahili

Hatua hii inapaswa kufanywa baada ya uthibitisho wa source, data, ruhusa na deployment. Lengo si kuongeza madai ya bidhaa, bali kuimarisha ushahidi wa kazi iliyopo: Extract safe, high-risk boundaries and code-split the large dashboard only after persistence and live blockers are addressed.

16 Part XV — Swahili Reference

This section is a second, complete Tanzanian Swahili reference. Technical English terms remain in parentheses where they reduce ambiguity.

16.1 XV.1 Ukurasa wa Umma wa Chapa na Masoko

Ukurasa wa mwanzo unaeleza thamani ya Smart Manager, unaunganisha mgeni na kuingia kwenye mfumo, na unaonyesha kwa uaminifu kwamba mwonekano wa masoko si rekodi ya biashara ya kampuni. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Ukurasa wa Umma wa Chapa na Masoko ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha companies. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Discover product → review capabilities → choose launch or authentication → enter the secure workspace. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Marketing delivery and conversion analytics are not the same as a signed-in company workflow; external campaign channels require configuration. Ukurasa wa mwanzo unaeleza thamani ya Smart Manager, unaunganisha mgeni na kuingia kwenye mfumo, na unaonyesha kwa uaminifu kwamba mwonekano wa masoko si rekodi ya biashara ya kampuni. Hali ya sasa: IMPLEMENTED. Mipaka: Marketing delivery and conversion analytics are not the same as a signed-in company workflow; external campaign channels require configuration.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Public visitor; prospective customer	Verified profile + role	Controlled acceptance
Workflow	Discover product → review capabilities → choose launch or authentication → enter the secure workspace.	Server-confirmed result	Role-specific test
Database	companies	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.2 XV.2 Uthibitishaji na Uanzishaji Salama

Gateway ya umma inasaidia kuingia kwa nenosiri, kurejesha nenosiri, kuthibitisha barua pepe, OAuth iliyokubaliwa, passkeys na uhifadhi wa kikao kabla ya kuingia kwenye nafasi ya kampuni. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Uthibitishaji na Uanzishaji Salama ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha companies, company\u{005F}memberships, profiles. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Load public Supabase configuration → authenticate or recover → persist token/session → resolve verified profile → enter /app. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Actual OAuth delivery, email delivery, and passkey availability remain dependent on Supabase/provider configuration and browser capability. Gateway ya umma inasaidia kuingia kwa nenosiri, kurejesha nenosiri, kuthibitisha barua pepe, OAuth iliyokubaliwa, passkeys na uhifadhi wa kikao kabla ya kuingia kwenye nafasi ya kampuni. Hali ya sasa: IMPLEMENTED. Mipaka: Actual OAuth delivery, email delivery, and passkey availability remain dependent on Supabase/provider configuration and browser capability.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Unauthenticated visitor; verified user; company owner	Verified profile + role	Controlled acceptance
Workflow	Load public Supabase configuration → authenticate or recover → persist token/session → resolve verified profile → enter /app.	Server-confirmed result	Role-specific test
Database	companies, company_memberships, profiles	RF \$/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.3 XV.3 Ganda Kuu la Programu na Urambazaji

Ganda la programu linaunganisha routes, uthibitishaji, lugha, mandhari, mapendeleo ya dashibodi, lazy loading na urambazaji unaozingatia jukumu pamoja na entitlement. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Ganda Kuu la Programu na Urambazaji ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `company\{005F}modules`, `workspaces`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Route match → auth decision → lazy workspace load → role/entitlement-aware navigation → module action. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. The dashboard remains a large legacy single-file boundary and produces a non-fatal bundle-size warning; safe decomposition is a future engineering workstream. Ganda la programu linaunganisha routes, uthibitishaji, lugha, mandhari, mapendeleo ya dashibodi, lazy loading na urambazaji unaozingatia jukumu pamoja na entitlement. Hali ya sasa: IMPLEMENTED. Mipaka: The dashboard remains a large legacy single-file boundary and produces a non-fatal bundle-size warning; safe decomposition is a future engineering workstream.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	All authenticated users, filtered by role and entitlement	Verified profile + role	Controlled acceptance
Workflow	Route match → auth decision → lazy workspace load → role/entitlement-aware navigation → module action.	Server-confirmed result	Role-specific test
Database	company_modules, workspaces	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.4 XV.4 Kituo cha Utambulisho wa Wasifu

Kituo cha wasifu hutenganisha utambulisho binafsi, kazi, usalama, mapendeleo, arifa na shughuli. Mtumiaji hubadilisha tu sehemu anazoruhusiwa na server ndiyo mamlaka ya mwisho. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Kituo cha Utambulisho wa Wasifu ni kufanya kazi inayorudiwa ione kane na idhibitwe. Chagamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha profiles, user\{005F}table\{005F}preferences. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Open profile → read verified profile → update permitted self-service fields → upload or remove avatar through controlled storage → refresh confirmed state. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Device/session listing and workspace switching are explicitly unavailable unless supported by a verified backend contract. Kituo cha wasifu hutenganisha utambulisho binafsi, kazi, usalama, mapendeleo, arifa na shughuli. Mtumiaji hubadilisha tu sehemu anazoruhusiwa na server ndiyo mamlaka ya mwisho. Hali ya sasa: IMPLEMENTED. Mipaka: Device/session listing and workspace switching are explicitly unavailable unless supported by a verified backend contract.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Authenticated user; workspace administrator for linked context	Verified profile + role	Controlled acceptance
Workflow	Open profile → read verified profile → update permitted self-service fields → upload or remove avatar through controlled storage → refresh confirmed state.	Server-confirmed result	Role-specific test
Database	profiles, user_table_preferences	File/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.5 XV.5 Dashibodi ya Uongozi

Dashibodi ya uongozi hukusanya ishara za uendeshaji kutoka moduli zilizounganishwa. Ni ya mapitio na kupanga kipaumbele, si chanzo tofauti cha ukweli wa fedha. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Dashibodi ya Uongozi ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `custom_kpis`, `financial_benchmarks`, `scheduled_reports`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Select company context → load confirmed metrics → inspect trends/exceptions → navigate to source module → approve or act through the protected boundary. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. A dashboard is only as complete as the connected records, schedules, and integrations that feed it; empty or unavailable states must remain visible. Dashibodi ya uongozi hukusanya ishara za uendeshaji kutoka moduli zilizounganishwa. Ni ya mapitio na kupanga kipaumbele, si chanzo tofauti cha ukweli wa fedha. Hali ya sasa: IMPLEMENTED. Mipaka: A dashboard is only as complete as the connected records, schedules, and integrations that feed it; empty or unavailable states must remain visible.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	CEO; COO; CFO; organization owner; manager roles	Verified profile + role	Controlled acceptance
Workflow	Select company context → load confirmed metrics → inspect trends/exceptions → navigate to source module → approve or act through the protected boundary.	Server-confirmed result	Role-specific test
Database	custom_kpis, financial_scheduled_reports	Per-company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.6 XV.6 CRM na Mfuatano wa Wateja

CRM huweka pamoja mawasiliano, leads, mwingiliano na mrejesho wa wateja. Lengo ni kuunganisha mazungumzo ya kibiashara na mauzo pamoja na huduma kwa mteja. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la CRM na Mfuatano wa Wateja ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Chagamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazonekana kwa module hii zinajumuisha `crm\{005F}contacts`, `crm\{005F}interactions`, `crm\{005F}leads`, `customer\{005F}feedback`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Create or import lead → qualify → record interaction → convert or follow up → connect to quotation/invoice/support. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. The repository contains strong persistence and truthfulness contracts, but broader authenticated CRUD walkthroughs remain required for every legacy screen. CRM huweka pamoja mawasiliano, leads, mwingiliano na mrejesho wa wateja. Lengo ni kuunganisha mazungumzo ya kibiashara na mauzo pamoja na huduma kwa mteja. Hali ya sasa: TESTING. Mipaka: The repository contains strong persistence and truthfulness contracts, but broader authenticated CRUD walkthroughs remain required for every legacy screen.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Sales Manager; sales user; account owner; support user	Verified profile + role	Controlled acceptance
Workflow	Create or import lead → qualify → record interaction → convert or follow up → connect to quotation/invoice/support.	Server-confirmed result	Role-specific test
Database	<code>crm_contacts</code> , <code>crm_interactions</code> , <code>crm_leads</code> , <code>customer_feedback</code>	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.7 XV.7 Mauzo na Utozaji

Sehemu ya mauzo inahusu quotations, ankara, subscriptions, malipo, returns na mwingiliano wa wateja. Bei na hali ya mwisho vinatoka server na database, si thamani ya browser. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Mauzo na Utozaji ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `ecommerce\orders`, `ecommerce\products`, `sales\invoice\items`, `sales\invoices`, `sales\order\items`, `sales\order\return\items`, `sales\order\returns`, `sales\orders`, `sales\payments`, `sales\quotation\items`, `sales\quotations`, `sales\subscriptions`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Customer → quotation/order → invoice → payment or collection → receipt/ledger/report → follow-up. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Legacy invoice payment posting still requires an atomic database RPC and durable idempotency constraint before concurrent-safe payment posting can be claimed. Sehemu ya mauzo inahusu quotations, ankara, subscriptions, malipo, returns na mwingiliano wa wateja. Bei na hali ya mwisho vinatoka server na database, si thamani ya browser. Hali ya sasa: TESTING. Mipaka: Legacy invoice payment posting still requires an atomic database RPC and durable idempotency constraint before concurrent-safe payment posting can be claimed.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Sales Manager; finance manager; sales user; billing manager	Verified profile + role	Controlled acceptance
Workflow	Customer → quotation/order → invoice → payment or collection → receipt/ledger/report → follow-up.	Server-confirmed result	Role-specific test
Database	<code>ecommerce_orders</code> , <code>ecommerce_products</code> , <code>sales_invoice_items</code> , <code>sales_invoices</code> , <code>sales_order_items</code> , <code>sales_order_return_items</code> , <code>sales_order_returns</code> , <code>sales_orders</code> , <code>sales_payments</code> , <code>sales_quotation_items</code> ,	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	sales_quotations, sales_subscriptions		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.8 XV.8 Mauzo ya Moja kwa Moja (POS)

POS inasimamia register, shift, bidhaa, malipo, returns, kodi, promotions, loyalty, reconciliation na audit. Mstari wa usalama ni kuthibitisha muamala kabla ya kuuita umehifadhiwa. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Mauzo ya Moja kwa Moja (POS) ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `pos\{005F}cash\{005F}movements`, `pos\{005F}discount\{005F}rules`, `pos\{005F}loyalty\{005F}ledger`, `pos\{005F}loyalty\{005F}members`, `pos\{005F}loyalty\{005F}programs`, `pos\{005F}loyalty\{005F}redemptions`, `pos\{005F}loyalty\{005F}rewards`, `pos\{005F}promotion\{005F}items`, `pos\{005F}promotions`, `pos\{005F}registers`, `pos\{005F}return\{005F}commits`, `pos\{005F}return\{005F}headers`, `pos\{005F}return\{005F}items`, `pos\{005F}return\{005F}lines`, `pos\{005F}returns`, `pos\{005F}sale\{005F}adjustments`, `pos\{005F}sale\{005F}headers`, `pos\{005F}sale\{005F}lines`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Open register/shift → scan or select item → validate price and stock → collect tender → confirm sale → receipt → reconcile or return. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Controlled staging credentials and real device/provider behavior remain operational acceptance requirements; client pending queues are not treated as durable server saves. POS inasimamia register, shift, bidhaa, malipo, returns, kodi, promotions, loyalty, reconciliation na audit. Mstari wa usalama ni kuthibitisha muamala kabla ya kuuita umehifadhiwa. Hali ya sasa: PASSED. Mipaka: Controlled staging credentials and real device/provider behavior remain operational acceptance requirements; client pending queues are not treated as durable server saves.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Cashier; finance manager; warehouse manager; administrator	Verified profile + role	Controlled acceptance
Workflow	Open register/shift → scan or select item → validate price and stock → collect tender → confirm sale → receipt → reconcile or return.	Server-confirmed result	Role-specific test
Database	<code>pos_cash_movements</code> , <code>pos_discount_rules</code> , <code>pos_loyalty_ledger</code> , <code>pos_loyalty_members</code> , <code>pos_loyalty_programs</code> , <code>pos_loyalty_redemptions</code> , <code>pos_loyalty_rewards</code> , <code>pos_promotion_items</code> ,	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	pos_promotions, pos_registers, pos_return_commits, pos_return_headers		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.9 XV.9 Usimamizi wa Hesabu na Maghala

Hesabu na maghala huunganisha items, batches, movement, suppliers, warehouses, transfers na audits. Kila mabadiliko ya quantity inapaswa kuonekana kama movement yenye ushahidi. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Usimamizi wa Hesabu na Maghala ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `inventory\batches`, `inventory\items`, `inventory\stock`, `inventory\movements`, `inventory\suppliers`, `inventory\transfers`, `inventory\warehouses`, `stock\audit\items`, `stock\audits`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Create item → receive or adjust stock → transfer or sell → record movement → inspect balance/audit → replenish. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Full authenticated import/export and concurrency walkthroughs remain part of the broader acceptance backlog. Hesabu na maghala huunganisha items, batches, movement, suppliers, warehouses, transfers na audits. Kila mabadiliko ya quantity inapaswa kuonekana kama movement yenye ushahidi. Hali ya sasa: TESTING. Mipaka: Full authenticated import/export and concurrency walkthroughs remain part of the broader acceptance backlog.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Warehouse Manager; procurement officer; sales user; administrator	Verified profile + role	Controlled acceptance
Workflow	Create item → receive or adjust stock → transfer or sell → record movement → inspect balance/audit → replenish.	Server-confirmed result	Role-specific test
Database	<code>inventory_batches</code> , <code>inventory_items</code> , <code>inventory_stock</code> , <code>inventory_suppliers</code> , <code>inventory_transfers</code> , <code>inventory_warehouses</code> , <code>stock_audit_items</code> , <code>stock_audits</code>	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.10 XV.10 Ununuzi na Wasambazaji

Ununuzi unaunganisha wasambazaji, purchase orders, contracts, mapokezi, stock, approvals na madeni. Mfumo unapaswa kutenganisha kilichoombwa, kilichoidhinishwa, kilichopokelewa na kinachodaiwa. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Ununuzi na Wasambazaji ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `procurement\{contracts, procurement\{purchase\{orders, purchase\{order\{items`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Supplier → request/order → approval → receipt → stock update → payable/finance evidence → report. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Downstream integrations and external supplier channels are configuration-dependent; recorded integration evidence must not be replaced with fabricated acknowledgements. Ununuzi unaunganisha wasambazaji, purchase orders, contracts, mapokezi, stock, approvals na madeni. Mfumo unapaswa kutenganisha kilichoombwa, kilichoidhinishwa, kilichopokelewa na kinachodaiwa. Hali ya sasa: TESTING. Mipaka: Downstream integrations and external supplier channels are configuration-dependent; recorded integration evidence must not be replaced with fabricated acknowledgements.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Procurement Officer; warehouse manager; finance manager; administrator	Verified profile + role	Controlled acceptance
Workflow	Supplier → request/order → approval → receipt → stock update → payable/finance evidence → report.	Server-confirmed result	Role-specific test
Database	<code>procurement_contracts</code> , <code>procurement_purchase_order_items</code>	RLS/company suppliers,	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.11 XV.11 Fedha na Uhasibu

Fedha na uhasibu vina accounts, journal, mapato na matumizi, bajeti, kodi/VAT, kufunga vipindi, reconciliation ya benki, assets na ripoti. Data halisi ndiyo msingi wa ripoti. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Fedha na Uhasibu ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazonekana kwa module hii zinajumuisha `expense\budgets`, `fin\accounts`, `fin\approval\requests`, `fin\idempotency\keys`, `fin\journal\batches`, `fin\journal\lines`, `fin\periods`, `fin\posting\links`, `fin\reconciliation\batches`, `fin\reconciliation\items`, `finance\assets`, `finance\expenses`, `journal\entries`, `period\closes`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Capture transaction → validate company scope → post or approve → reconcile → close period → report. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Production reporting depends on intended schema alignment and real company data; tax filing itself is not implied by a local summary. Fedha na uhasibu vina accounts, journal, mapato na matumizi, bajeti, kodi/VAT, kufunga vipindi, reconciliation ya benki, assets na ripoti. Data halisi ndiyo msingi wa ripoti. Hali ya sasa: TESTING. Mipaka: Production reporting depends on intended schema alignment and real company data; tax filing itself is not implied by a local summary.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	CFO; Finance Manager; accountant; or organization owner	Verified profile + role	Controlled acceptance
Workflow	Capture transaction → validate company scope → post or approve → reconcile → close period → report.	Server-confirmed result	Role-specific test
Database	<code>expense_budgets</code> , <code>fin_accounts</code> , <code>fin_approval_requests</code> , <code>fin_idempotency_keys</code> , <code>fin_journal_batches</code> , <code>fin_journal_lines</code> , <code>fin_periods</code> , <code>fin_posting_links</code> , <code>fin_reconciliation_batches</code> , <code>fin_reconciliation_items</code> , <code>finance_assets</code> , <code>finance_expenses</code>	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.12 XV.12 Ripoti na Ripoti Zilizopangwa

Ripoti huonyesha mwenendo, exports, schedules na hali ya delivery. Ripoti inaonyesha data iliyorekodiwa; haimaanishi moja kwa moja kwamba email ya nje imefika. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Ripoti na Ripoti Zilizopangwa ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `scheduled_reports`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Choose report → apply company/time scope → compute from source records → export or schedule → inspect delivery history. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Email delivery remains provider-configured; CSV is the dependency-safe export path when spreadsheet dependencies are intentionally absent. Ripoti huonyesha mwenendo, exports, schedules na hali ya delivery. Ripoti inaonyesha data iliyorekodiwa; haimaanishi moja kwa moja kwamba email ya nje imefika. Hali ya sasa: TESTING. Mipaka: Email delivery remains provider-configured; CSV is the dependency-safe export path when spreadsheet dependencies are intentionally absent.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Executive roles; manager; administrator	Verified profile + role	Controlled acceptance
Workflow	Choose report → apply company/time scope → compute from source records → export or schedule → inspect delivery history.	Server-confirmed result	Role-specific test
Database	<code>scheduled_reports</code>	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.13 XV.13 Rasilimali Watu na Mishahara

HR inahusu wafanyakazi, mahudhurio, benefits, leave, malengo, approvals, payroll, announcements na invitations salama. Data ya mfanyakazi inapaswa kubaki ndani ya kampuni yake. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Rasilimali Watu na Mishahara ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `hr\announcement\reads`, `hr\announcements`, `hr\approval\requests`, `hr\approval\steps`, `hr\attendance`, `hr\benefit\enrollments`, `hr\benefit\plans`, `hr\benefits`, `hr\candidates`, `hr\duties`, `hr\employee\documents`, `hr\employees`, `hr\expense\claims`, `hr\goal\updates`, `hr\goals`, `hr\holidays`, `hr\invite\codes`, `hr\leave\balances`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Invite employee → accept securely → assign role → record attendance/leave → approve → calculate payroll → report. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Payroll and employee records need a full authenticated tenant-data walkthrough; external salary/payment rails are outside the repository's verified contract. HR inahusu wafanyakazi, mahudhurio, benefits, leave, malengo, approvals, payroll, announcements na invitations salama. Data ya mfanyakazi inapaswa kubaki ndani ya kampuni yake. Hali ya sasa: TESTING. Mipaka: Payroll and employee records need a full authenticated tenant-data walkthrough; external salary/payment rails are outside the repository's verified contract.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	HR Manager; manager; employee; payroll approver; administrator	Verified profile + role	Controlled acceptance
Workflow	Invite employee → accept securely → assign role → record attendance/leave → approve → calculate payroll → report.	Server-confirmed result	Role-specific test
Database	<code>hr_announcement_reads</code> , <code>hr_announcements</code> , <code>hr_approval_requests</code> , <code>hr_approval_steps</code> , <code>hr_attendance</code> , <code>hr_benefit_enrollments</code> , <code>hr_benefit_plans</code> , <code>hr_benefits</code> , <code>hr_candidates</code> ,	URL/company	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	hr_duties, hr_employee_documents, hr_employees		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.14 XV.14 Uzalishaji na Maagizo ya Kazi

Repository ina mipaka ya manufacturing na work orders, lakini kitabu kinatenganisha uwepo wa surface na uthibitisho wa workflow kamili wa uzalishaji. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Uzalishaji na Maagizo ya Kazi ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha manufacturing\bom\components, manufacturing\boms, manufacturing\machines, manufacturing\maintenance, manufacturing\qc\inspections, manufacturing\work\orders. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Define work order → allocate or consume material → update status → record output → review cost and exceptions. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. The visible surface is broader than the independently verified end-to-end contract for every manufacturing scenario. Repository ina mipaka ya manufacturing na work orders, lakini kitabu kinatenganisha uwepo wa surface na uthibitisho wa workflow kamili wa uzalishaji. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: The visible surface is broader than the independently verified end-to-end contract for every manufacturing scenario.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Production manager; warehouse manager; project manager	Verified profile + role	Controlled acceptance
Workflow	Define work order → allocate or consume material → update status → record output → review cost and exceptions.	Server-confirmed result	Role-specific test
Database	manufacturing_bom_components, manufacturing_boms, manufacturing_machines, manufacturing_maintenance, manufacturing_qc_inspections, manufacturing_work_orders	Repository	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.15 XV.15 Mnyororo wa Ugavi na Magari

Ugavi na fleet huunganisha shipments, magari, trips, madereva, mafuta, matengenezo, incidents, routes, telematics na alerts. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Mnyororo wa Ugavi na Magari ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `fleet\alerts`, `fleet\audit\events`, `fleet\driver\assignments`, `fleet\drivers`, `fleet\fuel\cards`, `fleet\fuel\transactions`, `fleet\incidents`, `fleet\maintenance\jobs`, `fleet\maintenance\plans`, `fleet\routes`, `fleet\service\records`, `fleet\spare\parts`, `fleet\telematics\events`, `fleet\trips`, `fleet\tyres`, `fleet\vehicle\categories`, `fleet\vehicle\documents`, `fleet\vehicles`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Plan route/trip → assign driver/vehicle → record movement or fuel → ingest event → inspect alert → close maintenance or incident. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Telematics and external vehicle providers require approved webhook configuration and secrets; no external movement is claimed from a local screen alone. Ugavi na fleet huunganisha shipments, magari, trips, madereva, mafuta, matengenezo, incidents, routes, telematics na alerts. Hali ya sasa: TESTING. Mipaka: Telematics and external vehicle providers require approved webhook configuration and secrets; no external movement is claimed from a local screen alone.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Fleet manager; logistics user; administrator	Verified profile + role	Controlled acceptance
Workflow	Plan route/trip → assign driver/vehicle → record movement or fuel → ingest event → inspect alert → close maintenance or incident.	Server-confirmed result	Role-specific test
Database	<code>fleet_alerts</code> , <code>fleet_audit_events</code> , <code>fleet_driver_assignments</code> , <code>fleet_drivers</code> , <code>fleet_fuel_cards</code> , <code>fleet_fuel_transactions</code> , <code>fleet_incidents</code> , <code>fleet_maintenance_jobs</code> , <code>fleet_maintenance_plans</code> , <code>fleet_routes</code> ,	RLS/company	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	fleet_service_records, fleet_spare_parts		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.16 XV.16 Kampeni za Masoko

Masoko huandaa segments, templates, ujumbe na analytics. Ujumbe wa draft unatenganishwa na dispatch halisi ya huduma ya nje. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Kampeni za Masoko ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `crm\contacts`, `crm\interactions`, `crm\leads`, `emails`, `marketing\campaigns`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Define audience → draft message → validate template/link → obtain approval → dispatch through configured channel → measure response. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. External email, SMS, WhatsApp, and campaign providers remain configuration-dependent; the UI must not claim dispatch when only a draft exists. Masoko huandaa segments, templates, ujumbe na analytics. Ujumbe wa draft unatenganishwa na dispatch halisi ya huduma ya nje. Hali ya sasa: TESTING. Mipaka: External email, SMS, WhatsApp, and campaign providers remain configuration-dependent; the UI must not claim dispatch when only a draft exists.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	CMO; marketing manager; sales manager	Verified profile + role	Controlled acceptance
Workflow	Define audience → draft message → validate template/link → obtain approval → dispatch through configured channel → measure response.	Server-confirmed result	Role-specific test
Database	<code>crm_contacts</code> , <code>crm_interactions</code> , <code>crm_leads</code> , <code>emails</code> , <code>marketing_campaigns</code>	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.17 XV.17 Duka la Kielektroniki

Mfumo una surfaces za bidhaa na oda za e-commerce, lakini haujadai kuwa Shopify au mtoa huduma mwingine amewezeshwa bila ushahidi wa configuration. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Duka la Kielektroniki ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `ecommerce\orders`, `ecommerce\products`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Publish catalog → receive order → validate customer/order → connect fulfilment and finance → report. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. External commerce provider credentials and production storefront activation are not verified by this book. Mfumo una surfaces za bidhaa na oda za e-commerce, lakini haujadai kuwa Shopify au mtoa huduma mwingine amewezeshwa bila ushahidi wa configuration. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: External commerce provider credentials and production storefront activation are not verified by this book.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Store manager; sales manager; customer	Verified profile + role	Controlled acceptance
Workflow	Publish catalog → receive order → validate customer/order → connect fulfilment and finance → report.	Server-confirmed result	Role-specific test
Database	<code>ecommerce_orders</code> , <code>ecommerce_products</code>	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.18 XV.18 Nyaraka na Mafaili Salama

Nyaraka huhifadhi metadata na reference ya storage, si file bytes ndani ya business table. Upload, kusaini na download vinahitaji mipaka ya ruhusa. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Nyaraka na Mafaili Salama ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha approval\{005F}signatures, digital\{005F}signatures, documents, signatures. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Create metadata → upload through storage boundary → verify reference → share or sign under role control → download/export evidence. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. S3/storage configuration is required for live file bytes; metadata persistence alone is not evidence that a file was uploaded successfully. Nyaraka huhifadhi metadata na reference ya storage, si file bytes ndani ya business table. Upload, kusaini na download vinahitaji mipaka ya ruhusa. Hali ya sasa: TESTING. Mipaka: S3/storage configuration is required for live file bytes; metadata persistence alone is not evidence that a file was uploaded successfully.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	All authenticated users within permitted company scope; approver	Verified profile + role	Controlled acceptance
Workflow	Create metadata → upload through storage boundary → verify reference → share or sign under role control → download/export evidence.	Server-confirmed result	Role-specific test
Database	approval_signatures, digital_signatures, documents, signatures	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.19 XV.19 Miradi na Usimamizi wa Kazi

Miradi huunganisha tasks, milestones, expenses na maendeleo. Uthibitisho wa kina wa kila operation unahitaji data halisi ya kampuni na majukumu sahihi. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Miradi na Usimamizi wa Kazi ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `project\expenses`, `project\milestones`, `project\tasks`, `projects`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Create project → add milestone/task → assign owner → update progress → record expense → review completion. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. The repository contains the persistence boundary, but complete project operations still depend on authenticated data and role-specific acceptance. Miradi huunganisha tasks, milestones, expenses na maendeleo. Uthibitisho wa kina wa kila operation unahitaji data halisi ya kampuni na majukumu sahihi. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: The repository contains the persistence boundary, but complete project operations still depend on authenticated data and role-specific acceptance.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Project Manager; team member; executive	Verified profile + role	Controlled acceptance
Workflow	Create project → add milestone/task → assign owner → update progress → record expense → review completion.	Server-confirmed result	Role-specific test
Database	project_expenses, project_milestones, project_tasks, projects	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.20 XV.20 Huduma kwa Wateja na Helpdesk

Helpdesk ina tickets, timelines, internal notes, chat/call evidence, SLA, assignment, resolution na metrics. Dispatch ya nje inategemea channel iliyosanidiwa. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Huduma kwa Wateja na Helpdesk ni kufanya kazi inayorudiwa ionekane na idhibitwi. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `support\{agents, support\{call\{log, support\{chat\{conversations, support\{chat\{messages, support\{message\{templates, support\{sla\{policies, support\{team\{members, support\{teams, support\{ticket\{activity, support\{ticket\{messages, support\{ticket\{notes, support\{tickets`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; haya-maanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Receive issue → create ticket → classify/SLA → assign → communicate → resolve → audit and measure. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Automated external support execution is intentionally not claimed without configured channels; internal ticket persistence remains the source of record. Helpdesk ina tickets, timelines, internal notes, chat/call evidence, SLA, assignment, resolution na metrics. Dispatch ya nje inategemea channel iliyosanidiwa. Hali ya sasa: TESTING. Mipaka: Automated external support execution is intentionally not claimed without configured channels; internal ticket persistence remains the source of record.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Support agent; manager; customer; administrator	Verified profile + role	Controlled acceptance
Workflow	Receive issue → create ticket → classify/SLA → assign → communicate → resolve → audit and measure.	Server-confirmed result	Role-specific test
Database	support_agents, support_call_log, support_chat_conversations, support_chat_messages, support_message_templates, support_sla_policies, support_team_members, support_teams, support_ticket_activity, support_ticket_messages, support_ticket_notes, support_tickets	RLS/company	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.21 XV.21 Uchambuzi wa Biashara na BI

BI huleta KPI, benchmarks, market na trend pamoja. Ubora wa uchambuzi unategemea data iliyokamilika na iliyowekwa ndani ya kampuni sahihi. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Uchambuzi wa Biashara na BI ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha competitors, custom_kpis, financial_benchmarks. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Select scope → read source metrics → compare or trend → investigate source record → document action. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Analytical quality depends on complete, correctly scoped input data and configured market services. BI huleta KPI, benchmarks, market na trend pamoja. Ubora wa uchambuzi unategemea data iliyokamilika na iliyowekwa ndani ya kampuni sahihi. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: Analytical quality depends on complete, correctly scoped input data and configured market services.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Executive; analyst; manager	Verified profile + role	Controlled acceptance
Workflow	Select scope → read source metrics → compare or trend → investigate source record → document action.	Server-confirmed result	Role-specific test
Database	competitors, custom_kpis, financial_benchmarks	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.22 XV.22 Arifa na Tahadhari

Arifa zinarekodi tukio, delivery history, hali ya kusoma na retries. Rekodi ya notification si uthibitisho wa delivery ya mtoa huduma wa nje. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Arifa na Tahadhari ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zina-jumuisha emails, notification\{005F}channels, notification\{005F}log, notification\{005F}rules, sms\{005F}group\{005F}members, sms\{005F}groups, sms\{005F}templates. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Detect event → create notification record → deliver through configured channel → mark read or retry → audit result. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. SMS, email, push, and WhatsApp delivery depend on external provider configuration and should show explicit unavailable states. Arifa zinarekodi tukio, delivery history, hali ya kusoma na retries. Rekodi ya notification si uthibitisho wa delivery ya mtoa huduma wa nje. Hali ya sasa: TESTING. Mipaka: SMS, email, push, and WhatsApp delivery depend on external provider configuration and should show explicit unavailable states.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Authenticated users; managers; administrators	Verified profile + role	Controlled acceptance
Workflow	Detect event → create notification record → deliver through configured channel → mark read or retry → audit result.	Server-confirmed result	Role-specific test
Database	emails, notification_channels, notification_log, notification_rules, sms_group_members, sms_groups, sms_templates	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.23 XV.23 Mtiririko wa Shughuli na Ushahidi wa Ukaguzi

Ushahidi wa audit unaonyesha nani alifanya kitendo, kwenye kampuni ipi na matokeo yake pale taarifa hiyo ipo. Rekodi za global admin zinalindwa tofauti. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Mtiririko wa Shughuli na Ushahidi wa Ukaguzi ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `audit\log`, `audit\log\entries`, `bank\audit\events`, `billing\plan\audit\log`, `community\group\audit\log`, `hospitality\audit\log`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Authenticated action → protected mutation → audit record → scoped search/export → review or incident follow-up. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. An audit record increases traceability but does not correct an unsafe write; mutation confirmation and database constraints remain primary controls. Ushahidi wa audit unaonyesha nani alifanya kitendo, kwenye kampuni ipi na matokeo yake pale taarifa hiyo ipo. Rekodi za global admin zinalindwa tofauti. Hali ya sasa: TESTING. Mipaka: An audit record increases traceability but does not correct an unsafe write; mutation confirmation and database constraints remain primary controls.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Auditor; administrator; company manager with scope	Verified profile + role	Controlled acceptance
Workflow	Authenticated action → protected mutation → audit record → scoped search/export → review or incident follow-up.	Server-confirmed result	Role-specific test
Database	<code>audit_log</code> , <code>audit_log_entries</code> , <code>bank_audit_events</code> , <code>billing_plan_audit_log</code> , <code>community_group_audit_log</code> , <code>hospitality_audit_log</code>	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.24 XV.24 Kitovu cha Miunganisho

Kitovu cha miunganisho huonyesha metadata, webhooks, readiness na delivery history. Bila configuration halali, mfumo haupaswi kuonyesha kana kwamba channel inafanya kazi. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Kitovu cha Miunganisho ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `integration\connections`, `notification\channels`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Define connection → validate configuration → test safely → enable dispatch → monitor delivery → retry or disable. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. No provider credential, endpoint, or delivery success is assumed when configuration is missing; controls must remain disabled or clearly unavailable. Kitovu cha miunganisho huonyesha metadata, webhooks, readiness na delivery history. Bila configuration halali, mfumo haupaswi kuonyesha kana kwamba channel inafanya kazi. Hali ya sasa: CONFIGURATION REQUIRED. Mipaka: No provider credential, endpoint, or delivery success is assumed when configuration is missing; controls must remain disabled or clearly unavailable.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Administrator; integration manager; finance manager	Verified profile + role	Controlled acceptance
Workflow	Define connection → validate configuration → test safely → enable dispatch → monitor delivery → retry or disable.	Server-confirmed result	Role-specific test
Database	<code>integration_connections</code> , <code>notification_channels</code>	URL/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.25 XV.25 Studio ya Mitiririko na Soko

Workflow Studio huandaa templates, steps, approvals, tasks, notifications na audit. Draft ya ujumbe si uthibitisho wa kutumwa kwa mteja. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Studio ya Mitiririko na Soko ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha workflow\ marketplace\ templates, workflows. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Choose template → inspect steps → configure target → approve or run → record task/result → review audit. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. A workflow step that drafts an email or notification is not proof of external dispatch; complete orchestration requires channel configuration. Workflow Studio huandaa templates, steps, approvals, tasks, notifications na audit. Draft ya ujumbe si uthibitisho wa kutumwa kwa mteja. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: A workflow step that drafts an email or notification is not proof of external dispatch; complete orchestration requires channel configuration.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Workflow administrator; approver; manager	Verified profile + role	Controlled acceptance
Workflow	Choose template → inspect steps → configure target → approve or run → record task/result → review audit.	Server-confirmed result	Role-specific test
Database	workflow_marketplace_workflows	Business company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.26 XV.26 Kitovu cha Ushirikiano

Ushirikiano huunganisha channels, messages, presence, calendar na read receipts pale huduma hizo zimesanidiwa. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Kitovu cha Ushirikiano ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `calendar\{005F}` events, `collab\{005F}` channels, `collab\{005F}` messages, emails. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Create channel/event → post message → notify participants → track read/presence → retain evidence. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. External collaboration delivery and realtime behavior depend on configuration and provider availability; local state is not a durable message authority. Ushirikiano huunganisha channels, messages, presence, calendar na read receipts pale huduma hizo zimesanidiwa. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: External collaboration delivery and realtime behavior depend on configuration and provider availability; local state is not a durable message authority.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Team member; manager; executive	Verified profile + role	Controlled acceptance
Workflow	Create channel/event → post message → notify participants → track read/presence → retain evidence.	Server-confirmed result	Role-specific test
Database	calendar_events, collab_channels, collab_messages, emails	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.27 XV.27 TRA, VFD na Fiscalization ya Tanzania

Mifumo ya fiscal ina profiles, receipts, retry queue, Z-reports na tax settings. Uwasilishaji halisi wa TRA unahitaji TIN/VRN, configuration ya branch, credentials na mtoa huduma. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la TRA, VFD na Fiscalization ya Tanzania ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha tables za shared company scope. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Configure TIN/VRN and branch → validate readiness → submit idempotent transaction → receive provider result → store receipt/retry/audit evidence. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Production fiscalization requires approved provider credentials, device/branch configuration, and provider availability; a local receipt screen is not a TRA acceptance claim. Mifumo ya fiscal ina profiles, receipts, retry queue, Z-reports na tax settings. Uwasilishaji halisi wa TRA unahitaji TIN/VRN, configuration ya branch, credentials na mtoa huduma. Hali ya sasa: CONFIGURATION REQUIRED. Mipaka: Production fiscalization requires approved provider credentials, device/branch configuration, and provider availability; a local receipt screen is not a TRA acceptance claim.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	CFO; Finance Manager; administrator; fiscal operator	Verified profile + role	Controlled acceptance
Workflow	Configure TIN/VRN and branch → validate readiness → submit idempotent transaction → receive provider result → store receipt/retry/audit evidence.	Server-confirmed result	Role-specific test
Database	shared scope	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.28 XV.28 Msaidizi wa AI na Akili ya Biashara

Msaidizi wa AI hupokea context yenye mipaka, hutumia response iliyopangwa, na huandaa proposal badala ya kubadilisha rekodi bila approval ya jukumu husika. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Msaidizi wa AI na Akili ya Biashara ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `custom_kpis`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Verified user → scoped context → structured assistant response → optional navigation/proposal → independent role approval → confirmed business action. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. AI model availability and response quality depend on configured service access and supplied evidence; the assistant must not claim a mutation it did not execute. Msaidizi wa AI hupokea context yenye mipaka, hutumia response iliyopangwa, na huandaa proposal badala ya kubadilisha rekodi bila approval ya jukumu husika. Hali ya sasa: EXTERNAL SERVICE REQUIRED. Mipaka: AI model availability and response quality depend on configured service access and supplied evidence; the assistant must not claim a mutation it did not execute.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Authenticated user; approver for proposed actions	Verified profile + role	Controlled acceptance
Workflow	Verified user → scoped context → structured assistant response → optional navigation/proposal → independent role approval → confirmed business action.	Server-confirmed result	Role-specific test
Database	custom_kpis	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.29 XV.29 Muunganisho wa WhatsApp Web

WhatsApp ina metadata ya account, contacts, messages na conversations. API key au token haipaswi kuhifadhiwa kwenye browser, na delivery inategemea provider. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Muunganisho wa WhatsApp Web ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha whatsapp\{005F}account\{005F}links, whatsapp\{005F}accounts, whatsapp\{005F}contacts, whatsapp\{005F}conversations, whatsapp\{005F}message\{005F}events, whatsapp\{005F}messages. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Configure provider → link account/contact → draft message → validate content → dispatch through provider → record status. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Provider configuration and message delivery are not assumed; credential persistence in browser storage is a security risk and must remain removed or blocked. WhatsApp ina metadata ya account, contacts, messages na conversations. API key au token haipaswi kuhifadhiwa kwenye browser, na delivery inategemea provider. Hali ya sasa: CONFIGURATION REQUIRED. Mipaka: Provider configuration and message delivery are not assumed; credential persistence in browser storage is a security risk and must remain removed or blocked.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Marketing manager; support agent; administrator	Verified profile + role	Controlled acceptance
Workflow	Configure provider → link account/contact → draft message → validate content → dispatch through provider → record status.	Server-confirmed result	Role-specific test
Database	whatsapp_account_links, whatsapp_accounts, whatsapp_contacts, whatsapp_conversations, whatsapp_message_events, whatsapp_messages	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.30 XV.30 Mikopo Midogo

Microfinance inahusu borrowers, vikundi, bidhaa, applications, approvals, disbursement, repayments, savings, cash sessions na collections. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Mikopo Midogo ni kufanya kazi inayorudiwa ione kane na idhibitawe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `mfi\audit\logs`, `mfi\cash\sessions`, `mfi\cash\transactions`, `mfi\clients`, `mfi\collateral`, `mfi\collections`, `mfi\credit\scorecards`, `mfi\credit\scoring\settings`, `mfi\groups`, `mfi\guarantors`, `mfi\loan\applications`, `mfi\loan\products`, `mfi\loans`, `mfi\notifications`, `mfi\par\escalation\settings`, `mfi\repayment\schedules`, `mfi\repayments`, `mfi\savings`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliohibitishwa ni: Register borrower/group → choose product → submit application → approve → disburse → collect repayment/savings → monitor PAR/escalation. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Financial persistence and controlled data must be exercised against a real tenant schema; seeded UI examples are not production records. Microfinance inahusu borrowers, vikundi, bidhaa, applications, approvals, disbursement, repayments, savings, cash sessions na collections. Hali ya sasa: TESTING. Mipaka: Financial persistence and controlled data must be exercised against a real tenant schema; seeded UI examples are not production records.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	MFI manager; loan officer; cashier; approver; auditor	Verified profile + role	Controlled acceptance
Workflow	Register borrower/group → choose product → submit application → approve → disburse → collect repayment/savings → monitor PAR/escalation.	Server-confirmed result	Role-specific test
Database	<code>mfi_audit_logs</code> , <code>mfi_cash_sessions</code> , <code>mfi_cash_transactions</code> , <code>mfi_clients</code> , <code>mfi_collateral</code> , <code>mfi_collections</code> , <code>mfi_credit_scorecards</code> , <code>mfi_credit_scoring_settings</code> ,	RLS/company schema	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	mfi_groups, mfi_guarantors, mfi_loan_applications, mfi_loan_products		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.31 XV.31 Wakala wa Fedha

Wakala wa fedha huunganisha mawakala, wateja, wallets, cash actions, commissions, ledger, approvals na reconciliation. Movement halisi ya mtoa huduma inahitaji acceptance ya kudhibitiwa. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Wakala wa Fedha ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `money\agent\agents`, `money\agent\alerts`, `money\agent\approvals`, `money\agent\audit\events`, `money\agent\branches`, `money\agent\commission\rules`, `money\agent\customers`, `money\agent\daily\summaries`, `money\agent\fee\rules`, `money\agent\ledger\entries`, `money\agent\limits`, `money\agent\notifications`, `money\agent\pin\credentials`, `money\agent\receipts`, `money\agent\reconciliations`, `money\agent\risk\events`, `money\agent\services`, `money\agent\settlements`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Register agent/customer → open cash/wallet context → initiate action → validate balance/role → record ledger → reconcile and audit. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. The live schema includes the migration history and tables, but real wallet/provider movement still requires controlled credentials and operational acceptance. Wakala wa fedha huunganisha mawakala, wateja, wallets, cash actions, commissions, ledger, approvals na reconciliation. Movement halisi ya mtoa huduma inahitaji acceptance ya kudhibitiwa. Hali ya sasa: IMPLEMENTED. Mipaka: The live schema includes the migration history and tables, but real wallet/provider movement still requires controlled credentials and operational acceptance.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Money agent; supervisor; cashier; auditor	Verified profile + role	Controlled acceptance
Workflow	Register agent/customer → open cash/wallet context → initiate action → validate balance/role → record ledger → reconcile and audit.	Server-confirmed result	Role-specific test
Database	<code>money_agent_agents</code> , <code>money_agent_alerts</code> , <code>money_agent_approvals</code> , <code>money_agent_audit_events</code> , <code>money_agent_branches</code> ,	RLS/company	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	money_agent_commission_rules, money_agent_customers, money_agent_daily_summaries, money_agent_fee_rules, money_agent_ledger_entries, money_agent_limits, money_agent_notifications		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.32 XV.32 VICOBA, SACCOS na Vikundi vya Jamii

Vikundi vina groups, members, michango, savings, loans, meetings, attendance, budgets, welfare, votes, approvals na audit. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la VICOBA, SACCOS na Vikundi vya Jamii ni kufanya kazi inayorudiwa ionekane na idhibitiwe Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha community\{contributions, community\group\{announcements, community\group\{approvals, community\group\{assets, community\group\{attendance, community\group\{audit\{log, community\group\{budgets, community\group\{committee\{members, community\group\{committees, community\group\{contributions, community\group\{documents, community\group\{events, community\group\{expenses, community\group\{fundraising, community\group\{income, community\group\{loan\{guarantors, community\group\{loan\{penalties, community\group\{loan\{repayments. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Create group → add member → record contribution/meeting → approve loan or welfare action → update ledger/report. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Relationship guards and approvals are present, but broader group-accounting walkthroughs should still be completed with controlled tenant data. Vikundi vina groups, members, michango, savings, loans, meetings, attendance, budgets, welfare, votes, approvals na audit. Hali ya sasa: TESTING. Mipaka: Relationship guards and approvals are present, but broader group-accounting walkthroughs should still be completed with controlled tenant data.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Group chairperson; treasurer; secretary; member; auditor	Verified profile + role	Controlled acceptance
Workflow	Create group → add member → record contribution/meeting → approve loan or welfare action → update ledger/report.	Server-confirmed result	Role-specific test
Database	community_contributions, community_group_announcements, community_group_approvals, community_group_assets, community_group_attendance,	RLS/company supplements,	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	community_group_audit_log, community_group_budgets, community_group_committee_members, community_group_committees, community_group_contributions, community_group_documents, community_group_events		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.33 XV.33 Afya na Kliniki

Kliniki ina patients, appointments, visits, vitals, prescriptions, labs, radiology, claims, reports, reminders na patient consent. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Afya na Kliniki ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `hc\appointments`, `hc\doctors`, `hc\insurance\claims`, `hc\invoices`, `hc\lab\orders`, `hc\notifications`, `hc\patients`, `hc\portal\reference\approvals`, `hc\portal\reference\imports`, `hc\portal\reference\summary\settings`, `hc\prescriptions`, `hc\radiology`, `hc\reminder\deliveries`, `hc\reminder\settings`, `hc\reports`, `hc\visits`, `hc\vitals`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Register patient → book appointment → record visit/vitals → prescribe/order → issue report or claim → notify with consent → audit. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. SMS and interoperability depend on provider configuration; clinical data requires strict least-privilege review and must not be exposed through generic tenant screens. Kliniki ina patients, appointments, visits, vitals, prescriptions, labs, radiology, claims, reports, reminders na patient consent. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: SMS and interoperability depend on provider configuration; clinical data requires strict least-privilege review and must not be exposed through generic tenant screens.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Clinician; nurse; receptionist; finance user; patient portal user	Verified profile + role	Controlled acceptance
Workflow	Register patient → book appointment → record visit/vitals → prescribe/order → issue report or claim → notify with consent → audit.	Server-confirmed result	Role-specific test
Database	<code>hc_appointments</code> , <code>hc_doctors</code> , <code>hc_insurance_claims</code> , <code>hc_invoices</code> , <code>hc_lab_orders</code> , <code>hc_notifications</code> , <code>hc_patients</code> , <code>hc_portal_reference_approvals</code> ,	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	hc_portal_reference_imports, hc_portal_reference_summary_settings, hc_prescriptions, hc_radiology		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.34 XV.34 Usimamizi wa Shule

Mfumo wa shule una miaka ya masomo, terms, departments, subjects, classes, admissions, attendance, assessments, report cards, fees, scholarships, transport, library na announcements. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Usimamizi wa Shule ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `sch\academic\years`, `sch\admissions`, `sch\announcements`, `sch\approval\requests`, `sch\assessment\scores`, `sch\assessments`, `sch\assignment\submissions`, `sch\assignments`, `sch\attendance\records`, `sch\attendance\sessions`, `sch\audit\logs`, `sch\books`, `sch\classes`, `sch\departments`, `sch\disciplinary\records`, `sch\documents`, `sch\enrollments`, `sch\exams`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Set academic structure → admit learner → assign teacher/class → record attendance/assessment → publish report → invoice/receive fees → communicate. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Automated browser coverage is strong for the surface, while full per-school CRUD remains a data-backed acceptance task. Mfumo wa shule una miaka ya masomo, terms, departments, subjects, classes, admissions, attendance, assessments, report cards, fees, scholarships, transport, library na announcements. Hali ya sasa: PASSED. Mipaka: Automated browser coverage is strong for the surface, while full per-school CRUD remains a data-backed acceptance task.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	School administrator; teacher; student/guardian portal user; finance user	Verified profile + role	Controlled acceptance
Workflow	Set academic structure → admit learner → assign teacher/class → record attendance/assessment → publish report → invoice/receive fees → communicate.	Server-confirmed result	Role-specific test
Database	<code>sch_academic_years</code> , <code>sch_admissions</code> , <code>sch_announcements</code> , <code>sch_approval_requests</code> , <code>sch_assessment_scores</code> , <code>sch_assessments</code> , <code>sch_assignment_submissions</code> ,	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	sch_assignments, sch_attendance_records, sch_attendance_sessions, sch_audit_logs, sch_books		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.35 XV.35 Usimamizi wa Famasi

Famasi inahifadhi dawa, categories, brands, suppliers, purchase orders, stock, dispensing, sales, malipo, claims, returns na audit. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Usimamizi wa Famasi ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `phm\audit\logs`, `phm\batches`, `phm\brands`, `phm\categories`, `phm\controlled\medicine\register`, `phm\dispense`, `phm\dispense\items`, `phm\drugs`, `phm\insurance\claims`, `phm\notifications`, `phm\payments`, `phm\purchase\order\items`, `phm\purchase\orders`, `phm\return\items`, `phm\returns`, `phm\sale\items`, `phm\sales`, `phm\stock`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Create medicine → receive stock → adjust/transfer → dispense or sell → collect payment → return/claim → reconcile. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Live dispensing and inventory walkthroughs remain dependent on controlled pharmacy records and appropriate clinical/financial permissions. Famasi inahifadhi dawa, categories, brands, suppliers, purchase orders, stock, dispensing, sales, malipo, claims, returns na audit. Hali ya sasa: TESTING. Mipaka: Live dispensing and inventory walkthroughs remain dependent on controlled pharmacy records and appropriate clinical/financial permissions.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Pharmacist; store-keeper; cashier; clinician; administrator	Verified profile + role	Controlled acceptance
Workflow	Create medicine → receive stock → adjust/transfer → dispense or sell → collect payment → return/claim → reconcile.	Server-confirmed result	Role-specific test
Database	<code>phm_audit_logs</code> , <code>phm_batches</code> , <code>phm_brands</code> , <code>phm_categories</code> , <code>phm_controlled_medicine_register</code> , <code>phm_dispense</code> , <code>phm_dispense_items</code> , <code>phm_drugs</code> , <code>phm_insurance_claims</code> , <code>phm_notifications</code> , <code>phm_payments</code> , <code>phm_purchase_order_items</code>	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.36 XV.36 Hoteli na Ukarimu

Hoteli inahusu properties, room types, rooms, rates, reservations, guests, KYC, folios, payments, housekeeping, laundry, events na reconciliation. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Hoteli na Ukarimu ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `hospitality\amenities`, `hospitality\audit\log`, `hospitality\complaints`, `hospitality\event\venues`, `hospitality\events`, `hospitality\finance\reconciliations`, `hospitality\folio\lines`, `hospitality\folios`, `hospitality\guest\kyc`, `hospitality\guest\requests`, `hospitality\guests`, `hospitality\housekeeping\tasks`, `hospitality\laundry\orders`, `hospitality\loyalty\accounts`, `hospitality\maintenance\requests`, `hospitality\menu\items`, `hospitality\menus`, `hospitality\minibar\postings`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Configure property/rooms → create reservation → check in → post folio/service → receive payment → housekeeping/check out → reconcile. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. External booking-channel and payment connections remain provider-dependent; local reservation state is not proof of channel synchronization. Hoteli inahusu properties, room types, rooms, rates, reservations, guests, KYC, folios, payments, housekeeping, laundry, events na reconciliation. Hali ya sasa: TESTING. Mipaka: External booking-channel and payment connections remain provider-dependent; local reservation state is not proof of channel synchronization.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Hotel manager; front desk; housekeeping; finance; guest	Verified profile + role	Controlled acceptance
Workflow	Configure property/rooms → create reservation → check in → post folio/service → receive payment → housekeeping/check out → reconcile.	Server-confirmed result	Role-specific test
Database	<code>hospitality_amenities</code> , <code>hospitality_audit_log</code> , <code>hospitality_complaints</code> , <code>hospitality_event_venues</code> , <code>hospitality_events</code> , <code>hospitality_finance_reconciliations</code> , <code>hospitality_folio_lines</code> , <code>hospitality_folios</code> , <code>hospitality_guest_kyc</code> ,	RLS/company	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	hospitality_guest_requests, hospitality_guests, hospitality_housekeeping_tasks		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.37 XV.37 Mgahawa na Chakula/Vinywaji

Mgahawa una menus, items, tables, orders, reservations, waiters, payments, receipts, refunds na fiscal configuration ya Tanzania. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Mgahawa na Chakula/Vinywaji ni kufanya kazi inayorudiwa ionekane na idhibitwi. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `restaurant\{alerts, restaurant\{audit\{events, restaurant\{bill\{splits, restaurant\{combo\{items, restaurant\{customers, restaurant\{dining\{areas, restaurant\{fiscal\{profiles, restaurant\{fiscal\{receipts, restaurant\{kitchen\{tickets, restaurant\{menu\{categories, restaurant\{menu\{items, restaurant\{mobile\{money\{intents, restaurant\{mobile\{money\{profiles, restaurant\{modifier\{groups, restaurant\{modifier\{options, restaurant\{order\{lines, restaurant\{orders, restaurant\{outlets`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliohibitishwa ni: Configure menu/table → open order → add items → prepare/serve → collect payment → issue receipt/fiscal evidence → reconcile. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Delivery and external payment channels require configuration; fiscal status is a readiness state until a provider confirms submission. Mgahawa una menus, items, tables, orders, reservations, waiters, payments, receipts, refunds na fiscal configuration ya Tanzania. Hali ya sasa: TESTING. Mipaka: Delivery and external payment channels require configuration; fiscal status is a readiness state until a provider confirms submission.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Restaurant manager; waiter; cashier; kitchen; finance	Verified profile + role	Controlled acceptance
Workflow	Configure menu/table → open order → add items → prepare/serve → collect payment → issue receipt/fiscal evidence → reconcile.	Server-confirmed result	Role-specific test
Database	restaurant_alerts, restaurant_audit_events, restaurant_bill_splits, restaurant_combo_items, restaurant_customers, restaurant_dining_areas, restaurant_fiscal_profiles,	Plus compliance	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	restaurant_fiscal_receipts, restaurant_kitchen_tickets, restaurant_menu_categories, restaurant_menu_items, restaurant_mobile_money_intents		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.38 XV.38 Benki na MFI

Benki na MFI zina institutions, branches, customers, KYC, accounts, beneficiaries, transactions, loans, schedules, repayments, AML, reconciliation na audit. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Benki na MFI ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha bank\{account\beneficiaries, bank\{account\types, bank\{accounts, bank\{agents, bank\{aml\alerts, bank\{audit\events, bank\{beneficial\owners, bank\{branches, bank\{cash\movements, bank\{collateral, bank\{customer\documents, bank\{customers, bank\{fixed\deposits, bank\{group\members, bank\{groups, bank\{guarantors, bank\{idempotency\keys, bank\{institutions. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Set institution → register customer/KYC → open account → post transaction → originate/approve loan → schedule/collect → reconcile/resolve AML. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. External banking rails are not connected without approved adapters; SECURITY DEFINER routines require endpoint-by-endpoint privilege review. Benki na MFI zina institutions, branches, customers, KYC, accounts, beneficiaries, transactions, loans, schedules, repayments, AML, reconciliation na audit. Hali ya sasa: TESTING. Mipaka: External banking rails are not connected without approved adapters; SECURITY DEFINER routines require endpoint-by-endpoint privilege review.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Bank administrator; teller; loan officer; compliance officer; auditor	Verified profile + role	Controlled acceptance
Workflow	Set institution → register customer/KYC → open account → post transaction → originate/approve loan → schedule/collect → reconcile/resolve AML.	Server-confirmed result	Role-specific test
Database	bank_account_beneficials, bank_account_types, bank_accounts, bank_agents, bank_aml_alerts,	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
	bank_audit_events, bank_beneficial_owners, bank_branches, bank_cash_movements, bank_collateral, bank_customer_documents, bank_customers		
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.39 XV.39 Portal ya Mfanyakazi

Portal ya mfanyakazi hutoa self-service ya wasifu, attendance, leave, announcements, benefits na nyaraka kwa mipaka ya kampuni na role. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Portal ya Mfanyakazi ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `hr\announcement\reads`, `hr\announcements`, `hr\approval\requests`, `hr\approval\steps`, `hr\attendance`, `hr\benefit\enrollments`, `hr\benefit\plans`, `hr\benefits`, `hr\candidates`, `hr\duties`, `hr\employee\documents`, `hr\employees`, `hr\expense\claims`, `hr\goal\updates`, `hr\goals`, `hr\holidays`, `hr\invite\codes`, `hr\leave\balances`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Accept secure invitation → verify profile → view or request employee action → manager/HR decision → record outcome. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. An employee view does not imply access to payroll or HR administration; role boundaries must be enforced by server and RLS. Portal ya mfanyakazi hutoa self-service ya wasifu, attendance, leave, announcements, benefits na nyaraka kwa mipaka ya kampuni na role. Hali ya sasa: IMPLEMENTED. Mipaka: An employee view does not imply access to payroll or HR administration; role boundaries must be enforced by server and RLS.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Employee; manager; HR administrator	Verified profile + role	Controlled acceptance
Workflow	Accept secure invitation → verify profile → view or request employee action → manager/HR decision → record outcome.	Server-confirmed result	Role-specific test
Database	<code>hr_announcement_reads</code> , <code>hr_announcements</code> , <code>hr_approval_requests</code> , <code>hr_approval_steps</code> , <code>hr_attendance</code> , <code>hr_benefit_enrollments</code> , <code>hr_benefit_plans</code> , <code>hr_benefits</code> , <code>hr_candidates</code> , <code>hr_duties</code> , <code>hr_employee_documents</code> , <code>hr_employees</code>	RLS/company	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.40 XV.40 Usimamizi wa Mali Isiyohamishika

Property Management ina portfolio, owners, buildings, plots, units, listings, agents, tenants, applications, leases, inspections, handover, rent, utilities, invoices, payments, maintenance, expenses, budgets, insurance, notices na audit. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Usimamizi wa Mali Isiyohamishika ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `property\agents`, `property\applications`, `property\approvals`, `property\audit\log`, `property\budgets`, `property\buildings`, `property\contractors`, `property\documents`, `property\expenses`, `property\handover\records`, `property\inspection\items`, `property\inspections`, `property\insurances`, `property\integration\events`, `property\invoice\lines`, `property\invoices`, `property\leases`, `property\ledger\entries`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Portfolio → building/unit → owner/tenant/application → lease → inspection/handover → rent/utility invoice → payment/receipt → maintenance/expense → reconciliation. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. The live audit shows the property migration and tables present; external payment, map, messaging, and contractor channels still require configuration and controlled acceptance. Property Management ina portfolio, owners, buildings, plots, units, listings, agents, tenants, applications, leases, inspections, handover, rent, utilities, invoices, payments, maintenance, expenses, budgets, insurance, notices na audit. Hali ya sasa: IMPLEMENTED. Mipaka: The live audit shows the property migration and tables present; external payment, map, messaging, and contractor channels still require configuration and controlled acceptance.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Property manager; landlord/owner; agent; tenant; finance manager; maintenance user	Verified profile + role	Controlled acceptance
Workflow	Portfolio → building/unit → owner/tenant/application → lease → inspection/handover → rent/utility invoice → payment/receipt → maintenance/expense → reconciliation.	Server-confirmed result	Role-specific test

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Database	property_agents, property_applications, property_approvals, property_audit_log, property_budgets, property_buildings, property_contractors, property_documents, property_expenses, property_handover_records, property_inspection_items, property_inspections	RL-Scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.41 XV.41 Usajili na Utozaji

Mfumo wa usajili una FREE_15 ya TZS 0 kwa siku 15 na packages sita za kila mwezi. Paid access hutolewa baada ya provider kuthibitisha malipo; server na database ndiyo mamlaka. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Usajili na Utozaji ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa ku-fuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `billing\plan\audit\log`, `billing\plans`, `billing\profiles`, `subscription\events`, `subscription\invoices`, `subscription\notifications`, `subscription\payments`, `subscription\usage`, `tenant\subscriptions`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Read catalog → start Free or create paid intent → send/verify provider request → reconcile payment → activate subscription/invoice → refresh access snapshot. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. HarakaPay account authorization and provider credentials remain deployment prerequisites; failed or cancelled requests must not be shown as active entitlement. Mfumo wa usajili una FREE_15 ya TZS 0 kwa siku 15 na packages sita za kila mwezi. Paid access hutolewa baada ya provider kuthibitisha malipo; server na database ndiyo mamlaka. Hali ya sasa: IMPLEMENTED. Mipaka: HarakaPay account authorization and provider credentials remain deployment prerequisites; failed or cancelled requests must not be shown as active entitlement.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Billing manager; owner; CEO; CFO; finance manager; administrator	Verified profile + role	Controlled acceptance
Workflow	Read catalog → start Free or create paid intent → send/verify provider request → reconcile payment → activate subscription/invoice → refresh access snapshot.	Server-confirmed result	Role-specific test
Database	<code>billing_plan_audit_log</code> , <code>billing_plans</code> , <code>billing_profiles</code> , <code>subscription_events</code> , <code>subscription_invoices</code> , <code>subscription_notifications</code> , <code>subscription_payments</code> , <code>subscription_usage</code> , <code>tenant_subscriptions</code>	RLS/company scope	Schema verification

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.42 XV.42 Kituo cha Udhhibiti wa Msimamizi wa Mfumo

Global Admin ni control plane tofauti ya kampuni, watumiaji, subscriptions, usalama, integrations, health na audit. Si ruhusa pana kwa mtumiaji wa tenant. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Kituo cha Udhhibiti wa Msimamizi wa Mfumo ni kufanya kazi inayorudiwa ione kane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha `billing\plan\audit\log`, `companies`, `platform\admin\actions`, `profiles`. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Verify platform-admin authority → inspect scoped snapshot → perform explicit action → record platform audit → review outcome. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. The `platform_admin_actions` table is intentionally protected from a broad authenticated policy; access uses the protected platform-admin RPC boundary. Global Admin ni control plane tofauti ya kampuni, watumiaji, subscriptions, usalama, integrations, health na audit. Si ruhusa pana kwa mtumiaji wa tenant. Hali ya sasa: IMPLEMENTED. Mipaka: The `platform_admin_actions` table is intentionally protected from a broad authenticated policy; access uses the protected platform-admin RPC boundary.

Sehemu	Maelezo	Udhhibiti	Hatua inayofuata
Watumiaji	Super Administrator; platform administrator	Verified profile + role	Controlled acceptance
Workflow	Verify platform-admin authority → inspect scoped snapshot → perform explicit action → record platform audit → review outcome.	Server-confirmed result	Role-specific test
Database	<code>billing_plan_audit_log</code> , <code>companies</code> , <code>platform_admin_actions</code> , <code>profiles</code>	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.43 XV.43 Mipangilio na Udhibiti wa Usalama

Mipangilio inahusu kampuni, branding, lugha, timezone/currency, receipts, idle timeout, modules, branches, departments, preferences, passkeys na role approvals. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Mipangilio na Udhibiti wa Usalama ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha branches, company\{005F}modules, company\{005F}profile\{005F}settings, departments, user\{005F}table\{005F}preferences. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Open settings → read confirmed configuration → edit permitted field → server validates and persists → refresh UI → audit sensitive change. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Provider-specific email, SMS, API-key, and storage configuration remains environment-dependent; secret values must be redacted. Mipangilio inahusu kampuni, branding, lugha, timezone/currency, receipts, idle timeout, modules, branches, departments, preferences, passkeys na role approvals. Hali ya sasa: IMPLEMENTED. Mipaka: Provider-specific email, SMS, API-key, and storage configuration remains environment-dependent; secret values must be redacted.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Organization owner; administrator; manager; user with permitted settings	Verified profile + role	Controlled acceptance
Workflow	Open settings → read confirmed configuration → edit permitted field → server validates and persists → refresh UI → audit sensitive change.	Server-confirmed result	Role-specific test
Database	branches, company_modules, company_profile_settings, departments, user_table_preferences	RLS/company	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

16.44 XV.44 Uchambuzi wa Utabiri

Utabiri ni msaada wa maamuzi unaotegemea data ya kutosha na huduma iliyosanidiwa; si ahadi ya matokeo ya biashara. Sura hii imetokana na UI ya repository, mpaka wa server, migration history na inventory ya Supabase iliyosomwa bila kubadilisha data. Pale ushahidi unategemea configuration, data ya kampuni au provider wa nje, mpaka huo umeelezwa wazi. Lengo la Uchambuzi wa Utabiri ni kufanya kazi inayorudiwa ionekane na idhibitiwe. Changamoto si kutokuwepo kwa screen pekee; ni hatari ya kuandika ukweli uleule mara nyingi, role kuona zaidi ya inavyoruhusiwa, au write iliyokataliwa kuonekana kana kwamba imefanikiwa. Kwa hiyo module inapaswa kufuata input, validation, persistence, ushahidi na review ya uongozi. Sehemu za persistence zinazoonekana kwa module hii zinajumuisha competitors, custom_kpis, financial_benchmarks. Majina haya ni ushahidi wa inventory ya sasa au contract ya repository; hayamaanishi kwamba kila table inatumika na kila deployment au kwamba kila table ina data ya production. Mtiririko uliothibitishwa ni: Select evidence set → compute signal → present confidence/limitation → inspect source metrics → decide manually. Mtumiaji athibitishe hali iliyorejeshwa na atumie empty/error state badala ya kukisia. Predictions are decision support, not guarantees. Sparse or stale data, configuration, and model availability constrain their reliability. Utabiri ni msaada wa maamuzi unaotegemea data ya kutosha na huduma iliyosanidiwa; si ahadi ya matokeo ya biashara. Hali ya sasa: PARTIALLY IMPLEMENTED. Mipaka: Predictions are decision support, not guarantees. Sparse or stale data, configuration, and model availability constrain their reliability.

Sehemu	Maelezo	Udhibiti	Hatua inayofuata
Watumiaji	Executive; analyst; manager	Verified profile + role	Controlled acceptance
Workflow	Select evidence set → compute signal → present confidence/limitation → inspect source metrics → decide manually.	Server-confirmed result	Role-specific test
Database	competitors, custom_kpis, financial_benchmarks	RLS/company scope	Schema verification
Integration	Provider where configured	Server-side secrets	Readiness evidence
Mobile	Responsive review and entry	Same authorization	Browser journey

17 Appendices

17.1 Appendix A — Role and permission matrix

Role	Dashboard	Primary modules	Boundary
Super Administrator	Platform control plane; not ordinary tenant access	Global admin; security; health; audit	Explicit platform-admin boundary
Organization Owner	Company-wide visibility and approvals	All configured company modules	Company scope plus sensitive settings
CEO / COO / CFO	Executive and financial review	Reports, finance, sales, operational modules by assignment	Manager checks and company scope
Finance Manager / Accountant	Finance, billing, reconciliation	Finance, sales billing, subscriptions, tax evidence	Billing-manager roles and RLS
HR Manager / Manager	Workforce and approvals	HR, employee portal, leave, payroll as assigned	Role-aware HR operations
Sales Manager / Sales User	Commercial pipeline and sales	CRM, sales, POS as assigned	Tenant scope and confirmed writes
Warehouse Manager / Procurement Officer	Stock and supply chain	Inventory, procurement, fleet as assigned	Movement, approval, and company scope
Specialist operator	Assigned industry workflow	Healthcare, school, pharmacy, property, hospitality, banking, or MFI	Verified profile, role, and module entitlement
Member / Employee	Self-service and assigned tasks	Permitted records only	RLS and server permission checks

17.2 Appendix B — Master feature status matrix

Module	Status	UI evidence	Server evidence
Public Brand and Marketing Entry	IMPLEMENTED	client/src/pages/Home.tsx	server/_core/apiApp.ts
Authentication and Secure Onboarding	IMPLEMENTED	client/src/components/PublicAuthGateway.jsx; client/src/App.tsx	server/_core/apiApp.ts; server/_core/oauth.ts; server/authHeaders.ts
Master Application Shell and Navigation	IMPLEMENTED	client/src/App.tsx; client/src/BusinessSphereDashboard.jsx	server/routers.ts; server/_core/apiApp.ts
Profile Identity Center	IMPLEMENTED	client/src/components/ProfileIdentityCenter.jsx	server/profileIdentity.ts; server/routers.ts
Executive Dashboard	IMPLEMENTED	client/src/components/ExecutiveCommandCenter.jsx; client/src/BusinessSphereDashboard.jsx	server/dashboardReports.ts; server/routers.ts
CRM and Customer Pipeline	TESTING	client/src/components/CommercialCommandCenters.jsx; client/src/BusinessSphereDashboard.jsx	server/commercialCommandCenters.ts; server/salesInteractions.ts
Sales and Billing	TESTING	client/src/components/SalesDetailWorkspace.jsx; client/src/components/CommercialCommandCenters.jsx	server/routers.ts; server/subscriptionBilling.ts
Point of Sale	PASSED	client/src/components/OperationsCommandCenters.jsx; client/src/BusinessSphereDashboard.jsx	server/posWorkforceRpcAdapters.ts; server/posTransactionEngine.ts
Inventory and Warehouse Management	TESTING	client/src/components/OperationsCommandCenters.jsx	server/inventoryPersistence.ts; server/routers.ts
Procurement and Vendor Management	TESTING	client/src/components/OperationsCommandCenters.jsx	server/procurementPersistence.ts
Finance and Accounting	TESTING	client/src/components/FinanceCommandCenters.jsx	server/financeCommandCenters.ts; server/financePersistence.ts
Reports and Scheduled Reporting	TESTING	client/src/components/FinanceCommandCenters.jsx; client/src/BusinessSphereDashboard.jsx	server/dashboardReports.ts; server/reportSchedules.ts
Human Resources and Payroll	TESTING	client/src/components/PeopleCommandCenters.jsx; client/src/components/EmployeePortalWorkspace.jsx	server/teamWorkforce.ts; server/tanzaniaPayroll.ts; server/routers.ts
Manufacturing and Work Orders	PARTIALLY IMPLEMENTED	client/src/BusinessSphereDashboard.jsx; client/src/components/OperationsCommandCenters.jsx	server/manufacturingPersistence.ts

Module	Status	UI evidence	Server evidence
Supply Chain and Fleet	TESTING	client/src/components/OperationsCommandCenters.jsx; client/src/components/ FleetWorkspace.jsx	server/operationsCommandCenters.ts
Marketing Campaigns	TESTING	client/src/components/CommercialCommandCenters.jsx	server/ emailTemplateWorkflow.ts
E-Commerce Storefront	PARTIALLY IM- PLEMENTED	client/src/components/CommercialCommandCenters.jsx	server/operationsCommandCenters.ts
Documents and Secure Files	TESTING	client/src/components/ PeopleCommandCenters.jsx; client/src/components/ ProfileIdentityCenter.jsx	server/storage.ts; server/documentDoc
Projects and Task Management	PARTIALLY IM- PLEMENTED	client/src/components/ PeopleCommandCenters.jsx	server/routers.ts; server/ projectsPersistence.ts
Customer Support and Helpdesk	TESTING	client/src/components/IntelligenceCommandCenters.jsx; client/src/components/ PeopleCommandCenters.jsx	server/supportOperations.ts; server/supportMetrics.ts
Enterprise Analytics and BI	PARTIALLY IM- PLEMENTED	client/src/components/IntelligenceCommandCenters.jsx; client/src/components/PredictiveAnalyticsWidgetIntelligence.ts	server/analysisAndReports.ts; server/analysisWidgetIntelligence.ts
Notifications and Alerting	TESTING	client/src/ BusinessSphereDashboard.jsx; module workspaces	server/notificationHistory.ts; server/transactionalEmail.ts
Activity Stream and Audit Evidence	TESTING	client/src/components/ ComplianceAuditLogView.tsx; dashboard views	server/auditLogs.ts; server/ tenantAuditViewer.ts
Integration Hub	CONFIGURA- TION REQUIRED	client/src/components/ FinanceCommandCenters.jsx; client/src/components/ PeopleCommandCenters.jsx	server/webhooks.ts; server/ transactionalEmail.ts; server/in- tegration connections
Workflow Studio and Marketplace	PARTIALLY IM- PLEMENTED	client/src/components/ PeopleCommandCenters.jsx; dashboard registry	server/workflows.ts; server/ aiApprovals.ts
Collaboration Hub	PARTIALLY IM- PLEMENTED	client/src/components/ PeopleCommandCenters.jsx	server/collaborationEmailLinkCheck server/ collaborationPersistence.ts
TRA, VFD, and Tanzania Fiscalization	CONFIGURA- TION REQUIRED	client/src/components/ TraPortalModule.jsx	server/traFiscalRouter.ts; server/traFiscal.ts; server/ traZReportArchive.ts
AI Assistant and Smart Intelligence	EXTERNAL SER- VICE REQUIRED	client/src/components/ AIChatBox.tsx; client/src/com- ponents/IntelligenceCommandCenters.jsx	server/smartAssistant.ts; server/ aiApprovals.ts
WhatsApp Web Integra- tion	CONFIGURA- TION REQUIRED	client/src/ BusinessSphereDashboard.jsx; messaging surfaces	server/whatsappProvider.ts; server/whatsappSecurity.ts

Module	Status	UI evidence	Server evidence
Microfinance	TESTING	client/src/components/ MicrofinanceWorkspace.jsx; client/src/components/MicrofinanceGovernanceDialogs.jsx	server/ microfinanceOperations.ts
Money Agent	IMPLEMENTED	client/src/components/ MoneyAgentWorkspace.jsx; client/src/components/ SectorCommandCenters.jsx	server/ moneyAgentOperations.ts
VICOBA, SACCOS, and Community Groups	TESTING	client/src/components/ SectorCommandCenters.jsx; community group surfaces	server/communityGroups.ts; server/routers.ts
Healthcare and Clinic	PARTIALLY IM- PLEMENTED	client/src/components/ HealthcareClinicWorkspace.jsx	server/ healthcareOperations.ts; server/ healthcareInteroperability.ts; server/healthcareReminders.ts
School Management	PASSED	client/src/components/ SchoolWorkspace.jsx	server/schoolOperations.ts
Pharmacy Management	TESTING	client/src/components/ PharmacyWorkspace.jsx	server/pharmacyOperations.ts
Hotel and Hospitality	TESTING	client/src/components/ HospitalityWorkspace.jsx; client/src/components/ VerticalCommandCenters.jsx	server/hospitalityOperations.ts
Restaurant and Food & Beverage	TESTING	client/src/components/ RestaurantWorkspace.jsx; client/src/components/RestaurantTower/FiscalPanel.jsx	server/ restaurantManagement.ts;
Banking and MFI	TESTING	client/src/components/ BankMfiWorkspace.jsx; client/src/components/ SectorCommandCenters.jsx	server/bankMfiOperations.ts
Employee Portal	IMPLEMENTED	client/src/components/ EmployeePortalWorkspace.jsx	server/teamWorkforce.ts; server/employeePortal.ts
Property Management	IMPLEMENTED	client/src/components/PropertyManagementWorkspace.jsx	server/propertyManagementOperations.ts; server/propertyManagement.ts
Subscription and Billing	IMPLEMENTED	client/src/components/SubscriptionBillingWorkspace.jsx; client/src/ lib/subscriptionAccess.js	server/SubscriptionBilling.ts
Global Admin Control Center	IMPLEMENTED	client/src/components/ GlobalAdminControlCenter.tsx	server/globalAdmin.ts; server/ routers.ts
Enterprise Settings and Security Control Center	IMPLEMENTED	client/src/components/DashboardPreferences/EnterpriseSettings.ts; client/src/components/ ProfileIdentityCenter.jsx	server/workspaceBranding.ts; server/roleChangeApprovals.ts
Predictive Analytics	PARTIALLY IM- PLEMENTED	client/src/components/PredictiveAnalytics/SmartIntelligence.ts	server/smartAssistant.ts

17.3 Appendix C — Glossary

Term	English	Kiswahili
RLS	Row Level Security; database policy enforcement over which rows a role can read or change.	Usalama wa kiwango cha mstari; sera za database zinazoamua rows ambazo role inaweza kusoma au kubadilisha.
Company scope	The organization boundary used to separate tenant operations.	Mpaka wa kampuni unaotenganisha shughuli za tenant.
Entitlement	A server/database-confirmed permission to use a package or module.	Ruhusa iliyothibitishwa na server/database ya kutumia package au module.
Idempotency	Repeating a request with the same key does not create a duplicate operation.	Kurudia request yenye key ileile hakutengenezi operation ya pili.
SECURITY DE-FINER	A database function that runs with owner privileges and therefore requires narrow grants and safe search_path.	Function ya database inayotumia mam-laka ya owner; inahitaji grants finyu na search_path salama.
FREE_15	The free subscription package at TZS 0 for 15 days, with no automatic charge.	Kifurushi cha bure cha TZS 0 kwa siku 15 bila automatic charge.
Calendar month	A billing period calculated with calendar-month arithmetic rather than a fixed number of days.	Kipindi kinachohesabiwa kwa miezi ya kalenda, si idadi ya siku isiyobadilika.
Fail closed	Deny access or mutation when verification is missing rather than guessing success.	Kukataa access au mutation pale uthibitisho haupo badala ya kukisia mafanikio.

17.4 Appendix D — Verification record and references

Repository version: 1.0.0 from package.json. Documentation date: 24 August 2026. Live Supabase audit: read-only. Tables: 542; public: 519; auth: 23; RLS enabled: 535; RLS not enabled: 7; live migration records: 133; security advisor lints: 119; performance advisor lints: 851.

References: (1) client/src/App.tsx and the public authentication gateway; (2) client/src/BusinessSphereDashboard.jsx and the authenticated dashboard; (3) server/_core/apiApp.ts and server/routers.ts; (4) server/subscriptionBilling.ts; (5) server/propertyManagementOperations.ts; (6) server/smartAssistant.ts; (7) server/traFiscalRouter.ts; (8) FULL_SYSTEM_AUDIT_REPORT.md; (9) FULL_SYSTEM_IMPLEMENTATION_MATRIX.md; (10) package.json; (11) Supabase Row Level Security documentation (the official URL is preserved in the Markdown reference list). Documentation QA statement: the book was generated from repository files and read-only live evidence. It does not expose credentials, does not fabricate screenshots, does not create a duplicate auth/users architecture, and explicitly records known limitations and external prerequisites.